

# Bd. 02 - Theoreme der Logik

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# Kapitel 1

## Strukturgesetze der Junktoren

### 1.1 Selbstbezüglichkeit und Idempotenz

Im Kapitel verwendete Symbole

Aussagevariablen:  $P$

**Theorem 2.1.1.1 (Reflexivität der Implikation).**

$$P \vdash P$$

$$\begin{array}{l} 1 \quad 1 \quad P \\ A \end{array}$$

**Theorem 2.1.1.2 (Reflexivität der Implikation).**

$$P \rightarrow P$$

$$\begin{array}{l} 1 \quad 1 \quad P \\ 2 \quad P \rightarrow P \\ A \\ \rightarrow I(1,1) \end{array}$$

**Theorem 2.1.1.3 (Reflexivität des Bikonditionals).**

$$P \leftrightarrow P$$

$$\begin{array}{l} 1 \quad P \rightarrow P \\ 2 \quad P \leftrightarrow P \\ \text{Th. 2.1.1.2} \\ \leftrightarrow I(1,1) \end{array}$$

**Theorem 2.1.1.4 (Idempotenzgesetz für  $\vee$ ).**

$$P \vee P \dashv\vdash P$$

*Beweis.*

$\vdash$ :

|   |   |            |                         |
|---|---|------------|-------------------------|
| 1 | 1 | $P \vee P$ | $A$                     |
| 2 | 2 | $P$        | $A$                     |
| 1 | 3 | $P$        | $\vee E(1, 2, 2, 2, 2)$ |

$\dashv$ :

|   |   |            |              |
|---|---|------------|--------------|
| 1 | 1 | $P$        | $A$          |
| 1 | 2 | $P \vee P$ | $\vee I1(1)$ |

□

**Theorem 2.1.1.5 (Idempotenzgesetz für  $\wedge$ ).**

$$P \wedge P \dashv\vdash P$$

*Beweis.*

$\vdash$ :

|   |   |              |                |
|---|---|--------------|----------------|
| 1 | 1 | $P \wedge P$ | $A$            |
| 1 | 2 | $P$          | $\wedge E1(1)$ |

$\dashv$ :

|   |   |              |                               |
|---|---|--------------|-------------------------------|
| 1 | 1 | $P$          | $A$                           |
| 1 | 2 | $P \wedge P$ | $\wedge I^*(\text{Wdh}_2(1))$ |

□

## 1.2 Kommutativgesetze

**Im Kapitel verwendete Symbole**

**Aussagevariablen:**  $P, Q$

**Theorem 2.1.2.1 (Kommutativgesetz für  $\vee$ ).**

$$P \vee Q \vdash Q \vee P$$

|   |   |            |                         |
|---|---|------------|-------------------------|
| 1 | 1 | $P \vee Q$ | $A$                     |
| 2 | 2 | $P$        | $A$                     |
| 2 | 3 | $Q \vee P$ | $\vee I2(2)$            |
| 4 | 4 | $Q$        | $A$                     |
| 4 | 5 | $Q \vee P$ | $\vee I1(4)$            |
| 1 | 6 | $Q \vee P$ | $\vee E(1, 2, 3, 4, 5)$ |

**Theorem 2.1.2.2 (Kommutativgesetz für  $\vee$ ).**

$$P \vee Q \dashv\vdash Q \vee P$$

*Beweis.*

Hinrichtung:

|   |   |                                 |                       |
|---|---|---------------------------------|-----------------------|
| 1 | 1 | $P \vee Q$                      | $A$                   |
| 1 | 2 | $Q \vee P$                      | Th. 2.1.2.1(1)        |
|   | 3 | $P \vee Q \rightarrow Q \vee P$ | $\rightarrow I(1, 2)$ |

Rückrichtung:

|   |   |                                 |                       |
|---|---|---------------------------------|-----------------------|
| 4 | 4 | $Q \vee P$                      | $A$                   |
| 4 | 5 | $P \vee Q$                      | Th. 2.1.2.1(4)        |
|   | 6 | $Q \vee P \rightarrow P \vee Q$ | $\rightarrow I(4, 5)$ |

Zusammenführung:

$$7 \quad P \vee Q \leftrightarrow Q \vee P \quad \leftrightarrow I(3, 6)$$

□

**Theorem 2.1.2.3 (Kommutativgesetz für  $\wedge$ ).**

$$P \wedge Q \vdash Q \wedge P$$

|   |   |              |                  |
|---|---|--------------|------------------|
| 1 | 1 | $P \wedge Q$ | $A$              |
| 1 | 2 | $P$          | $\wedge E1(1)$   |
| 1 | 3 | $Q$          | $\wedge E2(1)$   |
| 1 | 4 | $Q \wedge P$ | $\wedge I(3, 2)$ |

**Theorem 2.1.2.4 (Kommutativgesetz für  $\wedge$ ).**

$$P \wedge Q \dashv\vdash Q \wedge P$$

*Beweis.*

$\vdash$ :

|   |   |              |                |
|---|---|--------------|----------------|
| 1 | 1 | $P \wedge Q$ | $A$            |
| 1 | 2 | $Q \wedge P$ | Th. 2.1.2.3(1) |

$\dashv$ :

|   |   |              |                |
|---|---|--------------|----------------|
| 1 | 1 | $Q \wedge P$ | $A$            |
| 1 | 2 | $P \wedge Q$ | Th. 2.1.2.3(1) |

□

**Theorem 2.1.2.5 (Kommutativgesetz für  $\leftrightarrow$ ).**

$$P \leftrightarrow Q \vdash Q \leftrightarrow P$$

|   |   |                       |                           |
|---|---|-----------------------|---------------------------|
| 1 | 1 | $P \leftrightarrow Q$ | $A$                       |
| 1 | 2 | $P \rightarrow Q$     | $\leftrightarrow E1(1)$   |
| 1 | 3 | $Q \rightarrow P$     | $\leftrightarrow E2(1)$   |
| 1 | 4 | $Q \leftrightarrow P$ | $\leftrightarrow I(3, 2)$ |

**Theorem 2.1.2.6 (Kommutativgesetz für  $\leftrightarrow$ ).**

$$P \leftrightarrow Q \dashv\vdash Q \leftrightarrow P$$

*Beweis.*

$\vdash$ :

|   |   |                       |                |
|---|---|-----------------------|----------------|
| 1 | 1 | $P \leftrightarrow Q$ | $A$            |
| 1 | 2 | $Q \leftrightarrow P$ | Th. 2.1.2.5(1) |

$\dashv$ :

|   |   |                       |                |
|---|---|-----------------------|----------------|
| 1 | 1 | $Q \leftrightarrow P$ | $A$            |
| 1 | 2 | $P \leftrightarrow Q$ | Th. 2.1.2.5(1) |

□

## 1.3 Assoziativgesetze

Im Kapitel verwendete Symbole

Aussagevariablen:  $P, Q, R$

**Theorem 2.1.3.1 (Assoziativgesetz für  $\vee$ ).**

$$P \vee (Q \vee R) \dashv\vdash (P \vee Q) \vee R$$

*Beweis.*

$\vdash$ :

|   |    |                     |                          |
|---|----|---------------------|--------------------------|
| 1 | 1  | $P \vee (Q \vee R)$ | $A$                      |
| 2 | 2  | $P$                 | $A$                      |
| 2 | 3  | $P \vee Q$          | $\vee I1(2)$             |
| 2 | 4  | $(P \vee Q) \vee R$ | $\vee I1(3)$             |
| 5 | 5  | $Q \vee R$          | $A$                      |
| 6 | 6  | $Q$                 | $A$                      |
| 6 | 7  | $P \vee Q$          | $\vee I2(6)$             |
| 6 | 8  | $(P \vee Q) \vee R$ | $\vee I1(7)$             |
| 9 | 9  | $R$                 | $A$                      |
| 9 | 10 | $(P \vee Q) \vee R$ | $\vee I2(9)$             |
| 5 | 11 | $(P \vee Q) \vee R$ | $\vee E(5, 6, 8, 9, 10)$ |
| 1 | 12 | $(P \vee Q) \vee R$ | $\vee E(1, 2, 4, 5, 11)$ |

$\dashv\vdash$ :

|   |    |                     |                          |
|---|----|---------------------|--------------------------|
| 1 | 1  | $(P \vee Q) \vee R$ | $A$                      |
| 2 | 2  | $P \vee Q$          | $A$                      |
| 3 | 3  | $P$                 | $A$                      |
| 3 | 4  | $P \vee (Q \vee R)$ | $\vee I1(3)$             |
| 5 | 5  | $Q$                 | $A$                      |
| 5 | 6  | $Q \vee R$          | $\vee I1(5)$             |
| 5 | 7  | $P \vee (Q \vee R)$ | $\vee I2(6)$             |
| 2 | 8  | $P \vee (Q \vee R)$ | $\vee E(2, 3, 4, 5, 7)$  |
| 9 | 9  | $R$                 | $A$                      |
| 9 | 10 | $Q \vee R$          | $\vee I2(9)$             |
| 9 | 11 | $P \vee (Q \vee R)$ | $\vee I2(10)$            |
| 1 | 12 | $P \vee (Q \vee R)$ | $\vee E(1, 2, 8, 9, 11)$ |

□

**Theorem 2.1.3.2 (Assoziativgesetz für  $\wedge$ ).**

$$P \wedge (Q \wedge R) \dashv\vdash (P \wedge Q) \wedge R$$

*Beweis.*

$\vdash$ :

|   |   |                         |                  |
|---|---|-------------------------|------------------|
| 1 | 1 | $P \wedge (Q \wedge R)$ | $A$              |
| 1 | 2 | $P$                     | $\wedge E1(1)$   |
| 1 | 3 | $Q \wedge R$            | $\wedge E2(1)$   |
| 1 | 4 | $Q$                     | $\wedge E1(3)$   |
| 1 | 5 | $R$                     | $\wedge E2(3)$   |
| 1 | 6 | $P \wedge Q$            | $\wedge I(2, 4)$ |
| 1 | 7 | $(P \wedge Q) \wedge R$ | $\wedge I(6, 5)$ |

$\dashv$ :

|   |   |                         |                  |
|---|---|-------------------------|------------------|
| 1 | 1 | $(P \wedge Q) \wedge R$ | $A$              |
| 1 | 2 | $P \wedge Q$            | $\wedge E1(1)$   |
| 1 | 3 | $R$                     | $\wedge E2(1)$   |
| 1 | 4 | $P$                     | $\wedge E1(2)$   |
| 1 | 5 | $Q$                     | $\wedge E2(2)$   |
| 1 | 6 | $Q \wedge R$            | $\wedge I(5, 3)$ |
| 1 | 7 | $P \wedge (Q \wedge R)$ | $\wedge I(4, 6)$ |

□

*Bemerkung.* Da  $\vee$  und  $\wedge$  assoziativ sind, ist jede Klammerung einer endlichen Kette gleichartiger Verknüpfungen äquivalent. Wir vereinbaren die *rechte* Klammerkonvention und lassen Klammern künftig weg, wo keine Mehrdeutigkeit entsteht.

**Definition 2.1.3.1 (Meta-Notation für dreifache Konjunktion).**

$$P \wedge Q \wedge R := P \wedge (Q \wedge R)$$

**Definition 2.1.3.2 (Meta-Notation für dreifache Disjunktion).**

$$P \vee Q \vee R := P \vee (Q \vee R)$$

## 1.4 Dreifach-Formeln

**Im Kapitel verwendete Symbole**

**Aussagevariablen:**  $P, Q, R, P_1, P_2, P_3$

### 1.4.1 Elimination dreifacher Konjugationen

**Theorem 2.1.4.1.**

$$P \wedge Q \wedge R \vdash Q$$

|   |   |                       |                |
|---|---|-----------------------|----------------|
| 1 | 1 | $P \wedge Q \wedge R$ | $A$            |
| 1 | 2 | $Q \wedge R$          | $\wedge E2(1)$ |
| 1 | 3 | $Q$                   | $\wedge E1(2)$ |

**Theorem 2.1.4.2.**

$$P \wedge Q \wedge R \vdash R$$

|   |   |                       |                |
|---|---|-----------------------|----------------|
| 1 | 1 | $P \wedge Q \wedge R$ | $A$            |
| 1 | 2 | $Q \wedge R$          | $\wedge E2(1)$ |
| 1 | 3 | $R$                   | $\wedge E2(2)$ |

**Theorem 2.1.4.3.**

$$P, Q, R \vdash P \wedge Q \wedge R$$

|       |   |                       |                 |
|-------|---|-----------------------|-----------------|
| 1     | 1 | $P$                   | $A$             |
| 2     | 2 | $Q$                   | $A$             |
| 3     | 3 | $R$                   | $A$             |
| 2,3   | 4 | $Q \wedge R$          | $\wedge I(2,3)$ |
| 1,2,3 | 5 | $P \wedge Q \wedge R$ | $\wedge I(1,4)$ |

### 1.4.2 Einführung dreifacher Disjunktion

**Theorem 2.1.4.4.**

$$Q \vdash P \vee Q \vee R$$

|   |   |                   |              |
|---|---|-------------------|--------------|
| 1 | 1 | $Q$               | $A$          |
| 1 | 2 | $Q \vee R$        | $\vee I1(1)$ |
| 1 | 3 | $P \vee Q \vee R$ | $\vee I2(2)$ |

**Theorem 2.1.4.5.**

$$P \vdash P \vee Q \vee R$$

|   |   |                   |              |
|---|---|-------------------|--------------|
| 1 | 1 | $P$               | $A$          |
| 1 | 2 | $P \vee Q \vee R$ | $\vee I1(1)$ |

**Theorem 2.1.4.6.**

$$R \vdash P \vee Q \vee R$$

|   |   |                   |              |
|---|---|-------------------|--------------|
| 1 | 1 | $R$               | $A$          |
| 1 | 2 | $Q \vee R$        | $\vee I2(1)$ |
| 1 | 3 | $P \vee Q \vee R$ | $\vee I2(2)$ |

### 1.4.3 Elimination mehrfacher Disjunktion

**Theorem 2.1.4.7.**

$$P_1 \vee P_2 \vee P_3, [P_1] \dot{:} R, [P_2] \dot{:} R, [P_3] \dot{:} R \vdash R$$

|   |    |                         |                         |
|---|----|-------------------------|-------------------------|
| 1 | 1  | $P_1 \vee P_2 \vee P_3$ | $A$                     |
| 2 | 2  | $P_1$                   | $A$                     |
| 2 | 3  | $R$                     | $[P_1] \dot{:} R(2)$    |
| 4 | 4  | $P_2 \vee P_3$          | $A$                     |
| 5 | 5  | $P_2$                   | $A$                     |
| 5 | 6  | $R$                     | $[P_2] \dot{:} R(5)$    |
| 7 | 7  | $P_3$                   | $A$                     |
| 7 | 8  | $R$                     | $[P_3] \dot{:} R(7)$    |
| 4 | 9  | $R$                     | $\vee E(4, 5, 6, 7, 8)$ |
| 1 | 10 | $R$                     | $\vee E(1, 2, 3, 4, 9)$ |

**Theorem 2.1.4.8.**

$$P_1 \vee P_2 \vee P_3 \vee P_4, [P_1] \dot{:} R, [P_2] \dot{:} R, [P_3] \dot{:} R, [P_4] \dot{:} R \vdash R$$

|   |   |                                  |                      |
|---|---|----------------------------------|----------------------|
| 1 | 1 | $P_1 \vee P_2 \vee P_3 \vee P_4$ | $A$                  |
| 2 | 2 | $P_1$                            | $A$                  |
| 2 | 3 | $R$                              | $[P_1] \dot{:} R(2)$ |
| 4 | 4 | $P_2 \vee P_3 \vee P_4$          | $A$                  |
| 5 | 5 | $P_2$                            | $A$                  |
| 5 | 6 | $R$                              | $[P_2] \dot{:} R(5)$ |
| 7 | 7 | $P_3 \vee P_4$                   | $A$                  |
| 8 | 8 | $P_3$                            | $A$                  |

|    |    |       |                           |
|----|----|-------|---------------------------|
| 8  | 9  | $R$   | $[P_3]:R(8)$              |
| 10 | 10 | $P_4$ | $A$                       |
| 10 | 11 | $R$   | $[P_4]:R(10)$             |
| 7  | 12 | $R$   | $\vee E(7, 8, 9, 10, 11)$ |
| 4  | 13 | $R$   | $\vee E(4, 5, 6, 7, 12)$  |
| 1  | 14 | $R$   | $\vee E(1, 2, 3, 4, 13)$  |

**Theorem 2.1.4.9 (Elimination einer fünffachen Disjunktion).**

**Aussagevariablen:**  $P_1, P_2, P_3, P_4, P_5, R$

$$P_1 \vee P_2 \vee P_3 \vee P_4 \vee P_5, [P_1]:R, [P_2]:R, [P_3]:R, \\ [P_4]:R, [P_5]:R \vdash R$$

|    |    |                                           |                              |
|----|----|-------------------------------------------|------------------------------|
| 1  | 1  | $P_1 \vee P_2 \vee P_3 \vee P_4 \vee P_5$ | $A$                          |
| 2  | 2  | $P_1$                                     | $A$                          |
| 2  | 3  | $R$                                       | $[P_1]:R(2)$                 |
| 4  | 4  | $P_2 \vee P_3 \vee P_4 \vee P_5$          | $A$                          |
| 5  | 5  | $P_2$                                     | $A$                          |
| 5  | 6  | $R$                                       | $[P_2]:R(5)$                 |
| 7  | 7  | $P_3 \vee P_4 \vee P_5$                   | $A$                          |
| 8  | 8  | $P_3$                                     | $A$                          |
| 8  | 9  | $R$                                       | $[P_3]:R(8)$                 |
| 10 | 10 | $P_4 \vee P_5$                            | $A$                          |
| 11 | 11 | $P_4$                                     | $A$                          |
| 11 | 12 | $R$                                       | $[P_4]:R(11)$                |
| 13 | 13 | $P_5$                                     | $A$                          |
| 13 | 14 | $R$                                       | $[P_5]:R(13)$                |
| 10 | 15 | $R$                                       | $\vee E(10, 11, 12, 13, 14)$ |
| 7  | 16 | $R$                                       | $\vee E(7, 8, 9, 10, 15)$    |
| 4  | 17 | $R$                                       | $\vee E(4, 5, 6, 7, 16)$     |
| 1  | 18 | $R$                                       | $\vee E(1, 2, 3, 4, 17)$     |

## 1.5 Distributivgesetze von $\wedge$ und $\vee$

**Im Kapitel verwendete Symbole**

**Aussagevariablen:**  $P, Q, R, S$

**Theorem 2.1.5.1 (Distributivgesetz).**

$$P \wedge (Q \vee R) \dashv\vdash (P \wedge Q) \vee (P \wedge R)$$

*Beweis.*

$\vdash$ :

|     |    |                                  |                     |
|-----|----|----------------------------------|---------------------|
| 1   | 1  | $P \wedge (Q \vee R)$            | $A$                 |
| 1   | 2  | $P$                              | $\wedge E1(1)$      |
| 1   | 3  | $Q \vee R$                       | $\wedge E2(1)$      |
| 4   | 4  | $Q$                              | $A$                 |
| 1,4 | 5  | $P \wedge Q$                     | $\wedge I(2,4)$     |
| 1,4 | 6  | $(P \wedge Q) \vee (P \wedge R)$ | $\vee I1(5)$        |
| 7   | 7  | $R$                              | $A$                 |
| 1,7 | 8  | $P \wedge R$                     | $\wedge I(2,7)$     |
| 1,7 | 9  | $(P \wedge Q) \vee (P \wedge R)$ | $\vee I2(8)$        |
| 1   | 10 | $(P \wedge Q) \vee (P \wedge R)$ | $\vee E(3,4,6,7,9)$ |

$\dashv\vdash$ :

|   |    |                                  |                      |
|---|----|----------------------------------|----------------------|
| 1 | 1  | $(P \wedge Q) \vee (P \wedge R)$ | $A$                  |
| 2 | 2  | $P \wedge Q$                     | $A$                  |
| 2 | 3  | $P$                              | $\wedge E1(2)$       |
| 2 | 4  | $Q$                              | $\wedge E2(2)$       |
| 2 | 5  | $Q \vee R$                       | $\vee I1(4)$         |
| 2 | 6  | $P \wedge (Q \vee R)$            | $\wedge I(3,5)$      |
| 7 | 7  | $P \wedge R$                     | $A$                  |
| 7 | 8  | $P$                              | $\wedge E1(7)$       |
| 7 | 9  | $R$                              | $\wedge E2(7)$       |
| 7 | 10 | $Q \vee R$                       | $\vee I2(9)$         |
| 7 | 11 | $P \wedge (Q \vee R)$            | $\wedge I(8,10)$     |
| 1 | 12 | $P \wedge (Q \vee R)$            | $\vee E(1,2,6,7,11)$ |

□

**Theorem 2.1.5.2 (Distributivgesetz).**

$$(P \vee Q) \wedge R \dashv\vdash (P \wedge R) \vee (Q \wedge R)$$

*Beweis.*

$\vdash$ :

|   |   |                       |                |
|---|---|-----------------------|----------------|
| 1 | 1 | $(P \vee Q) \wedge R$ | $A$            |
| 1 | 2 | $R$                   | $\wedge E2(1)$ |

|     |    |                                  |                         |
|-----|----|----------------------------------|-------------------------|
| 1   | 3  | $P \vee Q$                       | $\wedge E1(1)$          |
| 4   | 4  | $P$                              | $A$                     |
| 1,4 | 5  | $P \wedge R$                     | $\wedge I(4, 2)$        |
| 1,4 | 6  | $(P \wedge R) \vee (Q \wedge R)$ | $\vee I1(5)$            |
| 7   | 7  | $Q$                              | $A$                     |
| 1,7 | 8  | $Q \wedge R$                     | $\wedge I(7, 2)$        |
| 1,7 | 9  | $(P \wedge R) \vee (Q \wedge R)$ | $\vee I2(8)$            |
| 1   | 10 | $(P \wedge R) \vee (Q \wedge R)$ | $\vee E(3, 4, 6, 7, 9)$ |

$\vdash$ :

|   |    |                                  |                          |
|---|----|----------------------------------|--------------------------|
| 1 | 1  | $(P \wedge R) \vee (Q \wedge R)$ | $A$                      |
| 2 | 2  | $P \wedge R$                     | $A$                      |
| 2 | 3  | $P$                              | $\wedge E1(2)$           |
| 2 | 4  | $R$                              | $\wedge E2(2)$           |
| 2 | 5  | $P \vee Q$                       | $\vee I1(3)$             |
| 2 | 6  | $(P \vee Q) \wedge R$            | $\wedge I(5, 4)$         |
| 7 | 7  | $Q \wedge R$                     | $A$                      |
| 7 | 8  | $Q$                              | $\wedge E1(7)$           |
| 7 | 9  | $R$                              | $\wedge E2(7)$           |
| 7 | 10 | $P \vee Q$                       | $\vee I2(8)$             |
| 7 | 11 | $(P \vee Q) \wedge R$            | $\wedge I(10, 9)$        |
| 1 | 12 | $(P \vee Q) \wedge R$            | $\vee E(1, 2, 6, 7, 11)$ |

□

**Theorem 2.1.5.3 (Erweitertes Distributivgesetz).**

$$(P \vee Q) \wedge (R \vee S) \dashv\vdash (P \wedge R) \vee (P \wedge S) \vee (Q \wedge R) \vee (Q \wedge S)$$

|   |                                                    |                                            |
|---|----------------------------------------------------|--------------------------------------------|
| 1 | $(P \vee Q) \wedge (R \vee S)$                     | Th. 2.1.5.2                                |
|   | $\leftrightarrow (P \wedge (R \vee S))$            |                                            |
|   | $\vee (Q \wedge (R \vee S))$                       |                                            |
| 2 | $(P \vee Q) \wedge (R \vee S)$                     | $\leftrightarrow S(\text{Th. 2.1.2.4}, 1)$ |
|   | $\leftrightarrow ((R \vee S) \wedge P)$            |                                            |
|   | $\vee (Q \wedge (R \vee S))$                       |                                            |
| 3 | $(P \vee Q) \wedge (R \vee S)$                     | $\leftrightarrow S(\text{Th. 2.1.5.2}, 2)$ |
|   | $\leftrightarrow ((R \wedge P) \vee (S \wedge P))$ |                                            |
|   | $\vee (Q \wedge (R \vee S))$                       |                                            |
| 4 | $(P \vee Q) \wedge (R \vee S)$                     | $\leftrightarrow S(\text{Th. 2.1.2.4}, 3)$ |
|   | $\leftrightarrow ((P \wedge R) \vee (P \wedge S))$ |                                            |
|   | $\vee (Q \wedge (R \vee S))$                       |                                            |

$$\begin{array}{ll}
5 & (P \vee Q) \wedge (R \vee S) \leftrightarrow S(\text{Th. 2.1.2.4, 4}) \\
& \leftrightarrow ((P \wedge R) \vee (P \wedge S)) \\
& \quad \vee ((R \vee S) \wedge Q) \\
6 & (P \vee Q) \wedge (R \vee S) \leftrightarrow S(\text{Th. 2.1.5.2, 5}) \\
& \leftrightarrow ((P \wedge R) \vee (P \wedge S)) \\
& \quad \vee ((R \wedge Q) \vee (S \wedge Q)) \\
7 & (P \vee Q) \wedge (R \vee S) \leftrightarrow S(\text{Th. 2.1.2.4, 6}) \\
& \leftrightarrow ((P \wedge R) \vee (P \wedge S)) \\
& \quad \vee ((Q \wedge R) \vee (Q \wedge S)) \\
8 & (P \vee Q) \wedge (R \vee S) \leftrightarrow S(\text{Th. 2.1.3.1, 7}) \\
& \leftrightarrow (P \wedge R) \vee ((P \wedge S) \vee \\
& \quad ((Q \wedge R) \vee (Q \wedge S)))
\end{array}$$

**Theorem 2.1.5.4 (Distributivgesetz).**

$$P \vee (Q \wedge R) \dashv\vdash (P \vee Q) \wedge (P \vee R)$$

*Beweis.*

$\vdash$ :

|   |    |                                |                          |
|---|----|--------------------------------|--------------------------|
| 1 | 1  | $P \vee (Q \wedge R)$          | $A$                      |
| 2 | 2  | $P$                            | $A$                      |
| 2 | 3  | $P \vee Q$                     | $\vee I1(2)$             |
| 2 | 4  | $P \vee R$                     | $\vee I1(2)$             |
| 2 | 5  | $(P \vee Q) \wedge (P \vee R)$ | $\wedge I(3, 4)$         |
| 6 | 6  | $Q \wedge R$                   | $A$                      |
| 6 | 7  | $Q$                            | $\wedge E1(6)$           |
| 6 | 8  | $R$                            | $\wedge E2(6)$           |
| 6 | 9  | $P \vee Q$                     | $\vee I2(7)$             |
| 6 | 10 | $P \vee R$                     | $\vee I2(8)$             |
| 6 | 11 | $(P \vee Q) \wedge (P \vee R)$ | $\wedge I(9, 10)$        |
| 1 | 12 | $(P \vee Q) \wedge (P \vee R)$ | $\vee E(1, 2, 5, 6, 11)$ |

$\dashv\vdash$ :

|   |   |                                |                |
|---|---|--------------------------------|----------------|
| 1 | 1 | $(P \vee Q) \wedge (P \vee R)$ | $A$            |
| 1 | 2 | $P \vee Q$                     | $\wedge E1(1)$ |
| 1 | 3 | $P \vee R$                     | $\wedge E2(1)$ |
| 4 | 4 | $P$                            | $A$            |
| 4 | 5 | $P \vee (Q \wedge R)$          | $\vee I1(4)$   |
| 6 | 6 | $Q$                            | $A$            |
| 7 | 7 | $P$                            | $A$            |
| 7 | 8 | $P \vee (Q \wedge R)$          | $\vee I1(7)$   |
| 9 | 9 | $R$                            | $A$            |

|     |    |                       |                          |
|-----|----|-----------------------|--------------------------|
| 6,9 | 10 | $Q \wedge R$          | $\wedge I(6, 9)$         |
| 6,9 | 11 | $P \vee (Q \wedge R)$ | $\vee I2(10)$            |
| 1,6 | 12 | $P \vee (Q \wedge R)$ | $\vee E(3, 7, 8, 9, 11)$ |
| 1   | 13 | $P \vee (Q \wedge R)$ | $\vee E(2, 4, 5, 6, 12)$ |

□

**Theorem 2.1.5.5 (Distributivgesetz).**

$$(P \wedge Q) \vee R \dashv\vdash (P \vee R) \wedge (Q \vee R)$$

*Beweis.*

$\vdash$ :

|   |    |                                |                          |
|---|----|--------------------------------|--------------------------|
| 1 | 1  | $(P \wedge Q) \vee R$          | $A$                      |
| 2 | 2  | $R$                            | $A$                      |
| 2 | 3  | $P \vee R$                     | $\vee I2(2)$             |
| 2 | 4  | $Q \vee R$                     | $\vee I2(2)$             |
| 2 | 5  | $(P \vee R) \wedge (Q \vee R)$ | $\wedge I(3, 4)$         |
| 6 | 6  | $P \wedge Q$                   | $A$                      |
| 6 | 7  | $P$                            | $\wedge E1(6)$           |
| 6 | 8  | $Q$                            | $\wedge E2(6)$           |
| 6 | 9  | $P \vee R$                     | $\vee I1(7)$             |
| 6 | 10 | $Q \vee R$                     | $\vee I1(8)$             |
| 6 | 11 | $(P \vee R) \wedge (Q \vee R)$ | $\wedge I(9, 10)$        |
| 1 | 12 | $(P \vee R) \wedge (Q \vee R)$ | $\vee E(1, 2, 5, 6, 11)$ |

$\dashv\vdash$ :

|     |    |                                |                            |
|-----|----|--------------------------------|----------------------------|
| 1   | 1  | $(P \vee R) \wedge (Q \vee R)$ | $A$                        |
| 1   | 2  | $P \vee R$                     | $\wedge E1(1)$             |
| 1   | 3  | $Q \vee R$                     | $\wedge E2(1)$             |
| 4   | 4  | $P$                            | $A$                        |
| 5   | 5  | $Q$                            | $A$                        |
| 4,5 | 6  | $P \wedge Q$                   | $\wedge I(4, 5)$           |
| 4,5 | 7  | $(P \wedge Q) \vee R$          | $\vee I1(6)$               |
| 8   | 8  | $R$                            | $A$                        |
| 8   | 9  | $(P \wedge Q) \vee R$          | $\vee I2(8)$               |
| 1,4 | 10 | $(P \wedge Q) \vee R$          | $\vee E(3, 5, 7, 8, 9)$    |
| 11  | 11 | $R$                            | $A$                        |
| 11  | 12 | $(P \wedge Q) \vee R$          | $\vee I2(11)$              |
| 1   | 13 | $(P \wedge Q) \vee R$          | $\vee E(2, 4, 10, 11, 12)$ |

□

**Theorem 2.1.5.6 (Erweitertes Distributivgesetz).**

$$(P \wedge Q) \vee (R \wedge S) \dashv\vdash (P \vee R) \wedge (P \vee S) \wedge (Q \vee R) \wedge (Q \vee S)$$

|   |                                                         |                                            |
|---|---------------------------------------------------------|--------------------------------------------|
| 1 | $(P \wedge Q) \vee (R \wedge S)$                        | Th. 2.1.5.4                                |
|   | $\leftrightarrow ((P \wedge Q) \vee R)$                 |                                            |
|   | $\wedge ((P \wedge Q) \vee S)$                          |                                            |
| 2 | $(P \wedge Q) \vee (R \wedge S)$                        | $\leftrightarrow S(\text{Th. 2.1.5.5}, 1)$ |
|   | $\leftrightarrow ((P \vee R) \wedge (Q \vee R))$        |                                            |
|   | $\wedge ((P \wedge Q) \vee S)$                          |                                            |
| 3 | $(P \wedge Q) \vee (R \wedge S)$                        | $\leftrightarrow S(\text{Th. 2.1.5.5}, 2)$ |
|   | $\leftrightarrow ((P \vee R) \wedge (Q \vee R))$        |                                            |
|   | $\wedge ((P \vee S) \wedge (Q \vee S))$                 |                                            |
| 4 | $(P \wedge Q) \vee (R \wedge S)$                        | $\leftrightarrow S(\text{Th. 2.1.3.2}, 3)$ |
|   | $\leftrightarrow (P \vee R) \wedge ((Q \vee R) \wedge$  |                                            |
|   | $((P \vee S) \wedge (Q \vee S)))$                       |                                            |
| 5 | $(P \wedge Q) \vee (R \wedge S)$                        | $\leftrightarrow S(\text{Th. 2.1.3.2}, 4)$ |
|   | $\leftrightarrow (P \vee R) \wedge (((Q \vee R) \wedge$ |                                            |
|   | $(P \vee S)) \wedge (Q \vee S))$                        |                                            |
| 6 | $(P \wedge Q) \vee (R \wedge S)$                        | $\leftrightarrow S(\text{Th. 2.1.2.4}, 5)$ |
|   | $\leftrightarrow (P \vee R) \wedge (((P \vee S) \wedge$ |                                            |
|   | $(Q \vee R)) \wedge (Q \vee S))$                        |                                            |
| 7 | $(P \wedge Q) \vee (R \wedge S)$                        | $\leftrightarrow S(\text{Th. 2.1.3.2}, 6)$ |
|   | $\leftrightarrow (P \vee R) \wedge ((P \vee S) \wedge$  |                                            |
|   | $((Q \vee R) \wedge (Q \vee S)))$                       |                                            |

## 1.6 Doppelte Negation

**Im Kapitel verwendete Symbole**

**Aussagevariablen:**  $P, Q$

**Theorem 2.1.6.1 (Regel der doppelten Negation).**

$$P \dashv\vdash \neg\neg P$$

*Beweis.*

$\vdash$ :

|     |   |              |                 |
|-----|---|--------------|-----------------|
| 1   | 1 | $P$          | $A$             |
|     | 2 | $\neg P$     | $A$             |
| 1,2 | 3 | $\perp$      | $\perp I(1, 2)$ |
| 1   | 4 | $\neg\neg P$ | $\neg I(2, 3)$  |

$\dashv\vdash$ :

|     |   |              |                |
|-----|---|--------------|----------------|
| 1   | 1 | $\neg\neg P$ | $A$            |
|     | 2 | $\neg P$     | $A$            |
| 1,2 | 3 | $\perp$      | $\perp I(2,1)$ |
|     | 4 | $P$          | $\neg E(2,3)$  |

□

**Theorem 2.1.6.2 (Stabilität von  $\vee$  unter doppelter Negation).**

$$P \vee Q \dashv\vdash \neg\neg P \vee \neg\neg Q$$

*Beweis.*

$\vdash$ :

|   |   |                              |                     |
|---|---|------------------------------|---------------------|
| 1 | 1 | $P \vee Q$                   | $A$                 |
|   | 2 | $P$                          | $A$                 |
|   | 3 | $\neg\neg P$                 | Th. 2.1.6.1(2)      |
|   | 4 | $\neg\neg P \vee \neg\neg Q$ | $\vee I1(3)$        |
|   | 5 | $Q$                          | $A$                 |
|   | 6 | $\neg\neg Q$                 | Th. 2.1.6.1(5)      |
|   | 7 | $\neg\neg P \vee \neg\neg Q$ | $\vee I2(6)$        |
| 1 | 8 | $\neg\neg P \vee \neg\neg Q$ | $\vee E(1,2,4,5,7)$ |

$\dashv\vdash$ :

|   |   |                              |                     |
|---|---|------------------------------|---------------------|
| 1 | 1 | $\neg\neg P \vee \neg\neg Q$ | $A$                 |
|   | 2 | $\neg\neg P$                 | $A$                 |
|   | 3 | $P$                          | Th. 2.1.6.1(2)      |
|   | 4 | $P \vee Q$                   | $\vee I1(3)$        |
|   | 5 | $\neg\neg Q$                 | $A$                 |
|   | 6 | $Q$                          | Th. 2.1.6.1(5)      |
|   | 7 | $P \vee Q$                   | $\vee I2(6)$        |
| 1 | 8 | $P \vee Q$                   | $\vee E(1,2,4,5,7)$ |

□

**Theorem 2.1.6.3 (Stabilität von  $\wedge$  unter doppelter Negation).**

$$P \wedge Q \dashv\vdash \neg\neg P \wedge \neg\neg Q$$

*Beweis.*

$\vdash$ :

|   |   |              |     |
|---|---|--------------|-----|
| 1 | 1 | $P \wedge Q$ | $A$ |
|---|---|--------------|-----|

|   |   |                                |                  |
|---|---|--------------------------------|------------------|
| 1 | 2 | $P$                            | $\wedge E1(1)$   |
| 1 | 3 | $Q$                            | $\wedge E2(1)$   |
| 1 | 4 | $\neg\neg P$                   | Th. 2.1.6.1(2)   |
| 1 | 5 | $\neg\neg Q$                   | Th. 2.1.6.1(3)   |
| 1 | 6 | $\neg\neg P \wedge \neg\neg Q$ | $\wedge I(4, 5)$ |

$\vdash$ :

|   |   |                                |                  |
|---|---|--------------------------------|------------------|
| 1 | 1 | $\neg\neg P \wedge \neg\neg Q$ | $A$              |
| 1 | 2 | $\neg\neg P$                   | $\wedge E1(1)$   |
| 1 | 3 | $\neg\neg Q$                   | $\wedge E2(1)$   |
| 1 | 4 | $P$                            | Th. 2.1.6.1(2)   |
| 1 | 5 | $Q$                            | Th. 2.1.6.1(3)   |
| 1 | 6 | $P \wedge Q$                   | $\wedge I(4, 5)$ |

□

## 1.7 Gesetz vom ausgeschlossenen dritten

**Theorem 2.1.7.1 (Satz vom ausgeschlossenen Dritten).**

$$P \vee \neg P$$

|     |   |                       |                 |
|-----|---|-----------------------|-----------------|
| 1   | 1 | $\neg(P \vee \neg P)$ | $A$             |
| 2   | 2 | $P$                   | $A$             |
| 2   | 3 | $P \vee \neg P$       | $\vee I1(2)$    |
| 1,2 | 4 | $\perp$               | $\perp I(1, 3)$ |
| 1   | 5 | $\neg P$              | $\neg I(2, 4)$  |
| 1   | 6 | $P \vee \neg P$       | $\vee I2(5)$    |
| 1   | 7 | $\perp$               | $\perp I(1, 6)$ |
|     | 8 | $P \vee \neg P$       | $\neg E(1, 7)$  |

**Theorem 2.1.7.2 (Satz der Explosion).**

$$P \wedge \neg P \vdash Q$$

|   |   |                   |                 |
|---|---|-------------------|-----------------|
| 1 | 1 | $P \wedge \neg P$ | $A$             |
| 1 | 2 | $P$               | $\wedge E1(1)$  |
| 1 | 3 | $\neg P$          | $\wedge E2(1)$  |
| 1 | 4 | $\perp$           | $\perp I(3, 2)$ |
| 5 | 5 | $\neg Q$          | $A$             |
| 1 | 6 | $Q$               | $\neg E(5, 4)$  |

**Theorem 2.1.7.3 (Explosion ohne Konjunktion).**

$$P, \neg P \vdash Q$$

|     |   |                   |                  |
|-----|---|-------------------|------------------|
| 1   | 1 | $P$               | $A$              |
| 2   | 2 | $\neg P$          | $A$              |
| 1,2 | 3 | $P \wedge \neg P$ | $\wedge I(1, 2)$ |
| 1,2 | 4 | $Q$               | Th. 2.1.7.2(3)   |

**Theorem 2.1.7.4 (Explosion aus Falschheit).**

$$\perp \vdash P$$

|   |   |          |                |
|---|---|----------|----------------|
| 1 | 1 | $\perp$  | $A$            |
| 2 | 2 | $\neg P$ | $A$            |
| 1 | 3 | $P$      | $\neg E(2, 1)$ |

## 1.8 De-morgansche Gesetze

**Im Kapitel verwendete Symbole**

**Aussagevariablen:**  $P, Q$

**Theorem 2.1.8.1 (De Morgan).**

$$\neg(P \vee Q) \dashv\vdash \neg P \wedge \neg Q$$

*Beweis.*

Negation der ersten Aussage:

|     |   |                  |                 |
|-----|---|------------------|-----------------|
| 1   | 1 | $\neg(P \vee Q)$ | $A$             |
| 2   | 2 | $P$              | $A$             |
| 2   | 3 | $P \vee Q$       | $\vee I(2)$     |
| 1,2 | 4 | $\perp$          | $\perp I(1, 3)$ |
| 1   | 5 | $\neg P$         | $\neg I(2, 4)$  |

Negation der zweiten Aussage:

|     |   |            |                 |
|-----|---|------------|-----------------|
| 6   | 6 | $Q$        | $A$             |
| 6   | 7 | $P \vee Q$ | $\vee I(6)$     |
| 1,6 | 8 | $\perp$    | $\perp I(1, 7)$ |
| 1   | 9 | $\neg Q$   | $\neg I(6, 8)$  |

Zusammenführung der Hinrichtung:

|                             |                  |
|-----------------------------|------------------|
| 1 10 $\neg P \wedge \neg Q$ | $\wedge I(5, 9)$ |
|-----------------------------|------------------|

$\dashv$ :

|                            |                         |
|----------------------------|-------------------------|
| 1 1 $\neg P \wedge \neg Q$ | $A$                     |
| 1 2 $\neg P$               | $\wedge E1(1)$          |
| 1 3 $\neg Q$               | $\wedge E2(1)$          |
| 4 4 $P \vee Q$             | $A$                     |
| 5 5 $P$                    | $A$                     |
| 1,5 6 $\perp$              | $\perp I(2, 5)$         |
| 7 7 $Q$                    | $A$                     |
| 1,7 8 $\perp$              | $\perp I(3, 7)$         |
| 1,4 9 $\perp$              | $\vee E(4, 5, 6, 7, 8)$ |
| 1 10 $\neg(P \vee Q)$      | $\neg I(4, 9)$          |

□

**Theorem 2.1.8.2 (De Morgan 2).**

$$\neg(P \wedge Q) \dashv\vdash \neg P \vee \neg Q$$

*Beweis.*

$\vdash$ :

|                            |                          |
|----------------------------|--------------------------|
| 1 1 $\neg(P \wedge Q)$     | $A$                      |
| 2 $P \vee \neg P$          | Th. 2.1.7.1              |
| 3 3 $P$                    | $A$                      |
| 4 4 $Q$                    | $A$                      |
| 3,4 5 $P \wedge Q$         | $\wedge I(3, 4)$         |
| 1,3,4 6 $\perp$            | $\perp I(1, 5)$          |
| 1,3 7 $\neg Q$             | $\neg I(4, 6)$           |
| 1,3 8 $\neg P \vee \neg Q$ | $\vee I2(7)$             |
| 9 9 $\neg P$               | $A$                      |
| 9 10 $\neg P \vee \neg Q$  | $\vee I1(9)$             |
| 1 11 $\neg P \vee \neg Q$  | $\vee E(2, 3, 8, 9, 10)$ |

$\dashv$ :

|                          |                 |
|--------------------------|-----------------|
| 1 1 $\neg P \vee \neg Q$ | $A$             |
| 2 2 $P \wedge Q$         | $A$             |
| 2 3 $P$                  | $\wedge E1(2)$  |
| 2 4 $Q$                  | $\wedge E2(2)$  |
| 5 5 $\neg P$             | $A$             |
| 2,5 6 $\perp$            | $\perp I(3, 5)$ |

|     |    |                    |                          |
|-----|----|--------------------|--------------------------|
| 5   | 7  | $\neg(P \wedge Q)$ | $\neg I(2, 6)$           |
| 8   | 8  | $\neg Q$           | $A$                      |
| 2,8 | 9  | $\perp$            | $\perp I(4, 8)$          |
| 8   | 10 | $\neg(P \wedge Q)$ | $\neg I(2, 9)$           |
| 1   | 11 | $\neg(P \wedge Q)$ | $\vee E(1, 5, 7, 8, 10)$ |

□

**Theorem 2.1.8.3 (Fallunterscheidung für zwei Bedingungen).**

$$(P \wedge Q) \vee (\neg P \vee \neg Q)$$

|   |                                          |                         |
|---|------------------------------------------|-------------------------|
| 1 | $(P \wedge Q) \vee \neg(P \wedge Q)$     | Th. 2.1.7.1             |
| 2 | $P \wedge Q$                             | $A$                     |
| 2 | $(P \wedge Q) \vee (\neg P \vee \neg Q)$ | $\vee I1(2)$            |
| 4 | $\neg(P \wedge Q)$                       | $A$                     |
| 4 | $\neg P \vee \neg Q$                     | Th. 2.1.8.2(4)          |
| 4 | $(P \wedge Q) \vee (\neg P \vee \neg Q)$ | $\vee I2(5)$            |
| 7 | $(P \wedge Q) \vee (\neg P \vee \neg Q)$ | $\vee E(1, 2, 3, 4, 6)$ |

**Theorem 2.1.8.4.**

$$\neg(\neg P \wedge \neg Q) \dashv\vdash P \vee Q$$

*Beweis.*

$\vdash$ :

|   |                              |                |
|---|------------------------------|----------------|
| 1 | $\neg(\neg P \wedge \neg Q)$ | $A$            |
| 1 | $\neg\neg P \vee \neg\neg Q$ | Th. 2.1.8.2(1) |
| 1 | $P \vee Q$                   | Th. 2.1.6.2(2) |

$\dashv\vdash$ :

|   |                              |                |
|---|------------------------------|----------------|
| 1 | $P \vee Q$                   | $A$            |
| 1 | $\neg\neg P \vee \neg\neg Q$ | Th. 2.1.6.2(1) |
| 1 | $\neg(\neg P \wedge \neg Q)$ | Th. 2.1.8.2(2) |

□

**Theorem 2.1.8.5.**

$$\neg(\neg P \vee \neg Q) \dashv\vdash P \wedge Q$$

*Beweis.*

$\vdash$ :

|   |   |                                |                |
|---|---|--------------------------------|----------------|
| 1 | 1 | $\neg(\neg P \vee \neg Q)$     | $A$            |
| 1 | 2 | $\neg\neg P \wedge \neg\neg Q$ | Th. 2.1.8.1(1) |
| 1 | 3 | $P \wedge Q$                   | Th. 2.1.6.3(2) |

$\dashv$ :

|   |   |                                |                |
|---|---|--------------------------------|----------------|
| 1 | 1 | $P \wedge Q$                   | $A$            |
| 1 | 2 | $\neg\neg P \wedge \neg\neg Q$ | Th. 2.1.6.3(1) |
| 1 | 3 | $\neg(\neg P \vee \neg Q)$     | Th. 2.1.8.1(2) |

□

## Kapitel 2

# Implikation und Bikonditional

### 2.1 Implikation und Konjunktion im Konsequens

Im Kapitel verwendete Symbole

Aussagevariablen:  $P, Q, R$

**Theorem 2.2.1.1 (Konjunktion im Konsequens).**

$$P \rightarrow Q, P \rightarrow R \vdash P \rightarrow (Q \wedge R)$$

|       |   |                              |                       |
|-------|---|------------------------------|-----------------------|
| 1     | 1 | $P \rightarrow Q$            | $A$                   |
| 2     | 2 | $P \rightarrow R$            | $A$                   |
| 3     | 3 | $P$                          | $A$                   |
| 1,3   | 4 | $Q$                          | $\rightarrow E(1, 3)$ |
| 2,3   | 5 | $R$                          | $\rightarrow E(2, 3)$ |
| 1,2,3 | 6 | $Q \wedge R$                 | $\wedge I(4, 5)$      |
| 1,2   | 7 | $P \rightarrow (Q \wedge R)$ | $\rightarrow I(3, 6)$ |

**Theorem 2.2.1.2.**

$$P \rightarrow Q, P \rightarrow R \vdash P \rightarrow (R \wedge Q)$$

|       |   |                   |                       |
|-------|---|-------------------|-----------------------|
| 1     | 1 | $P \rightarrow Q$ | $A$                   |
| 2     | 2 | $P \rightarrow R$ | $A$                   |
| 3     | 3 | $P$               | $A$                   |
| 1,3   | 4 | $Q$               | $\rightarrow E(1, 3)$ |
| 2,3   | 5 | $R$               | $\rightarrow E(2, 3)$ |
| 1,2,3 | 6 | $R \wedge Q$      | $\wedge I(5, 4)$      |

$$1,2 \quad 7 \quad P \rightarrow (R \wedge Q) \quad \rightarrow I(3, 6)$$

**Theorem 2.2.1.3.**

$$P \rightarrow Q \vdash P \leftrightarrow (P \wedge Q)$$

*Beweis.*

Hinrichtung:

$$\begin{array}{lll} 1 & 1 & P \rightarrow Q & A \\ & 2 & P \rightarrow P & \text{Th. 2.1.1.2} \\ 1 & 3 & P \rightarrow (P \wedge Q) & \text{Th. 2.2.1.1(2, 1)} \end{array}$$

Rückrichtung:

$$\begin{array}{lll} 4 & 4 & P \wedge Q & A \\ 4 & 5 & P & \wedge E1(4) \\ & 6 & (P \wedge Q) \rightarrow P & \rightarrow I(4, 5) \end{array}$$

Zusammenführung:

$$1 \quad 7 \quad P \leftrightarrow (P \wedge Q) \quad \leftrightarrow I(3, 6)$$

□

**Theorem 2.2.1.4.**

$$P \rightarrow Q \vdash P \leftrightarrow (Q \wedge P)$$

*Beweis.*

Hinrichtung:

$$\begin{array}{lll} 1 & 1 & P \rightarrow Q & A \\ & 2 & P \rightarrow P & \text{Th. 2.1.1.2} \\ 1 & 3 & P \rightarrow (Q \wedge P) & \text{Th. 2.2.1.2(2, 1)} \end{array}$$

Rückrichtung:

$$\begin{array}{lll} 4 & 4 & Q \wedge P & A \\ 4 & 5 & P & \wedge E2(4) \\ & 6 & (Q \wedge P) \rightarrow P & \rightarrow I(4, 5) \end{array}$$

Zusammenführung:

$$1 \quad 7 \quad P \leftrightarrow (Q \wedge P) \quad \leftrightarrow I(3, 6)$$

□

**Theorem 2.2.1.5.**

$$P \rightarrow Q \vdash (P \wedge Q) \leftrightarrow P$$

*Beweis.*

Rückrichtung:

$$\begin{array}{lll} 1 & 1 & P \rightarrow Q & A \\ 2 & 2 & P & A \\ 1,2 & 3 & Q & \rightarrow E(1, 2) \\ 1,2 & 4 & P \wedge Q & \wedge I(2, 3) \\ 1 & 5 & P \rightarrow (P \wedge Q) & \rightarrow I(2, 4) \end{array}$$

Hinrichtung:

$$\begin{array}{lll} 6 & 6 & P \wedge Q & A \\ 6 & 7 & P & \wedge E1(6) \\ & 8 & (P \wedge Q) \rightarrow P & \rightarrow I(6, 7) \end{array}$$

Zusammenführung:

$$1 \quad 9 \quad (P \wedge Q) \leftrightarrow P \quad \leftrightarrow I(8, 5)$$

□

**Theorem 2.2.1.6.**

$$P \rightarrow Q \vdash (Q \wedge P) \leftrightarrow P$$

*Beweis.*

Rückrichtung:

$$\begin{array}{lll} 1 & 1 & P \rightarrow Q & A \\ 2 & 2 & P & A \\ 1,2 & 3 & Q & \rightarrow E(1, 2) \\ 1,2 & 4 & Q \wedge P & \wedge I(3, 2) \\ 1 & 5 & P \rightarrow (Q \wedge P) & \rightarrow I(2, 4) \end{array}$$

Hinrichtung:

|   |   |                              |                       |
|---|---|------------------------------|-----------------------|
| 6 | 6 | $Q \wedge P$                 | $A$                   |
| 6 | 7 | $P$                          | $\wedge E2(6)$        |
|   | 8 | $(Q \wedge P) \rightarrow P$ | $\rightarrow I(6, 7)$ |

Zusammenführung:

|   |   |                                  |                           |
|---|---|----------------------------------|---------------------------|
| 1 | 9 | $(Q \wedge P) \leftrightarrow P$ | $\leftrightarrow I(8, 5)$ |
|---|---|----------------------------------|---------------------------|

□

**Theorem 2.2.1.7.**

$$P \rightarrow P \vee Q$$

|   |   |                          |                       |
|---|---|--------------------------|-----------------------|
| 1 | 1 | $P$                      | $A$                   |
| 1 | 2 | $P \vee Q$               | $\vee I1(1)$          |
|   | 3 | $P \rightarrow P \vee Q$ | $\rightarrow I(1, 2)$ |

**Theorem 2.2.1.8.**

$$P \rightarrow Q \vee P$$

|   |   |                          |                       |
|---|---|--------------------------|-----------------------|
| 1 | 1 | $P$                      | $A$                   |
| 1 | 2 | $Q \vee P$               | $\vee I2(1)$          |
|   | 3 | $P \rightarrow Q \vee P$ | $\rightarrow I(1, 2)$ |

**Theorem 2.2.1.9 (Monotonie der Disjunktion im Konsequens).**

$$P \rightarrow Q, P \wedge R \vdash Q \wedge R$$

|     |   |                   |                       |
|-----|---|-------------------|-----------------------|
| 1   | 1 | $P \rightarrow Q$ | $A$                   |
| 2   | 2 | $P \wedge R$      | $A$                   |
| 2   | 3 | $P$               | $\wedge E1(2)$        |
| 2   | 4 | $R$               | $\wedge E2(2)$        |
| 1,2 | 5 | $Q$               | $\rightarrow E(1, 3)$ |
| 1,2 | 6 | $Q \wedge R$      | $\wedge I(5, 4)$      |

**Theorem 2.2.1.10 (Monotonie der Disjunktion im Konsequens).**

$$P \rightarrow Q, R \wedge P \vdash R \wedge Q$$

|     |   |                   |                       |
|-----|---|-------------------|-----------------------|
| 1   | 1 | $P \rightarrow Q$ | $A$                   |
| 2   | 2 | $R \wedge P$      | $A$                   |
| 2   | 3 | $R$               | $\wedge E1(2)$        |
| 2   | 4 | $P$               | $\wedge E2(2)$        |
| 1,2 | 5 | $Q$               | $\rightarrow E(1, 4)$ |
| 1,2 | 6 | $R \wedge Q$      | $\wedge I(3, 5)$      |

## 2.2 Modus Tollens und Kontraposition

Im Kapitel verwendete Symbole

Aussagevariablen:  $P, Q$

**Theorem 2.2.2.1** (Modus Tollens).

$$P \rightarrow Q, \neg Q \vdash \neg P$$

|       |   |                   |                       |
|-------|---|-------------------|-----------------------|
| 1     | 1 | $P \rightarrow Q$ | $A$                   |
| 2     | 2 | $\neg Q$          | $A$                   |
| 3     | 3 | $P$               | $A$                   |
| 1,3   | 4 | $Q$               | $\rightarrow E(1, 3)$ |
| 1,2,3 | 5 | $\perp$           | $\perp I(2, 4)$       |
| 1,2   | 6 | $\neg P$          | $\neg I(3, 5)$        |

**Theorem 2.2.2.2.**

$$\neg P \rightarrow \neg Q, Q \vdash P$$

|       |   |                             |                       |
|-------|---|-----------------------------|-----------------------|
| 1     | 1 | $\neg P \rightarrow \neg Q$ | $A$                   |
| 2     | 2 | $Q$                         | $A$                   |
| 3     | 3 | $\neg P$                    | $A$                   |
| 1,3   | 4 | $\neg Q$                    | $\rightarrow E(1, 3)$ |
| 1,2,3 | 5 | $\perp$                     | $\perp I(2, 4)$       |
| 1,2   | 6 | $P$                         | $\neg E(3, 5)$        |

**Theorem 2.2.2.3.**

$$P \leftrightarrow Q, \neg Q \vdash \neg P$$

|     |   |                       |                         |
|-----|---|-----------------------|-------------------------|
| 1   | 1 | $P \leftrightarrow Q$ | $A$                     |
| 1   | 2 | $P \rightarrow Q$     | $\leftrightarrow E1(1)$ |
| 3   | 3 | $\neg Q$              | $A$                     |
| 1,3 | 4 | $\neg P$              | Th. 2.2.2.1(2, 3)       |

**Theorem 2.2.2.4.**

$$P \leftrightarrow Q, \neg P \vdash \neg Q$$

|     |   |                       |                         |
|-----|---|-----------------------|-------------------------|
| 1   | 1 | $P \leftrightarrow Q$ | $A$                     |
| 1   | 2 | $Q \rightarrow P$     | $\leftrightarrow E2(1)$ |
| 3   | 3 | $\neg P$              | $A$                     |
| 1,3 | 4 | $\neg Q$              | Th. 2.2.2.1(2, 3)       |

**Theorem 2.2.2.5.**

$$\neg P \leftrightarrow \neg Q, Q \vdash P$$

|     |   |                                 |                         |
|-----|---|---------------------------------|-------------------------|
| 1   | 1 | $\neg P \leftrightarrow \neg Q$ | $A$                     |
| 1   | 2 | $\neg P \rightarrow \neg Q$     | $\leftrightarrow E1(1)$ |
| 3   | 3 | $Q$                             | $A$                     |
| 1,3 | 4 | $P$                             | Th. 2.2.2.2(2, 3)       |

**Theorem 2.2.2.6.**

$$\neg P \leftrightarrow \neg Q, P \vdash Q$$

|     |   |                                 |                         |
|-----|---|---------------------------------|-------------------------|
| 1   | 1 | $\neg P \leftrightarrow \neg Q$ | $A$                     |
| 1   | 2 | $\neg Q \rightarrow \neg P$     | $\leftrightarrow E2(1)$ |
| 3   | 3 | $P$                             | $A$                     |
| 1,3 | 4 | $Q$                             | Th. 2.2.2.2(2, 3)       |

**Theorem 2.2.2.7 (Kontraposition).**

$$P \rightarrow Q \dashv\vdash \neg Q \rightarrow \neg P$$

*Beweis.*

$\vdash$ :

|     |   |                             |                       |
|-----|---|-----------------------------|-----------------------|
| 1   | 1 | $P \rightarrow Q$           | $A$                   |
| 2   | 2 | $\neg Q$                    | $A$                   |
| 1,2 | 3 | $\neg P$                    | Th. 2.2.2.1(1, 2)     |
| 1   | 4 | $\neg Q \rightarrow \neg P$ | $\rightarrow I(2, 3)$ |

$\dashv\vdash$ :

|   |   |                             |     |
|---|---|-----------------------------|-----|
| 1 | 1 | $\neg Q \rightarrow \neg P$ | $A$ |
| 2 | 2 | $P$                         | $A$ |

|     |   |                   |
|-----|---|-------------------|
| 1,2 | 3 | $Q$               |
| 1   | 4 | $P \rightarrow Q$ |

Th. 2.2.2.2(1, 2)  
 $\rightarrow I(2, 3)$

□

## 2.3 Transitivitäten

**Im Kapitel verwendete Symbole**

**Aussagevariablen:**  $P, Q, R$

**Theorem 2.2.3.1.**

$$P \rightarrow Q, Q \rightarrow R, P \vdash R$$

|       |   |                   |                       |
|-------|---|-------------------|-----------------------|
| 1     | 1 | $P \rightarrow Q$ | $A$                   |
| 2     | 2 | $Q \rightarrow R$ | $A$                   |
| 3     | 3 | $P$               | $A$                   |
| 1,3   | 4 | $Q$               | $\rightarrow E(1, 3)$ |
| 1,2,3 | 5 | $R$               | $\rightarrow E(2, 4)$ |

**Theorem 2.2.3.2 (Implikationstransitivität).**

$$P \rightarrow Q, Q \rightarrow R \vdash P \rightarrow R$$

|       |   |                   |                       |
|-------|---|-------------------|-----------------------|
| 1     | 1 | $P \rightarrow Q$ | $A$                   |
| 2     | 2 | $Q \rightarrow R$ | $A$                   |
| 3     | 3 | $P$               | $A$                   |
| 1,2,3 | 4 | $R$               | Th. 2.2.3.1(1, 2, 3)  |
| 1,2   | 5 | $P \rightarrow R$ | $\rightarrow I(3, 4)$ |

**Theorem 2.2.3.3.**

$$P \leftrightarrow Q, Q \leftrightarrow R \vdash P \rightarrow R$$

|     |   |                       |                         |
|-----|---|-----------------------|-------------------------|
| 1   | 1 | $P \leftrightarrow Q$ | $A$                     |
| 2   | 2 | $Q \leftrightarrow R$ | $A$                     |
| 1   | 3 | $P \rightarrow Q$     | $\leftrightarrow E1(1)$ |
| 2   | 4 | $Q \rightarrow R$     | $\leftrightarrow E1(2)$ |
| 1,2 | 5 | $P \rightarrow R$     | Th. 2.2.3.2(3, 4)       |

**Theorem 2.2.3.4.**

$$P \leftrightarrow Q, Q \leftrightarrow R \vdash R \rightarrow P$$

|     |   |                       |                         |
|-----|---|-----------------------|-------------------------|
| 1   | 1 | $P \leftrightarrow Q$ | $A$                     |
| 2   | 2 | $Q \leftrightarrow R$ | $A$                     |
| 1   | 3 | $Q \rightarrow P$     | $\leftrightarrow E2(1)$ |
| 2   | 4 | $R \rightarrow Q$     | $\leftrightarrow E2(2)$ |
| 1,2 | 5 | $R \rightarrow P$     | Th. 2.2.3.2(4, 3)       |

**Theorem 2.2.3.5 (Äquivalenztransitivität).**

$$P \leftrightarrow Q, Q \leftrightarrow R \vdash P \leftrightarrow R$$

|     |   |                       |                           |
|-----|---|-----------------------|---------------------------|
| 1   | 1 | $P \leftrightarrow Q$ | $A$                       |
| 2   | 2 | $Q \leftrightarrow R$ | $A$                       |
| 1,2 | 3 | $P \rightarrow R$     | Th. 2.2.3.3(1, 2)         |
| 1,2 | 4 | $R \rightarrow P$     | Th. 2.2.3.4(1, 2)         |
| 1,2 | 5 | $P \leftrightarrow R$ | $\leftrightarrow I(3, 4)$ |

**Theorem 2.2.3.6 (Äquivalenztransitivität).**

$$P \leftrightarrow Q, R \leftrightarrow Q \vdash P \leftrightarrow R$$

|     |   |                       |                   |
|-----|---|-----------------------|-------------------|
| 1   | 1 | $P \leftrightarrow Q$ | $A$               |
| 2   | 2 | $R \leftrightarrow Q$ | $A$               |
| 2   | 3 | $Q \leftrightarrow R$ | Th. 2.1.2.5(2)    |
| 1,2 | 4 | $P \leftrightarrow R$ | Th. 2.2.3.5(1, 3) |

**Theorem 2.2.3.7 (gemischte Transitivität).**

$$P \rightarrow Q, Q \leftrightarrow R \vdash P \rightarrow R$$

|     |   |                       |                         |
|-----|---|-----------------------|-------------------------|
| 1   | 1 | $P \rightarrow Q$     | $A$                     |
| 2   | 2 | $Q \leftrightarrow R$ | $A$                     |
| 2   | 3 | $Q \rightarrow R$     | $\leftrightarrow E1(2)$ |
| 1,2 | 4 | $P \rightarrow R$     | Th. 2.2.3.2(1, 3)       |

## 2.4 Äquivalenz als Folgerung

**Im Kapitel verwendete Symbole**

**Aussagevariablen:**  $P, Q$

**Theorem 2.2.4.1 (Bikonditional-Elimination).**

$$P \leftrightarrow Q, P \vdash Q$$

|     |   |                       |                         |
|-----|---|-----------------------|-------------------------|
| 1   | 1 | $P \leftrightarrow Q$ | $A$                     |
| 2   | 2 | $P$                   | $A$                     |
| 1   | 3 | $P \rightarrow Q$     | $\leftrightarrow E1(1)$ |
| 1,2 | 4 | $Q$                   | $\rightarrow E(2, 3)$   |

**Theorem 2.2.4.2 (Bikonditional-Elimination).**

$$P \leftrightarrow Q, Q \vdash P$$

|     |   |                       |                         |
|-----|---|-----------------------|-------------------------|
| 1   | 1 | $P \leftrightarrow Q$ | $A$                     |
| 2   | 2 | $Q$                   | $A$                     |
| 1   | 3 | $Q \rightarrow P$     | $\leftrightarrow E2(1)$ |
| 1,2 | 4 | $P$                   | $\rightarrow E(2, 3)$   |

## 2.5 Grundlegende Äquivalenzen der Implikation

Im Kapitel verwendete Symbole

Aussagevariablen:  $P, Q$

**Theorem 2.2.5.1 (Materielle Implikation).**

$$P \rightarrow Q \dashv\vdash \neg P \vee Q$$

*Beweis.*

$\vdash$ :

|     |   |                   |                         |
|-----|---|-------------------|-------------------------|
| 1   | 1 | $P \rightarrow Q$ | $A$                     |
|     | 2 | $P \vee \neg P$   | Th. 2.1.7.1             |
| 3   | 3 | $P$               | $A$                     |
| 1,3 | 4 | $Q$               | $\rightarrow E(1, 3)$   |
| 1,3 | 5 | $\neg P \vee Q$   | $\vee I2(4)$            |
| 6   | 6 | $\neg P$          | $A$                     |
| 6   | 7 | $\neg P \vee Q$   | $\vee I1(6)$            |
| 1   | 8 | $\neg P \vee Q$   | $\vee E(2, 3, 5, 6, 7)$ |

$\dashv\vdash$ :

|     |   |                            |                  |
|-----|---|----------------------------|------------------|
| 1   | 1 | $\neg P \vee Q$            | $A$              |
| 2   | 2 | $P$                        | $A$              |
| 3   | 3 | $\neg Q$                   | $A$              |
| 2   | 4 | $\neg\neg P$               | Th. 2.1.6.1(2)   |
| 2,3 | 5 | $\neg\neg P \wedge \neg Q$ | $\wedge I(4, 3)$ |

|       |   |                       |                       |
|-------|---|-----------------------|-----------------------|
| 2,3   | 6 | $\neg(\neg P \vee Q)$ | Th. 2.1.8.1(5)        |
| 1,2,3 | 7 | $\perp$               | $\perp I(1, 6)$       |
| 1,2   | 8 | $Q$                   | $\neg E(3, 7)$        |
| 1     | 9 | $P \rightarrow Q$     | $\rightarrow I(2, 8)$ |

□

**Theorem 2.2.5.2.**

$$\neg P \rightarrow Q \dashv\vdash P \vee Q$$

$$\begin{array}{l} 1 \quad \neg P \rightarrow Q \leftrightarrow \neg\neg P \vee Q \text{ Th. 2.2.5.1} \\ 2 \quad \quad \quad \leftrightarrow P \vee Q \quad \leftrightarrow S(1, \text{Th. 2.1.6.1}) \end{array}$$

**Theorem 2.2.5.3.**

$$\neg Q \rightarrow P \dashv\vdash P \vee Q$$

$$\begin{array}{l} 1 \quad \neg Q \rightarrow P \leftrightarrow Q \vee P \text{ Th. 2.2.5.1} \\ 2 \quad \quad \quad \leftrightarrow P \vee Q \text{ Th. 2.1.2.2(1)} \end{array}$$

**Theorem 2.2.5.4.**

$$P \vee Q, \neg P \vdash Q$$

|     |   |                                 |                       |
|-----|---|---------------------------------|-----------------------|
| 1   | 1 | $P \vee Q$                      | $A$                   |
| 2   | 2 | $\neg P$                        | $A$                   |
| 1   | 3 | $\neg\neg P \vee \neg\neg Q$    | Th. 2.1.6.2(1)        |
| 1   | 4 | $\neg P \rightarrow \neg\neg Q$ | Th. 2.2.5.1(3)        |
| 1,2 | 5 | $\neg\neg Q$                    | $\rightarrow E(2, 4)$ |
| 1,2 | 6 | $Q$                             | Th. 2.1.6.1(5)        |

**Theorem 2.2.5.5 (Implikation-Negations-Regel).**

$$\neg(P \rightarrow Q) \dashv\vdash P \wedge \neg Q$$

*Beweis.*

$\vdash$ :

|   |   |                                                 |                   |
|---|---|-------------------------------------------------|-------------------|
| 1 | 1 | $\neg(P \rightarrow Q)$                         | $A$               |
|   | 2 | $P \rightarrow Q \leftrightarrow \neg P \vee Q$ | Th. 2.2.5.1       |
| 1 | 3 | $\neg(\neg P \vee Q)$                           | Th. 2.2.2.4(2, 1) |
| 1 | 4 | $\neg\neg P \wedge \neg Q$                      | Th. 2.1.8.1(3)    |

|   |   |                   |                  |
|---|---|-------------------|------------------|
| 1 | 5 | $\neg\neg P$      | $\wedge E1(4)$   |
| 1 | 6 | $P$               | Th. 2.1.6.1(5)   |
| 1 | 7 | $\neg Q$          | $\wedge E2(4)$   |
| 1 | 8 | $P \wedge \neg Q$ | $\wedge I(6, 7)$ |

$\dashv$ :

|   |   |                                                 |                   |
|---|---|-------------------------------------------------|-------------------|
| 1 | 1 | $P \wedge \neg Q$                               | $A$               |
| 1 | 2 | $P$                                             | $\wedge E1(1)$    |
| 1 | 3 | $\neg Q$                                        | $\wedge E2(1)$    |
| 1 | 4 | $\neg\neg P$                                    | Th. 2.1.6.1(2)    |
| 1 | 5 | $\neg\neg P \wedge \neg Q$                      | $\wedge I(4, 3)$  |
| 1 | 6 | $\neg(\neg P \vee Q)$                           | Th. 2.1.8.1(5)    |
|   | 7 | $P \rightarrow Q \leftrightarrow \neg P \vee Q$ | Th. 2.2.5.1       |
| 1 | 8 | $\neg(P \rightarrow Q)$                         | Th. 2.2.2.3(7, 6) |

□

## 2.6 Charakterisierungen des Bikonditionals

**Im Kapitel verwendete Symbole**

**Aussagevariablen:**  $P, Q$

**Theorem 2.2.6.1 (Negationsäquivalenz).**

$$P \leftrightarrow Q \dashv\vdash \neg P \leftrightarrow \neg Q$$

*Beweis.*

$\vdash$ :

|   |   |                                 |                           |
|---|---|---------------------------------|---------------------------|
| 1 | 1 | $P \leftrightarrow Q$           | $A$                       |
| 1 | 2 | $P \rightarrow Q$               | $\leftrightarrow E1(1)$   |
| 1 | 3 | $Q \rightarrow P$               | $\leftrightarrow E2(1)$   |
| 1 | 4 | $\neg Q \rightarrow \neg P$     | Th. 2.2.2.7(2)            |
| 1 | 5 | $\neg P \rightarrow \neg Q$     | Th. 2.2.2.7(3)            |
| 1 | 6 | $\neg P \leftrightarrow \neg Q$ | $\leftrightarrow I(5, 4)$ |

$\dashv$ :

|   |   |                                 |                         |
|---|---|---------------------------------|-------------------------|
| 1 | 1 | $\neg P \leftrightarrow \neg Q$ | $A$                     |
| 1 | 2 | $\neg P \rightarrow \neg Q$     | $\leftrightarrow E1(1)$ |
| 1 | 3 | $\neg Q \rightarrow \neg P$     | $\leftrightarrow E2(1)$ |
| 1 | 4 | $Q \rightarrow P$               | Th. 2.2.2.7(2)          |

|   |   |                       |                           |
|---|---|-----------------------|---------------------------|
| 1 | 5 | $P \rightarrow Q$     | Th. 2.2.2.7(3)            |
| 1 | 6 | $P \leftrightarrow Q$ | $\leftrightarrow I(5, 4)$ |

□

**Theorem 2.2.6.2 (Negationsäquivalenz).**

$$P \leftrightarrow \neg Q \dashv\vdash \neg P \leftrightarrow Q$$

|   |                                                                                  |                           |
|---|----------------------------------------------------------------------------------|---------------------------|
| 1 | $(P \leftrightarrow \neg Q) \leftrightarrow (\neg P \leftrightarrow \neg\neg Q)$ | Th. 2.2.6.1               |
| 2 | $(Q \leftrightarrow \neg\neg Q)$                                                 | Th. 2.1.6.1               |
| 3 | $(\neg\neg Q \leftrightarrow Q)$                                                 | Th. 2.1.2.5(2)            |
| 4 | $(P \leftrightarrow \neg Q) \leftrightarrow (\neg P \leftrightarrow Q)$          | $\leftrightarrow S(3, 1)$ |

**Theorem 2.2.6.3.**

$$P \leftrightarrow Q \dashv\vdash (P \wedge Q) \vee (\neg P \wedge \neg Q)$$

*Beweis.*

⊢:

|     |    |                                            |                          |
|-----|----|--------------------------------------------|--------------------------|
| 1   | 1  | $P \leftrightarrow Q$                      | $A$                      |
| 1   | 2  | $P \rightarrow Q$                          | $\leftrightarrow E1(1)$  |
| 1   | 3  | $Q \rightarrow P$                          | $\leftrightarrow E2(1)$  |
|     | 4  | $P \vee \neg P$                            | Th. 2.1.7.1              |
| 5   | 5  | $P$                                        | $A$                      |
| 1,5 | 6  | $Q$                                        | $\rightarrow E(2, 5)$    |
| 1,5 | 7  | $P \wedge Q$                               | $\wedge I(5, 6)$         |
| 1,5 | 8  | $(P \wedge Q) \vee (\neg P \wedge \neg Q)$ | $\vee I1(7)$             |
| 9   | 9  | $\neg P$                                   | $A$                      |
| 1,9 | 10 | $\neg Q$                                   | Th. 2.2.2.1(3, 9)        |
| 1,9 | 11 | $\neg P \wedge \neg Q$                     | $\wedge I(9, 10)$        |
| 1,9 | 12 | $(P \wedge Q) \vee (\neg P \wedge \neg Q)$ | $\vee I2(11)$            |
| 1   | 13 | $(P \wedge Q) \vee (\neg P \wedge \neg Q)$ | $\vee E(4, 5, 8, 9, 12)$ |

⊢:

|   |   |                                            |                 |
|---|---|--------------------------------------------|-----------------|
| 1 | 1 | $(P \wedge Q) \vee (\neg P \wedge \neg Q)$ | $A$             |
| 2 | 2 | $P \wedge Q$                               | $A$             |
| 2 | 3 | $P$                                        | $\wedge E1(2)$  |
| 2 | 4 | $Q$                                        | $\wedge E2(2)$  |
| 2 | 5 | $P \rightarrow Q$                          | Th. 2.2.7.11(4) |
| 2 | 6 | $Q \rightarrow P$                          | Th. 2.2.7.11(3) |

|   |    |                        |                             |
|---|----|------------------------|-----------------------------|
| 2 | 7  | $P \leftrightarrow Q$  | $\leftrightarrow I(5, 6)$   |
| 8 | 8  | $\neg P \wedge \neg Q$ | $A$                         |
| 8 | 9  | $\neg P$               | $\wedge E1(8)$              |
| 8 | 10 | $\neg Q$               | $\wedge E2(8)$              |
| 8 | 11 | $P \rightarrow Q$      | Th. 2.2.7.12(9)             |
| 8 | 12 | $Q \rightarrow P$      | Th. 2.2.7.12(10)            |
| 8 | 13 | $P \leftrightarrow Q$  | $\leftrightarrow I(11, 12)$ |
| 1 | 14 | $P \leftrightarrow Q$  | $\vee E(1, 2, 7, 8, 13)$    |

□

**Theorem 2.2.6.4.**

$$P \leftrightarrow Q \dashv\vdash (P \rightarrow Q) \wedge (Q \rightarrow P)$$

*Beweis.*

$\vdash$ :

|   |   |                                              |                         |
|---|---|----------------------------------------------|-------------------------|
| 1 | 1 | $P \leftrightarrow Q$                        | $A$                     |
| 1 | 2 | $P \rightarrow Q$                            | $\leftrightarrow E1(1)$ |
| 1 | 3 | $Q \rightarrow P$                            | $\leftrightarrow E2(1)$ |
| 1 | 4 | $(P \rightarrow Q) \wedge (Q \rightarrow P)$ | $\wedge I(2, 3)$        |

$\dashv\vdash$ :

|   |   |                                              |                           |
|---|---|----------------------------------------------|---------------------------|
| 1 | 1 | $(P \rightarrow Q) \wedge (Q \rightarrow P)$ | $A$                       |
| 1 | 2 | $P \rightarrow Q$                            | $\wedge E1(1)$            |
| 1 | 3 | $Q \rightarrow P$                            | $\wedge E2(1)$            |
| 1 | 4 | $P \leftrightarrow Q$                        | $\leftrightarrow I(2, 3)$ |

□

**Theorem 2.2.6.5.**

$$P \leftrightarrow Q \dashv\vdash (\neg P \vee Q) \wedge (\neg Q \vee P)$$

*Beweis.*

$\vdash$ :

|   |   |                       |                         |
|---|---|-----------------------|-------------------------|
| 1 | 1 | $P \leftrightarrow Q$ | $A$                     |
| 1 | 2 | $P \rightarrow Q$     | $\leftrightarrow E1(1)$ |
| 1 | 3 | $Q \rightarrow P$     | $\leftrightarrow E2(1)$ |
| 1 | 4 | $\neg P \vee Q$       | Th. 2.2.5.1(2)          |
| 1 | 5 | $\neg Q \vee P$       | Th. 2.2.5.1(3)          |

|   |                                              |                  |
|---|----------------------------------------------|------------------|
| 1 | $6 \ (\neg P \vee Q) \wedge (\neg Q \vee P)$ | $\wedge I(4, 5)$ |
|---|----------------------------------------------|------------------|

$\dashv$ :

|   |                                              |                           |
|---|----------------------------------------------|---------------------------|
| 1 | $1 \ (\neg P \vee Q) \wedge (\neg Q \vee P)$ | $A$                       |
| 1 | $2 \ \neg P \vee Q$                          | $\wedge E1(1)$            |
| 1 | $3 \ \neg Q \vee P$                          | $\wedge E2(1)$            |
| 1 | $4 \ P \rightarrow Q$                        | $\text{Th. 2.2.5.1}(2)$   |
| 1 | $5 \ Q \rightarrow P$                        | $\text{Th. 2.2.5.1}(3)$   |
| 1 | $6 \ P \leftrightarrow Q$                    | $\leftrightarrow I(4, 5)$ |

□

**Theorem 2.2.6.6.**

$$\neg(P \leftrightarrow Q) \dashv\vdash (\neg P \wedge Q) \vee (P \wedge \neg Q)$$

*Beweis.*

$\vdash$ :

|   |                                                                                          |                            |
|---|------------------------------------------------------------------------------------------|----------------------------|
| 1 | $1 \ \neg(P \leftrightarrow Q)$                                                          | $A$                        |
|   | $2 \ (P \leftrightarrow Q) \leftrightarrow ((P \rightarrow Q) \wedge (Q \rightarrow P))$ | $\text{Th. 2.2.6.4}$       |
| 1 | $3 \ \neg((P \rightarrow Q) \wedge (Q \rightarrow P))$                                   | $\text{Th. 2.2.2.4}(2, 1)$ |
| 1 | $4 \ \neg(P \rightarrow Q) \vee \neg(Q \rightarrow P)$                                   | $\text{Th. 2.1.8.2}(3)$    |
| 5 | $5 \ \neg(P \rightarrow Q)$                                                              | $A$                        |
| 5 | $6 \ P \wedge \neg Q$                                                                    | $\text{Th. 2.2.5.5}(5)$    |
| 5 | $7 \ (\neg P \wedge Q) \vee (P \wedge \neg Q)$                                           | $\vee I2(6)$               |
| 8 | $8 \ \neg(Q \rightarrow P)$                                                              | $A$                        |
| 8 | $9 \ Q \wedge \neg P$                                                                    | $\text{Th. 2.2.5.5}(8)$    |
| 8 | $10 \ \neg P \wedge Q$                                                                   | $\text{Th. 2.1.2.3}(9)$    |
| 8 | $11 \ (\neg P \wedge Q) \vee (P \wedge \neg Q)$                                          | $\vee I1(10)$              |
| 1 | $12 \ (\neg P \wedge Q) \vee (P \wedge \neg Q)$                                          | $\vee E(4, 5, 7, 8, 11)$   |

$\dashv$ :

|   |                                                        |                         |
|---|--------------------------------------------------------|-------------------------|
| 1 | $1 \ (\neg P \wedge Q) \vee (P \wedge \neg Q)$         | $A$                     |
| 2 | $2 \ \neg P \wedge Q$                                  | $A$                     |
| 2 | $3 \ Q \wedge \neg P$                                  | $\text{Th. 2.1.2.3}(2)$ |
| 2 | $4 \ \neg(Q \rightarrow P)$                            | $\text{Th. 2.2.5.5}(3)$ |
| 2 | $5 \ \neg(P \rightarrow Q) \vee \neg(Q \rightarrow P)$ | $\vee I2(4)$            |
| 6 | $6 \ P \wedge \neg Q$                                  | $A$                     |
| 6 | $7 \ \neg(P \rightarrow Q)$                            | $\text{Th. 2.2.5.5}(6)$ |
| 6 | $8 \ \neg(P \rightarrow Q) \vee \neg(Q \rightarrow P)$ | $\vee I1(7)$            |
| 1 | $9 \ \neg(P \rightarrow Q) \vee \neg(Q \rightarrow P)$ | $\vee E(1, 2, 5, 6, 8)$ |

|   |    |                                                                                      |                     |
|---|----|--------------------------------------------------------------------------------------|---------------------|
| 1 | 10 | $\neg((P \rightarrow Q) \wedge (Q \rightarrow P))$                                   | Th. 2.1.8.2(9)      |
|   | 11 | $(P \leftrightarrow Q) \leftrightarrow ((P \rightarrow Q) \wedge (Q \rightarrow P))$ | Th. 2.2.6.4         |
| 1 | 12 | $\neg(P \leftrightarrow Q)$                                                          | Th. 2.2.2.3(11, 10) |

□

## 2.7 Weitere Implikationsschemata

**Im Kapitel verwendete Symbole**

**Aussagevariablen:**  $P, Q, R$

**Theorem 2.2.7.1 (Exportationsgesetz).**

**Aussagenvariablen:**  $P, Q, R$

$$(P \wedge Q) \rightarrow R \dashv\vdash P \rightarrow (Q \rightarrow R)$$

*Beweis.*

$\vdash$ :

|       |   |                                   |                       |
|-------|---|-----------------------------------|-----------------------|
| 1     | 1 | $(P \wedge Q) \rightarrow R$      | $A$                   |
| 2     | 2 | $P$                               | $A$                   |
| 3     | 3 | $Q$                               | $A$                   |
| 2,3   | 4 | $P \wedge Q$                      | $\wedge I(2, 3)$      |
| 1,2,3 | 5 | $R$                               | $\rightarrow E(1, 4)$ |
| 1,2   | 6 | $Q \rightarrow R$                 | $\rightarrow I(3, 5)$ |
| 1     | 7 | $P \rightarrow (Q \rightarrow R)$ | $\rightarrow I(2, 6)$ |

$\dashv\vdash$ :

|     |   |                                   |                       |
|-----|---|-----------------------------------|-----------------------|
| 1   | 1 | $P \rightarrow (Q \rightarrow R)$ | $A$                   |
| 2   | 2 | $P \wedge Q$                      | $A$                   |
| 2   | 3 | $P$                               | $\wedge E1(2)$        |
| 2   | 4 | $Q$                               | $\wedge E2(2)$        |
| 1,2 | 5 | $Q \rightarrow R$                 | $\rightarrow E(1, 3)$ |
| 1,2 | 6 | $R$                               | $\rightarrow E(5, 4)$ |
| 1   | 7 | $(P \wedge Q) \rightarrow R$      | $\rightarrow I(2, 6)$ |

□

**Theorem 2.2.7.2 (Exportation).**

**Aussagenvariablen:**  $P, Q, R$

$$(P \wedge Q) \rightarrow R \dashv\vdash Q \rightarrow (P \rightarrow R)$$

*Beweis.*

$\vdash$ :

|       |     |                                   |                      |
|-------|-----|-----------------------------------|----------------------|
| 1     | 1   | $(P \wedge Q) \rightarrow R$      | $A$                  |
|       | 2   | $P$                               | $A$                  |
|       | 3   | $Q$                               | $A$                  |
|       | 2,3 | $P \wedge Q$                      | $\wedge I(2,3)$      |
| 1,2,3 | 5   | $R$                               | $\rightarrow E(1,4)$ |
|       | 1,3 | $P \rightarrow R$                 | $\rightarrow I(2,5)$ |
|       | 1   | $Q \rightarrow (P \rightarrow R)$ | $\rightarrow I(3,6)$ |

$\dashv$ :

|     |   |                                   |                      |
|-----|---|-----------------------------------|----------------------|
|     | 1 | $Q \rightarrow (P \rightarrow R)$ | $A$                  |
|     | 2 | $P \wedge Q$                      | $A$                  |
|     | 2 | $P$                               | $\wedge E1(2)$       |
|     | 2 | $Q$                               | $\wedge E2(2)$       |
| 1,2 | 5 | $P \rightarrow R$                 | $\rightarrow E(1,4)$ |
| 1,2 | 6 | $R$                               | $\rightarrow E(5,3)$ |
|     | 1 | $(P \wedge Q) \rightarrow R$      | $\rightarrow I(2,6)$ |

□

**Theorem 2.2.7.3.**

$$P \rightarrow (Q \rightarrow R) \vdash Q \rightarrow (P \rightarrow R)$$

|   |   |                                   |                |
|---|---|-----------------------------------|----------------|
| 1 | 1 | $P \rightarrow (Q \rightarrow R)$ | $A$            |
| 1 | 2 | $(P \wedge Q) \rightarrow R$      | Th. 2.2.7.1(1) |
| 1 | 3 | $Q \rightarrow (P \rightarrow R)$ | Th. 2.2.7.2(2) |

**Theorem 2.2.7.4 (Vertauschung der Exportation).**

**Aussagenvariablen:**  $P, Q, R$

$$P \rightarrow (Q \rightarrow R) \dashv\vdash Q \rightarrow (P \rightarrow R)$$

*Beweis.*

$\vdash$ :

|   |   |                                   |                |
|---|---|-----------------------------------|----------------|
| 1 | 1 | $P \rightarrow (Q \rightarrow R)$ | $A$            |
| 1 | 2 | $(P \wedge Q) \rightarrow R$      | Th. 2.2.7.1(1) |
| 1 | 3 | $Q \rightarrow (P \rightarrow R)$ | Th. 2.2.7.2(2) |

$\dashv$ :

|   |   |                                   |                |
|---|---|-----------------------------------|----------------|
| 1 | 1 | $Q \rightarrow (P \rightarrow R)$ | $A$            |
| 1 | 2 | $(P \wedge Q) \rightarrow R$      | Th. 2.2.7.2(1) |
| 1 | 3 | $P \rightarrow (Q \rightarrow R)$ | Th. 2.2.7.1(2) |

□

**Theorem 2.2.7.5 (Äquivalenz-Exportation).**

**Aussagenvariablen:**  $P, Q, R$

$$(P \wedge Q) \leftrightarrow R \vdash P \rightarrow (Q \leftrightarrow R)$$

|     |    |                                       |                            |
|-----|----|---------------------------------------|----------------------------|
| 1   | 1  | $(P \wedge Q) \leftrightarrow R$      | $A$                        |
| 1   | 2  | $(P \wedge Q) \rightarrow R$          | $\leftrightarrow E1(1)$    |
| 1   | 3  | $P \rightarrow (Q \rightarrow R)$     | Th. 2.2.7.1(2)             |
| 1   | 4  | $R \rightarrow (P \wedge Q)$          | $\leftrightarrow E2(1)$    |
| 5   | 5  | $R$                                   | $A$                        |
| 1,5 | 6  | $P \wedge Q$                          | $\rightarrow E(4, 5)$      |
| 1,5 | 7  | $Q$                                   | $\wedge E2(6)$             |
| 1   | 8  | $R \rightarrow Q$                     | $\rightarrow I(5, 7)$      |
| 9   | 9  | $P$                                   | $A$                        |
| 1,9 | 10 | $Q \rightarrow R$                     | $\rightarrow E(3, 9)$      |
| 1,9 | 11 | $Q \leftrightarrow R$                 | $\leftrightarrow I(10, 8)$ |
| 1   | 12 | $P \rightarrow (Q \leftrightarrow R)$ | $\rightarrow I(9, 11)$     |

**Theorem 2.2.7.6 (Vertauschte Äquivalenz-Exportation).**

**Aussagenvariablen:**  $P, Q, R$

$$(P \wedge Q) \leftrightarrow R \vdash Q \rightarrow (P \leftrightarrow R)$$

|      |    |                                       |                            |
|------|----|---------------------------------------|----------------------------|
| 1    | 1  | $(P \wedge Q) \leftrightarrow R$      | $A$                        |
| 1    | 2  | $(P \wedge Q) \rightarrow R$          | $\leftrightarrow E1(1)$    |
| 1    | 3  | $P \rightarrow (Q \rightarrow R)$     | Th. 2.2.7.1(2)             |
| 1    | 4  | $Q \rightarrow (P \rightarrow R)$     | Th. 2.2.7.4(3)             |
| 1    | 5  | $R \rightarrow (P \wedge Q)$          | $\leftrightarrow E2(1)$    |
| 6    | 6  | $R$                                   | $A$                        |
| 1,6  | 7  | $P \wedge Q$                          | $\rightarrow E(5, 6)$      |
| 1,6  | 8  | $P$                                   | $\wedge E1(7)$             |
| 1    | 9  | $R \rightarrow P$                     | $\rightarrow I(6, 8)$      |
| 10   | 10 | $Q$                                   | $A$                        |
| 1,10 | 11 | $P \rightarrow R$                     | $\rightarrow E(4, 10)$     |
| 1,10 | 12 | $P \leftrightarrow R$                 | $\leftrightarrow I(11, 9)$ |
| 1    | 13 | $Q \rightarrow (P \leftrightarrow R)$ | $\rightarrow I(10, 12)$    |

**Theorem 2.2.7.7 (Äquivalenz-Exportation).**

**Aussagenvariablen:**  $P, Q, R$

$$R \leftrightarrow (P \wedge Q) \vdash P \rightarrow (Q \leftrightarrow R)$$

|   |   |                                       |                |
|---|---|---------------------------------------|----------------|
| 1 | 1 | $R \leftrightarrow (P \wedge Q)$      | $A$            |
| 1 | 2 | $(P \wedge Q) \leftrightarrow R$      | Th. 2.1.2.5(1) |
| 1 | 3 | $P \rightarrow (Q \leftrightarrow R)$ | Th. 2.2.7.5(2) |

**Theorem 2.2.7.8 (Vertauschte Äquivalenz-Exportation).**

**Aussagenvariablen:**  $P, Q, R$

$$R \leftrightarrow (P \wedge Q) \vdash Q \rightarrow (P \leftrightarrow R)$$

|   |   |                                       |                |
|---|---|---------------------------------------|----------------|
| 1 | 1 | $R \leftrightarrow (P \wedge Q)$      | $A$            |
| 1 | 2 | $(P \wedge Q) \leftrightarrow R$      | Th. 2.1.2.5(1) |
| 1 | 3 | $Q \rightarrow (P \leftrightarrow R)$ | Th. 2.2.7.6(2) |

**Theorem 2.2.7.9.**

$$P \leftrightarrow (Q \wedge R), Q \rightarrow R \vdash P \leftrightarrow Q$$

*Beweis.*

Hinrichtung:

|     |   |                                  |                       |
|-----|---|----------------------------------|-----------------------|
| 1   | 1 | $P \leftrightarrow (Q \wedge R)$ | $A$                   |
| 2   | 2 | $Q \rightarrow R$                | $A$                   |
| 3   | 3 | $P$                              | $A$                   |
| 1,3 | 4 | $Q \wedge R$                     | Th. 2.2.4.1(1, 3)     |
| 1,3 | 5 | $Q$                              | $\wedge E1(4)$        |
| 1   | 6 | $P \rightarrow Q$                | $\rightarrow I(3, 5)$ |

Rückrichtung:

|       |    |                   |                        |
|-------|----|-------------------|------------------------|
| 7     | 7  | $Q$               | $A$                    |
| 2,7   | 8  | $R$               | $\rightarrow E(2, 7)$  |
| 2,7   | 9  | $Q \wedge R$      | $\wedge I(7, 8)$       |
| 1,2,7 | 10 | $P$               | Th. 2.2.4.2(1, 9)      |
| 1,2   | 11 | $Q \rightarrow P$ | $\rightarrow I(7, 10)$ |

Zusammenführung:

$$1,2 \quad 12 \quad P \leftrightarrow Q$$

$$\leftrightarrow I(6, 11)$$

□

**Theorem 2.2.7.10.**

$$P \vee Q, \neg Q \vdash P$$

$$\begin{array}{ll} 1 & 1 \quad P \vee Q \\ 2 & 2 \quad \neg Q \\ 1 & 3 \quad Q \vee P \\ 1,2 & 4 \quad P \end{array}$$

$$\begin{array}{l} A \\ A \\ \text{Th. 2.1.2.1(1)} \\ \text{Th. 2.2.5.4(3, 2)} \end{array}$$

**Theorem 2.2.7.11.**

$$Q \vdash P \rightarrow Q$$

$$\begin{array}{ll} 1 & 1 \quad Q \\ 1 & 2 \quad \neg P \vee Q \\ 1 & 3 \quad P \rightarrow Q \end{array}$$

$$\begin{array}{l} A \\ \vee I2(1) \\ \text{Th. 2.2.5.1(2)} \end{array}$$

**Theorem 2.2.7.12.**

$$\neg P \vdash P \rightarrow Q$$

$$\begin{array}{ll} 1 & 1 \quad \neg P \\ 1 & 2 \quad \neg P \vee Q \\ 1 & 3 \quad P \rightarrow Q \end{array}$$

$$\begin{array}{l} A \\ \vee I1(1) \\ \text{Th. 2.2.5.1(2)} \end{array}$$

**Theorem 2.2.7.13.**

$$P \rightarrow Q, R \rightarrow Q, P \vee R \vdash Q$$

$$\begin{array}{ll} 1 & 1 \quad P \rightarrow Q \\ 2 & 2 \quad R \rightarrow Q \\ 3 & 3 \quad P \vee R \\ 4 & 4 \quad P \\ 1,4 & 5 \quad Q \\ 6 & 6 \quad R \\ 2,6 & 7 \quad Q \\ 1,2,3 & 8 \quad Q \end{array}$$

$$\begin{array}{l} A \\ A \\ A \\ A \\ \rightarrow E(1, 4) \\ A \\ \rightarrow E(2, 6) \\ \vee E(3, 4, 5, 6, 7) \end{array}$$

**Theorem 2.2.7.14.**

$$P \rightarrow Q, P \vee R \vdash Q \vee R$$

|     |   |                   |                         |
|-----|---|-------------------|-------------------------|
| 1   | 1 | $P \rightarrow Q$ | $A$                     |
| 2   | 2 | $P \vee R$        | $A$                     |
| 3   | 3 | $P$               | $A$                     |
| 1,3 | 4 | $Q$               | $\rightarrow E(1, 3)$   |
| 1,3 | 5 | $Q \vee R$        | $\vee I1(4)$            |
| 6   | 6 | $R$               | $A$                     |
| 6   | 7 | $Q \vee R$        | $\vee I2(6)$            |
| 1,2 | 8 | $Q \vee R$        | $\vee E(2, 3, 5, 6, 7)$ |

## 2.8 Axiome Metamath

**Theorem 2.2.8.1 (Axiom ax-1 (Metamath)).**

*Aussagenvariablen:*  $P, Q$

$$Q \rightarrow (P \rightarrow Q)$$

|   |   |                                   |                       |
|---|---|-----------------------------------|-----------------------|
| 1 | 1 | $Q$                               | $A$                   |
| 1 | 2 | $P \rightarrow Q$                 | Th. 2.2.7.11(1)       |
|   | 3 | $Q \rightarrow (P \rightarrow Q)$ | $\rightarrow I(1, 2)$ |

**Theorem 2.2.8.2 (Axiom ax-2 (Metamath)).**

*Aussagevariablen:*  $P, Q, R$

$$(P \rightarrow (Q \rightarrow R)) \rightarrow ((P \rightarrow Q) \rightarrow (P \rightarrow R))$$

|       |   |                                                                                                   |                       |
|-------|---|---------------------------------------------------------------------------------------------------|-----------------------|
| 1     | 1 | $P \rightarrow (Q \rightarrow R)$                                                                 | $A$                   |
| 2     | 2 | $P \rightarrow Q$                                                                                 | $A$                   |
| 3     | 3 | $P$                                                                                               | $A$                   |
| 1,3   | 4 | $Q \rightarrow R$                                                                                 | $\rightarrow E(1, 3)$ |
| 2,3   | 5 | $Q$                                                                                               | $\rightarrow E(2, 3)$ |
| 1,2,3 | 6 | $R$                                                                                               | $\rightarrow E(4, 5)$ |
| 1,2   | 7 | $P \rightarrow R$                                                                                 | $\rightarrow I(3, 6)$ |
| 1     | 8 | $(P \rightarrow Q) \rightarrow (P \rightarrow R)$                                                 | $\rightarrow I(2, 7)$ |
|       | 9 | $(P \rightarrow (Q \rightarrow R)) \rightarrow ((P \rightarrow Q) \rightarrow (P \rightarrow R))$ | $\rightarrow I(1, 8)$ |

**Theorem 2.2.8.3** (Satz ax-3 (Metamath)).

*Aussagenvariablen:*  $P, Q$

$$(\neg P \rightarrow \neg Q) \rightarrow (Q \rightarrow P)$$

|       |   |                                                             |                       |
|-------|---|-------------------------------------------------------------|-----------------------|
| 1     | 1 | $\neg P \rightarrow \neg Q$                                 | $A$                   |
| 2     | 2 | $Q$                                                         | $A$                   |
| 3     | 3 | $\neg P$                                                    | $A$                   |
| 1,3   | 4 | $\neg Q$                                                    | $\rightarrow E(1, 3)$ |
| 1,2,3 | 5 | $\perp$                                                     | $\perp I(2, 4)$       |
| 1,2   | 6 | $\neg\neg P$                                                | $\neg I(3, 5)$        |
| 1,2   | 7 | $P$                                                         | Th. 2.1.6.1(6)        |
| 1     | 8 | $Q \rightarrow P$                                           | $\rightarrow I(2, 7)$ |
|       | 9 | $(\neg P \rightarrow \neg Q) \rightarrow (Q \rightarrow P)$ | $\rightarrow I(1, 8)$ |

## 2.9 Kontradiktionen und Invarianzen

Im Kapitel verwendete Symbole

*Aussagevariablen:*  $P, Q, R$

**Theorem 2.2.9.1.**

$$\neg(P \leftrightarrow \neg P)$$

|     |   |                                  |                   |
|-----|---|----------------------------------|-------------------|
| 1   | 1 | $P \leftrightarrow \neg P$       | $A$               |
| 2   | 2 | $P$                              | $A$               |
| 1,2 | 3 | $\neg P$                         | Th. 2.2.4.1(1, 2) |
| 1,2 | 4 | $\perp$                          | $\perp I(2, 3)$   |
| 1   | 5 | $\neg P$                         | $\neg I(2, 4)$    |
| 1   | 6 | $P$                              | Th. 2.2.4.2(1, 5) |
| 1   | 7 | $\perp$                          | $\perp I(6, 5)$   |
|     | 8 | $\neg(P \leftrightarrow \neg P)$ | $\neg I(1, 7)$    |

**Theorem 2.2.9.2.**

$$\neg(\neg P \leftrightarrow P)$$

|     |   |                            |                   |
|-----|---|----------------------------|-------------------|
| 1   | 1 | $\neg P \leftrightarrow P$ | $A$               |
| 2   | 2 | $P$                        | $A$               |
| 1,2 | 3 | $\neg P$                   | Th. 2.2.4.2(1, 2) |

|     |   |                                  |                  |
|-----|---|----------------------------------|------------------|
| 1,2 | 4 | $\perp$                          | $\perp I(2,3)$   |
| 1   | 5 | $\neg P$                         | $\neg I(2,4)$    |
| 1   | 6 | $P$                              | Th. 2.2.4.1(1,5) |
| 1   | 7 | $\perp$                          | $\perp I(5,6)$   |
|     | 8 | $\neg(\neg P \leftrightarrow P)$ | $\neg I(1,7)$    |

### 2.9.1 Invarianzen

#### Theorem 2.2.9.3.

$$Q \vdash (P \leftrightarrow Q \wedge R) \leftrightarrow (P \leftrightarrow R)$$

*Beweis.*

Hinrichtung:

|       |    |                                                                    |                           |
|-------|----|--------------------------------------------------------------------|---------------------------|
| 1     | 1  | $Q$                                                                | $A$                       |
| 2     | 2  | $P \leftrightarrow Q \wedge R$                                     | $A$                       |
| 2     | 3  | $P \rightarrow Q \wedge R$                                         | $\leftrightarrow E1(2)$   |
| 2     | 4  | $Q \wedge R \rightarrow P$                                         | $\leftrightarrow E2(2)$   |
| 5     | 5  | $P$                                                                | $A$                       |
| 2,5   | 6  | $Q \wedge R$                                                       | $\rightarrow E(3,5)$      |
| 2,5   | 7  | $R$                                                                | $\wedge E2(6)$            |
| 2     | 8  | $P \rightarrow R$                                                  | $\rightarrow I(5,7)$      |
| 9     | 9  | $R$                                                                | $A$                       |
| 1,9   | 10 | $Q \wedge R$                                                       | $\wedge I(1,9)$           |
| 1,2,9 | 11 | $P$                                                                | $\rightarrow E(4,10)$     |
| 1,2   | 12 | $R \rightarrow P$                                                  | $\rightarrow I(9,11)$     |
| 1,2   | 13 | $P \leftrightarrow R$                                              | $\leftrightarrow I(8,12)$ |
| 1     | 14 | $(P \leftrightarrow Q \wedge R) \rightarrow (P \leftrightarrow R)$ | $\rightarrow I(2,13)$     |

Rückrichtung:

|         |    |                            |                          |
|---------|----|----------------------------|--------------------------|
| 15      | 15 | $P \leftrightarrow R$      | $A$                      |
| 15      | 16 | $P \rightarrow R$          | $\leftrightarrow E1(15)$ |
| 15      | 17 | $R \rightarrow P$          | $\leftrightarrow E2(15)$ |
| 18      | 18 | $P$                        | $A$                      |
| 15,18   | 19 | $R$                        | $\rightarrow E(16,18)$   |
| 1,15,18 | 20 | $Q \wedge R$               | $\wedge I(1,19)$         |
| 1,15    | 21 | $P \rightarrow Q \wedge R$ | $\rightarrow I(18,20)$   |
| 22      | 22 | $Q \wedge R$               | $A$                      |
| 22      | 23 | $R$                        | $\wedge E2(22)$          |
| 15,22   | 24 | $P$                        | $\rightarrow E(17,23)$   |
| 15      | 25 | $Q \wedge R \rightarrow P$ | $\rightarrow I(22,24)$   |

$$\begin{array}{lll}
1,15 & 26 & P \leftrightarrow Q \wedge R & \leftrightarrow I(21, 25) \\
1 & 27 & (P \leftrightarrow R) \rightarrow (P \leftrightarrow Q \wedge R) & \rightarrow I(15, 26)
\end{array}$$

Zusammenführung:

$$1 \quad 28 \quad (P \leftrightarrow Q \wedge R) \leftrightarrow (P \leftrightarrow R) \quad \leftrightarrow I(14, 27)$$

□

# Kapitel 3

## Exklusives Oder

### 3.1 Definition und Grundregeln

Im Kapitel verwendete Symbole

Aussagevariablen:  $P, Q$

**Definition 2.3.1.1 (Exklusives Oder).**

$$P \underline{\vee} Q := \neg(P \leftrightarrow Q)$$

**Theorem 2.3.1.1 (Definitionsäquivalenz für  $\underline{\vee}$ ).**

$$P \underline{\vee} Q \dashv\vdash \neg(P \leftrightarrow Q)$$

*Beweis.*

$\vdash$ :

|   |   |                             |                 |
|---|---|-----------------------------|-----------------|
| 1 | 1 | $P \underline{\vee} Q$      | $A$             |
| 1 | 2 | $\neg(P \leftrightarrow Q)$ | Def. 2.3.1.1(1) |

$\dashv$ :

|   |   |                             |                 |
|---|---|-----------------------------|-----------------|
| 1 | 1 | $\neg(P \leftrightarrow Q)$ | $A$             |
| 1 | 2 | $P \underline{\vee} Q$      | Def. 2.3.1.1(1) |

□

**Theorem 2.3.1.2 (Fallunterscheidung für  $\underline{\vee}$ ).**

$$P \underline{\vee} Q \dashv\vdash (\neg P \wedge Q) \vee (P \wedge \neg Q)$$

*Beweis.*

$\vdash$ :

|   |   |                                            |                |
|---|---|--------------------------------------------|----------------|
| 1 | 1 | $P \underline{\vee} Q$                     | $A$            |
| 1 | 2 | $\neg(P \leftrightarrow Q)$                | Th. 2.3.1.1(1) |
| 1 | 3 | $(\neg P \wedge Q) \vee (P \wedge \neg Q)$ | Th. 2.2.6.6(2) |

$\dashv$ :

|   |   |                                            |                |
|---|---|--------------------------------------------|----------------|
| 1 | 1 | $(\neg P \wedge Q) \vee (P \wedge \neg Q)$ | $A$            |
| 1 | 2 | $\neg(P \leftrightarrow Q)$                | Th. 2.2.6.6(1) |
| 1 | 3 | $P \underline{\vee} Q$                     | Th. 2.3.1.1(2) |

□

## 3.2 Einführungsregeln

**Theorem 2.3.2.1 (Einführung von  $\underline{\vee}$ , linker Fall).**

$$P, \neg Q \vdash P \underline{\vee} Q$$

|     |   |                                            |                 |
|-----|---|--------------------------------------------|-----------------|
| 1   | 1 | $P$                                        | $A$             |
|     | 2 | $\neg Q$                                   | $A$             |
| 1,2 | 3 | $P \wedge \neg Q$                          | $\wedge I(1,2)$ |
| 1,2 | 4 | $(\neg P \wedge Q) \vee (P \wedge \neg Q)$ | $\vee I2(3)$    |
| 1,2 | 5 | $P \underline{\vee} Q$                     | Th. 2.3.1.2(4)  |

**Theorem 2.3.2.2 (Einführung von  $\underline{\vee}$ , rechter Fall).**

$$\neg P, Q \vdash P \underline{\vee} Q$$

|     |   |                                            |                 |
|-----|---|--------------------------------------------|-----------------|
| 1   | 1 | $\neg P$                                   | $A$             |
|     | 2 | $Q$                                        | $A$             |
| 1,2 | 3 | $\neg P \wedge Q$                          | $\wedge I(1,2)$ |
| 1,2 | 4 | $(\neg P \wedge Q) \vee (P \wedge \neg Q)$ | $\vee I1(3)$    |
| 1,2 | 5 | $P \underline{\vee} Q$                     | Th. 2.3.1.2(4)  |

## 3.3 Eliminationsregeln

**Theorem 2.3.3.1 (Elimination von  $\underline{\vee}$  bei  $P$ ).**

$$P \underline{\vee} Q, P \vdash \neg Q$$

|       |   |                             |                          |
|-------|---|-----------------------------|--------------------------|
| 1     | 1 | $P \vee Q$                  | $A$                      |
| 2     | 2 | $P$                         | $A$                      |
| 1     | 3 | $\neg(P \leftrightarrow Q)$ | Th. 2.3.1.1(1)           |
| 4     | 4 | $Q$                         | $A$                      |
| 4     | 5 | $P \rightarrow Q$           | Th. 2.2.7.11(4)          |
| 2     | 6 | $Q \rightarrow P$           | Th. 2.2.7.11(2)          |
| 2,4   | 7 | $P \leftrightarrow Q$       | $\leftrightarrow I(5,6)$ |
| 1,2,4 | 8 | $\perp$                     | $\perp I(3,7)$           |
| 1,2   | 9 | $\neg Q$                    | $\neg I(4,8)$            |

**Theorem 2.3.3.2 (Elimination von  $\vee$  bei  $Q$ ).**

$$P \vee Q, Q \vdash \neg P$$

|       |   |                             |                          |
|-------|---|-----------------------------|--------------------------|
| 1     | 1 | $P \vee Q$                  | $A$                      |
| 2     | 2 | $Q$                         | $A$                      |
| 1     | 3 | $\neg(P \leftrightarrow Q)$ | Th. 2.3.1.1(1)           |
| 4     | 4 | $P$                         | $A$                      |
| 2     | 5 | $P \rightarrow Q$           | Th. 2.2.7.11(2)          |
| 4     | 6 | $Q \rightarrow P$           | Th. 2.2.7.11(4)          |
| 2,4   | 7 | $P \leftrightarrow Q$       | $\leftrightarrow I(5,6)$ |
| 1,2,4 | 8 | $\perp$                     | $\perp I(3,7)$           |
| 1,2   | 9 | $\neg P$                    | $\neg I(4,8)$            |

**Theorem 2.3.3.3 (Elimination von  $\vee$  bei  $\neg P$ ).**

$$P \vee Q, \neg P \vdash Q$$

|       |   |                             |                          |
|-------|---|-----------------------------|--------------------------|
| 1     | 1 | $P \vee Q$                  | $A$                      |
| 2     | 2 | $\neg P$                    | $A$                      |
| 1     | 3 | $\neg(P \leftrightarrow Q)$ | Th. 2.3.1.1(1)           |
| 4     | 4 | $\neg Q$                    | $A$                      |
| 2     | 5 | $P \rightarrow Q$           | Th. 2.2.7.12(2)          |
| 4     | 6 | $Q \rightarrow P$           | Th. 2.2.7.12(4)          |
| 2,4   | 7 | $P \leftrightarrow Q$       | $\leftrightarrow I(5,6)$ |
| 1,2,4 | 8 | $\perp$                     | $\perp I(3,7)$           |
| 1,2   | 9 | $Q$                         | $\neg E(4,8)$            |

**Theorem 2.3.3.4 (Elimination von  $\vee$  bei  $\neg Q$ ).**

$$P \vee Q, \neg Q \vdash P$$

|       |   |                             |                          |
|-------|---|-----------------------------|--------------------------|
| 1     | 1 | $P \vee Q$                  | $A$                      |
| 2     | 2 | $\neg Q$                    | $A$                      |
| 1     | 3 | $\neg(P \leftrightarrow Q)$ | Th. 2.3.1.1(1)           |
| 4     | 4 | $\neg P$                    | $A$                      |
| 4     | 5 | $P \rightarrow Q$           | Th. 2.2.7.12(4)          |
| 2     | 6 | $Q \rightarrow P$           | Th. 2.2.7.12(2)          |
| 2,4   | 7 | $P \leftrightarrow Q$       | $\leftrightarrow I(5,6)$ |
| 1,2,4 | 8 | $\perp$                     | $\perp I(3,7)$           |
| 1,2   | 9 | $P$                         | $\neg E(4,8)$            |

### 3.4 Strukturgesetze

**Theorem 2.3.4.1 (Kommutativgesetz für  $\vee$ ).**

$$P \vee Q \dashv\vdash Q \vee P$$

*Beweis.*

$\vdash$ :

|     |   |                             |                |
|-----|---|-----------------------------|----------------|
| 1   | 1 | $P \vee Q$                  | $A$            |
| 1   | 2 | $\neg(P \leftrightarrow Q)$ | Th. 2.3.1.1(1) |
| 3   | 3 | $Q \leftrightarrow P$       | $A$            |
| 3   | 4 | $P \leftrightarrow Q$       | Th. 2.1.2.5(3) |
| 1,3 | 5 | $\perp$                     | $\perp I(2,4)$ |
| 1   | 6 | $\neg(Q \leftrightarrow P)$ | $\neg I(3,5)$  |
| 1   | 7 | $Q \vee P$                  | Th. 2.3.1.1(6) |

$\dashv\vdash$ :

|     |   |                             |                |
|-----|---|-----------------------------|----------------|
| 1   | 1 | $Q \vee P$                  | $A$            |
| 1   | 2 | $\neg(Q \leftrightarrow P)$ | Th. 2.3.1.1(1) |
| 3   | 3 | $P \leftrightarrow Q$       | $A$            |
| 3   | 4 | $Q \leftrightarrow P$       | Th. 2.1.2.5(3) |
| 1,3 | 5 | $\perp$                     | $\perp I(2,4)$ |
| 1   | 6 | $\neg(P \leftrightarrow Q)$ | $\neg I(3,5)$  |
| 1   | 7 | $P \vee Q$                  | Th. 2.3.1.1(6) |

□

**Theorem 2.3.4.2 (Negation von  $\vee$ ).**

$$\neg(P \vee Q) \dashv\vdash P \leftrightarrow Q$$

*Beweis.*

$\vdash$ :

|     |   |                                 |                |
|-----|---|---------------------------------|----------------|
| 1   | 1 | $\neg(P \vee Q)$                | $A$            |
| 2   | 2 | $\neg(P \leftrightarrow Q)$     | $A$            |
| 2   | 3 | $P \vee Q$                      | Th. 2.3.1.1(2) |
| 1,2 | 4 | $\perp$                         | $\perp I(1,3)$ |
| 1   | 5 | $\neg\neg(P \leftrightarrow Q)$ | $\neg I(2,4)$  |
| 1   | 6 | $P \leftrightarrow Q$           | Th. 2.1.6.1(5) |

$\dashv$ :

|     |   |                             |                |
|-----|---|-----------------------------|----------------|
| 1   | 1 | $P \leftrightarrow Q$       | $A$            |
| 2   | 2 | $P \vee Q$                  | $A$            |
| 2   | 3 | $\neg(P \leftrightarrow Q)$ | Th. 2.3.1.1(2) |
| 1,2 | 4 | $\perp$                     | $\perp I(3,1)$ |
| 1   | 5 | $\neg(P \vee Q)$            | $\neg I(2,4)$  |

□

**Theorem 2.3.4.3 (Invarianz von  $\vee$  unter doppelter Negation).**

$$P \vee Q \dashv\vdash \neg P \vee \neg Q$$

*Beweis.*

$\vdash$ :

|     |   |                                       |                |
|-----|---|---------------------------------------|----------------|
| 1   | 1 | $P \vee Q$                            | $A$            |
| 1   | 2 | $\neg(P \leftrightarrow Q)$           | Th. 2.3.1.1(1) |
| 3   | 3 | $\neg P \leftrightarrow \neg Q$       | $A$            |
| 3   | 4 | $P \leftrightarrow Q$                 | Th. 2.2.6.1(3) |
| 1,3 | 5 | $\perp$                               | $\perp I(2,4)$ |
| 1   | 6 | $\neg(\neg P \leftrightarrow \neg Q)$ | $\neg I(3,5)$  |
| 1   | 7 | $\neg P \vee \neg Q$                  | Th. 2.3.1.1(6) |

$\dashv$ :

|     |   |                                       |                |
|-----|---|---------------------------------------|----------------|
| 1   | 1 | $\neg P \vee \neg Q$                  | $A$            |
| 1   | 2 | $\neg(\neg P \leftrightarrow \neg Q)$ | Th. 2.3.1.1(1) |
| 3   | 3 | $P \leftrightarrow Q$                 | $A$            |
| 3   | 4 | $\neg P \leftrightarrow \neg Q$       | Th. 2.2.6.1(3) |
| 1,3 | 5 | $\perp$                               | $\perp I(2,4)$ |
| 1   | 6 | $\neg(P \leftrightarrow Q)$           | $\neg I(3,5)$  |
| 1   | 7 | $P \vee Q$                            | Th. 2.3.1.1(6) |

□

**Theorem 2.3.4.4 (Irreflexivität von  $\underline{\vee}$ ).**

$$\neg(P \underline{\vee} P)$$

$$1 \ P \leftrightarrow P$$

Th. 2.1.1.3

$$2 \ \neg(P \underline{\vee} P)$$

Th. 2.3.4.2(1)

**Theorem 2.3.4.5 (Exklusives Drittengesetz).**

$$P \underline{\vee} \neg P$$

$$1 \ \neg(P \leftrightarrow \neg P)$$

Th. 2.2.9.1

$$2 \ P \underline{\vee} \neg P$$

Th. 2.3.1.1(1)

# Kapitel 4

## NAND

### 4.1 Definition und Grundregeln

Im Kapitel verwendete Symbole

Aussagevariablen:  $P, Q$

**Definition 2.4.1.1 (NAND).**

$$P \uparrow Q := \neg(P \wedge Q)$$

**Theorem 2.4.1.1 (Definitionsäquivalenz für  $\uparrow$ ).**

$$P \uparrow Q \dashv\vdash \neg(P \wedge Q)$$

*Beweis.*

$\vdash$ :

|   |   |                    |                 |
|---|---|--------------------|-----------------|
| 1 | 1 | $P \uparrow Q$     | $A$             |
| 1 | 2 | $\neg(P \wedge Q)$ | Def. 2.4.1.1(1) |

$\dashv$ :

|   |   |                    |                 |
|---|---|--------------------|-----------------|
| 1 | 1 | $\neg(P \wedge Q)$ | $A$             |
| 1 | 2 | $P \uparrow Q$     | Def. 2.4.1.1(1) |

□

**Theorem 2.4.1.2 (De-Morgan-Form für  $\uparrow$ ).**

$$P \uparrow Q \dashv\vdash \neg P \vee \neg Q$$

*Beweis.*

$\vdash$ :

|   |   |                      |                |
|---|---|----------------------|----------------|
| 1 | 1 | $P \uparrow Q$       | $A$            |
| 1 | 2 | $\neg(P \wedge Q)$   | Th. 2.4.1.1(1) |
| 1 | 3 | $\neg P \vee \neg Q$ | Th. 2.1.8.2(2) |

$\dashv$ :

|   |   |                      |                |
|---|---|----------------------|----------------|
| 1 | 1 | $\neg P \vee \neg Q$ | $A$            |
| 1 | 2 | $\neg(P \wedge Q)$   | Th. 2.1.8.2(1) |
| 1 | 3 | $P \uparrow Q$       | Th. 2.4.1.1(2) |

□

**Theorem 2.4.1.3 (Negation von  $\uparrow$ ).**

$$\neg(P \uparrow Q) \dashv\vdash P \wedge Q$$

*Beweis.*

$\vdash$ :

|     |   |                        |                |
|-----|---|------------------------|----------------|
| 1   | 1 | $\neg(P \uparrow Q)$   | $A$            |
| 2   | 2 | $\neg(P \wedge Q)$     | $A$            |
| 2   | 3 | $P \uparrow Q$         | Th. 2.4.1.1(2) |
| 1,2 | 4 | $\perp$                | $\perp I(1,3)$ |
| 1   | 5 | $\neg\neg(P \wedge Q)$ | $\neg I(2,4)$  |
| 1   | 6 | $P \wedge Q$           | Th. 2.1.6.1(5) |

$\dashv$ :

|     |   |                      |                |
|-----|---|----------------------|----------------|
| 1   | 1 | $P \wedge Q$         | $A$            |
| 2   | 2 | $P \uparrow Q$       | $A$            |
| 2   | 3 | $\neg(P \wedge Q)$   | Th. 2.4.1.1(2) |
| 1,2 | 4 | $\perp$              | $\perp I(3,1)$ |
| 1   | 5 | $\neg(P \uparrow Q)$ | $\neg I(2,4)$  |

□

## 4.2 Einführungsregeln

**Theorem 2.4.2.1 (Einführung von  $\uparrow$ , linker Fall).**

$$\neg P \vdash P \uparrow Q$$

|   |   |                      |                |
|---|---|----------------------|----------------|
| 1 | 1 | $\neg P$             | $A$            |
| 1 | 2 | $\neg P \vee \neg Q$ | $\vee I1(1)$   |
| 1 | 3 | $P \uparrow Q$       | Th. 2.4.1.2(2) |

**Theorem 2.4.2.2 (Einführung von  $\uparrow$ , rechter Fall).**

$$\neg Q \vdash P \uparrow Q$$

|   |   |                      |                |
|---|---|----------------------|----------------|
| 1 | 1 | $\neg Q$             | $A$            |
| 1 | 2 | $\neg P \vee \neg Q$ | $\vee I2(1)$   |
| 1 | 3 | $P \uparrow Q$       | Th. 2.4.1.2(2) |

## 4.3 Eliminationsregeln

**Theorem 2.4.3.1 (Elimination von  $\uparrow$  bei  $P$ ).**

$$P \uparrow Q, P \vdash \neg Q$$

|       |   |                    |                 |
|-------|---|--------------------|-----------------|
| 1     | 1 | $P \uparrow Q$     | $A$             |
| 2     | 2 | $P$                | $A$             |
| 1     | 3 | $\neg(P \wedge Q)$ | Th. 2.4.1.1(1)  |
| 4     | 4 | $Q$                | $A$             |
| 2,4   | 5 | $P \wedge Q$       | $\wedge I(2,4)$ |
| 1,2,4 | 6 | $\perp$            | $\perp I(3,5)$  |
| 1,2   | 7 | $\neg Q$           | $\neg I(4,6)$   |

**Theorem 2.4.3.2 (Elimination von  $\uparrow$  bei  $Q$ ).**

$$P \uparrow Q, Q \vdash \neg P$$

|       |   |                    |                 |
|-------|---|--------------------|-----------------|
| 1     | 1 | $P \uparrow Q$     | $A$             |
| 2     | 2 | $Q$                | $A$             |
| 1     | 3 | $\neg(P \wedge Q)$ | Th. 2.4.1.1(1)  |
| 4     | 4 | $P$                | $A$             |
| 2,4   | 5 | $P \wedge Q$       | $\wedge I(4,2)$ |
| 1,2,4 | 6 | $\perp$            | $\perp I(3,5)$  |

1,2 7  $\neg P$

$\neg I(4, 6)$

## 4.4 Strukturgesetze

**Theorem 2.4.4.1 (Kommutativgesetz für  $\uparrow$ ).**

$$P \uparrow Q \dashv\vdash Q \uparrow P$$

*Beweis.*

$\vdash$ :

|   |   |                      |                |
|---|---|----------------------|----------------|
| 1 | 1 | $P \uparrow Q$       | $A$            |
| 1 | 2 | $\neg P \vee \neg Q$ | Th. 2.4.1.2(1) |
| 1 | 3 | $\neg Q \vee \neg P$ | Th. 2.1.2.2(2) |
| 1 | 4 | $Q \uparrow P$       | Th. 2.4.1.2(3) |

$\dashv$ :

|   |   |                      |                |
|---|---|----------------------|----------------|
| 1 | 1 | $Q \uparrow P$       | $A$            |
| 1 | 2 | $\neg Q \vee \neg P$ | Th. 2.4.1.2(1) |
| 1 | 3 | $\neg P \vee \neg Q$ | Th. 2.1.2.2(2) |
| 1 | 4 | $P \uparrow Q$       | Th. 2.4.1.2(3) |

□

**Theorem 2.4.4.2 (Selbst-NAND als Negation).**

$$P \uparrow P \dashv\vdash \neg P$$

*Beweis.*

$\vdash$ :

|   |   |                      |                |
|---|---|----------------------|----------------|
| 1 | 1 | $P \uparrow P$       | $A$            |
| 1 | 2 | $\neg P \vee \neg P$ | Th. 2.4.1.2(1) |
| 1 | 3 | $\neg P$             | Th. 2.1.1.4(2) |

$\dashv$ :

|   |   |                |                |
|---|---|----------------|----------------|
| 1 | 1 | $\neg P$       | $A$            |
| 1 | 2 | $P \uparrow P$ | Th. 2.4.2.1(1) |

□

**Theorem 2.4.4.3 (Konjunktion durch  $\uparrow$ ).**

$$P \wedge Q \dashv\vdash (P \uparrow Q) \uparrow (P \uparrow Q)$$

*Beweis.*

$\vdash$ :

|   |   |                                          |                |
|---|---|------------------------------------------|----------------|
| 1 | 1 | $P \wedge Q$                             | $A$            |
| 1 | 2 | $\neg(P \uparrow Q)$                     | Th. 2.4.1.3(1) |
| 1 | 3 | $(P \uparrow Q) \uparrow (P \uparrow Q)$ | Th. 2.4.4.2(2) |

$\dashv$ :

|   |   |                                          |                |
|---|---|------------------------------------------|----------------|
| 1 | 1 | $(P \uparrow Q) \uparrow (P \uparrow Q)$ | $A$            |
| 1 | 2 | $\neg(P \uparrow Q)$                     | Th. 2.4.4.2(1) |
| 1 | 3 | $P \wedge Q$                             | Th. 2.4.1.3(2) |

□

**Theorem 2.4.4.4 (Disjunktion durch  $\uparrow$ ).**

$$P \vee Q \dashv\vdash (P \uparrow P) \uparrow (Q \uparrow Q)$$

*Beweis.*

$\vdash$ :

|   |    |                                              |                               |
|---|----|----------------------------------------------|-------------------------------|
| 1 | 1  | $P \vee Q$                                   | $A$                           |
| 2 | 2  | $P$                                          | $A$                           |
| 2 | 3  | $P \wedge P$                                 | $\wedge I^*(\text{Wdh}_2(2))$ |
| 2 | 4  | $\neg(P \uparrow P)$                         | Th. 2.4.1.3(3)                |
| 2 | 5  | $\neg(P \uparrow P) \vee \neg(Q \uparrow Q)$ | $\vee I1(4)$                  |
| 6 | 6  | $Q$                                          | $A$                           |
| 6 | 7  | $Q \wedge Q$                                 | $\wedge I^*(\text{Wdh}_2(6))$ |
| 6 | 8  | $\neg(Q \uparrow Q)$                         | Th. 2.4.1.3(7)                |
| 6 | 9  | $\neg(P \uparrow P) \vee \neg(Q \uparrow Q)$ | $\vee I2(8)$                  |
| 1 | 10 | $\neg(P \uparrow P) \vee \neg(Q \uparrow Q)$ | $\vee E(1, 2, 5, 6, 9)$       |
| 1 | 11 | $(P \uparrow P) \uparrow (Q \uparrow Q)$     | Th. 2.4.1.2(10)               |

$\dashv$ :

|   |   |                                              |                |
|---|---|----------------------------------------------|----------------|
| 1 | 1 | $(P \uparrow P) \uparrow (Q \uparrow Q)$     | $A$            |
| 1 | 2 | $\neg(P \uparrow P) \vee \neg(Q \uparrow Q)$ | Th. 2.4.1.2(1) |
| 3 | 3 | $\neg(P \uparrow P)$                         | $A$            |
| 3 | 4 | $P \wedge P$                                 | Th. 2.4.1.3(3) |

|   |    |                      |                          |
|---|----|----------------------|--------------------------|
| 3 | 5  | $P$                  | $\wedge E1(4)$           |
| 3 | 6  | $P \vee Q$           | $\vee I1(5)$             |
| 7 | 7  | $\neg(Q \uparrow Q)$ | $A$                      |
| 7 | 8  | $Q \wedge Q$         | Th. 2.4.1.3(7)           |
| 7 | 9  | $Q$                  | $\wedge E1(8)$           |
| 7 | 10 | $P \vee Q$           | $\vee I2(9)$             |
| 1 | 11 | $P \vee Q$           | $\vee E(2, 3, 6, 7, 10)$ |

□

**Theorem 2.4.4.5 (Implikation durch  $\uparrow$ ).**

$$P \rightarrow Q \dashv\vdash P \uparrow (Q \uparrow Q)$$

*Beweis.*

$\vdash$ :

|   |    |                                  |                               |
|---|----|----------------------------------|-------------------------------|
| 1 | 1  | $P \rightarrow Q$                | $A$                           |
| 1 | 2  | $\neg P \vee Q$                  | Th. 2.2.5.1(1)                |
| 3 | 3  | $\neg P$                         | $A$                           |
| 3 | 4  | $\neg P \vee \neg(Q \uparrow Q)$ | $\vee I1(3)$                  |
| 5 | 5  | $Q$                              | $A$                           |
| 5 | 6  | $Q \wedge Q$                     | $\wedge I^*(\text{Wdh}_2(5))$ |
| 5 | 7  | $\neg(Q \uparrow Q)$             | Th. 2.4.1.3(6)                |
| 5 | 8  | $\neg P \vee \neg(Q \uparrow Q)$ | $\vee I2(7)$                  |
| 1 | 9  | $\neg P \vee \neg(Q \uparrow Q)$ | $\vee E(2, 3, 4, 5, 8)$       |
| 1 | 10 | $P \uparrow (Q \uparrow Q)$      | Th. 2.4.1.2(9)                |

$\dashv\vdash$ :

|   |    |                                  |                         |
|---|----|----------------------------------|-------------------------|
| 1 | 1  | $P \uparrow (Q \uparrow Q)$      | $A$                     |
| 1 | 2  | $\neg P \vee \neg(Q \uparrow Q)$ | Th. 2.4.1.2(1)          |
| 3 | 3  | $\neg P$                         | $A$                     |
| 3 | 4  | $\neg P \vee Q$                  | $\vee I1(3)$            |
| 5 | 5  | $\neg(Q \uparrow Q)$             | $A$                     |
| 5 | 6  | $Q \wedge Q$                     | Th. 2.4.1.3(5)          |
| 5 | 7  | $Q$                              | $\wedge E1(6)$          |
| 5 | 8  | $\neg P \vee Q$                  | $\vee I2(7)$            |
| 1 | 9  | $\neg P \vee Q$                  | $\vee E(2, 3, 4, 5, 8)$ |
| 1 | 10 | $P \rightarrow Q$                | Th. 2.2.5.1(9)          |

□

# Kapitel 5

## NOR

### 5.1 Definition und Grundregeln

Im Kapitel verwendete Symbole

Aussagevariablen:  $P, Q$

**Definition 2.5.1.1 (NOR).**

$$P \downarrow Q := \neg(P \vee Q)$$

**Theorem 2.5.1.1 (Definitionsäquivalenz für  $\downarrow$ ).**

$$P \downarrow Q \dashv\vdash \neg(P \vee Q)$$

*Beweis.*

$\vdash$ :

|   |   |                  |                 |
|---|---|------------------|-----------------|
| 1 | 1 | $P \downarrow Q$ | $A$             |
| 1 | 2 | $\neg(P \vee Q)$ | Def. 2.5.1.1(1) |

$\dashv\vdash$ :

|   |   |                  |                 |
|---|---|------------------|-----------------|
| 1 | 1 | $\neg(P \vee Q)$ | $A$             |
| 1 | 2 | $P \downarrow Q$ | Def. 2.5.1.1(1) |

□

**Theorem 2.5.1.2 (De-Morgan-Form für  $\downarrow$ ).**

$$P \downarrow Q \dashv\vdash \neg P \wedge \neg Q$$

*Beweis.*

$\vdash$ :

|   |   |                        |                |
|---|---|------------------------|----------------|
| 1 | 1 | $P \downarrow Q$       | $A$            |
| 1 | 2 | $\neg(P \vee Q)$       | Th. 2.5.1.1(1) |
| 1 | 3 | $\neg P \wedge \neg Q$ | Th. 2.1.8.1(2) |

$\dashv$ :

|   |   |                        |                |
|---|---|------------------------|----------------|
| 1 | 1 | $\neg P \wedge \neg Q$ | $A$            |
| 1 | 2 | $\neg(P \vee Q)$       | Th. 2.1.8.1(1) |
| 1 | 3 | $P \downarrow Q$       | Th. 2.5.1.1(2) |

□

**Theorem 2.5.1.3 (Negation von  $\downarrow$ ).**

$$\neg(P \downarrow Q) \dashv\vdash P \vee Q$$

*Beweis.*

$\vdash$ :

|     |   |                        |                |
|-----|---|------------------------|----------------|
| 1   | 1 | $\neg(P \downarrow Q)$ | $A$            |
| 2   | 2 | $\neg(P \vee Q)$       | $A$            |
| 2   | 3 | $P \downarrow Q$       | Th. 2.5.1.1(2) |
| 1,2 | 4 | $\perp$                | $\perp I(1,3)$ |
| 1   | 5 | $\neg\neg(P \vee Q)$   | $\neg I(2,4)$  |
| 1   | 6 | $P \vee Q$             | Th. 2.1.6.1(5) |

$\dashv$ :

|     |   |                        |                |
|-----|---|------------------------|----------------|
| 1   | 1 | $P \vee Q$             | $A$            |
| 2   | 2 | $P \downarrow Q$       | $A$            |
| 2   | 3 | $\neg(P \vee Q)$       | Th. 2.5.1.1(2) |
| 1,2 | 4 | $\perp$                | $\perp I(3,1)$ |
| 1   | 5 | $\neg(P \downarrow Q)$ | $\neg I(2,4)$  |

□

## 5.2 Einführungsregel

**Theorem 2.5.2.1 (Einführung von  $\downarrow$ ).**

$$\neg P, \neg Q \vdash P \downarrow Q$$

|     |   |                        |                 |
|-----|---|------------------------|-----------------|
| 1   | 1 | $\neg P$               | $A$             |
|     | 2 | $\neg Q$               | $A$             |
| 1,2 | 3 | $\neg P \wedge \neg Q$ | $\wedge I(1,2)$ |
| 1,2 | 4 | $P \downarrow Q$       | Th. 2.5.1.2(3)  |

## 5.3 Eliminationsregeln

**Theorem 2.5.3.1 (Elimination von  $\downarrow$ , linker Teil).**

$$P \downarrow Q \vdash \neg P$$

|   |   |                        |                |
|---|---|------------------------|----------------|
| 1 | 1 | $P \downarrow Q$       | $A$            |
| 1 | 2 | $\neg P \wedge \neg Q$ | Th. 2.5.1.2(1) |
| 1 | 3 | $\neg P$               | $\wedge E1(2)$ |

**Theorem 2.5.3.2 (Elimination von  $\downarrow$ , rechter Teil).**

$$P \downarrow Q \vdash \neg Q$$

|   |   |                        |                |
|---|---|------------------------|----------------|
| 1 | 1 | $P \downarrow Q$       | $A$            |
| 1 | 2 | $\neg P \wedge \neg Q$ | Th. 2.5.1.2(1) |
| 1 | 3 | $\neg Q$               | $\wedge E2(2)$ |

## 5.4 Strukturgesetze

**Theorem 2.5.4.1 (Kommutativgesetz für  $\downarrow$ ).**

$$P \downarrow Q \dashv\vdash Q \downarrow P$$

*Beweis.*

$\vdash$ :

|   |   |                        |                |
|---|---|------------------------|----------------|
| 1 | 1 | $P \downarrow Q$       | $A$            |
| 1 | 2 | $\neg P \wedge \neg Q$ | Th. 2.5.1.2(1) |
| 1 | 3 | $\neg Q \wedge \neg P$ | Th. 2.1.2.4(2) |
| 1 | 4 | $Q \downarrow P$       | Th. 2.5.1.2(3) |

$\dashv$ :

|   |   |                        |                |
|---|---|------------------------|----------------|
| 1 | 1 | $Q \downarrow P$       | $A$            |
| 1 | 2 | $\neg Q \wedge \neg P$ | Th. 2.5.1.2(1) |
| 1 | 3 | $\neg P \wedge \neg Q$ | Th. 2.1.2.4(2) |
| 1 | 4 | $P \downarrow Q$       | Th. 2.5.1.2(3) |

□

**Theorem 2.5.4.2 (Selbst-NOR als Negation).**

$$P \downarrow P \dashv \vdash \neg P$$

*Beweis.*

$\vdash$ :

|   |   |                        |                |
|---|---|------------------------|----------------|
| 1 | 1 | $P \downarrow P$       | $A$            |
| 1 | 2 | $\neg P \wedge \neg P$ | Th. 2.5.1.2(1) |
| 1 | 3 | $\neg P$               | Th. 2.1.1.5(2) |

$\dashv$ :

|   |   |                        |                               |
|---|---|------------------------|-------------------------------|
| 1 | 1 | $\neg P$               | $A$                           |
| 1 | 2 | $\neg P \wedge \neg P$ | $\wedge I^*(\text{Wdh}_2(1))$ |
| 1 | 3 | $P \downarrow P$       | Th. 2.5.1.2(2)                |

□

**Theorem 2.5.4.3 (Disjunktion durch  $\downarrow$ ).**

$$P \vee Q \dashv \vdash (P \downarrow Q) \downarrow (P \downarrow Q)$$

*Beweis.*

$\vdash$ :

|   |   |                                                |                |
|---|---|------------------------------------------------|----------------|
| 1 | 1 | $P \vee Q$                                     | $A$            |
| 1 | 2 | $\neg(P \downarrow Q)$                         | Th. 2.5.1.3(1) |
| 1 | 3 | $(P \downarrow Q) \downarrow (P \downarrow Q)$ | Th. 2.5.4.2(2) |

$\dashv$ :

|   |   |                                                |                |
|---|---|------------------------------------------------|----------------|
| 1 | 1 | $(P \downarrow Q) \downarrow (P \downarrow Q)$ | $A$            |
| 1 | 2 | $\neg(P \downarrow Q)$                         | Th. 2.5.4.2(1) |
| 1 | 3 | $P \vee Q$                                     | Th. 2.5.1.3(2) |

□

**Theorem 2.5.4.4 (Konjunktion durch  $\downarrow$ ).**

$$P \wedge Q \dashv\vdash (P \downarrow P) \downarrow (Q \downarrow Q)$$

*Beweis.*

$\vdash$ :

|   |   |                                                    |                  |
|---|---|----------------------------------------------------|------------------|
| 1 | 1 | $P \wedge Q$                                       | $A$              |
| 1 | 2 | $P$                                                | $\wedge E1(1)$   |
| 1 | 3 | $Q$                                                | $\wedge E2(1)$   |
| 1 | 4 | $P \vee P$                                         | $\vee I1(2)$     |
| 1 | 5 | $Q \vee Q$                                         | $\vee I1(3)$     |
| 1 | 6 | $\neg(P \downarrow P)$                             | Th. 2.5.1.3(4)   |
| 1 | 7 | $\neg(Q \downarrow Q)$                             | Th. 2.5.1.3(5)   |
| 1 | 8 | $\neg(P \downarrow P) \wedge \neg(Q \downarrow Q)$ | $\wedge I(6, 7)$ |
| 1 | 9 | $(P \downarrow P) \downarrow (Q \downarrow Q)$     | Th. 2.5.1.2(8)   |

$\dashv\vdash$ :

|   |   |                                                    |                  |
|---|---|----------------------------------------------------|------------------|
| 1 | 1 | $(P \downarrow P) \downarrow (Q \downarrow Q)$     | $A$              |
| 1 | 2 | $\neg(P \downarrow P) \wedge \neg(Q \downarrow Q)$ | Th. 2.5.1.2(1)   |
| 1 | 3 | $\neg(P \downarrow P)$                             | $\wedge E1(2)$   |
| 1 | 4 | $\neg(Q \downarrow Q)$                             | $\wedge E2(2)$   |
| 1 | 5 | $P \vee P$                                         | Th. 2.5.1.3(3)   |
| 1 | 6 | $Q \vee Q$                                         | Th. 2.5.1.3(4)   |
| 1 | 7 | $P$                                                | Th. 2.1.1.4(5)   |
| 1 | 8 | $Q$                                                | Th. 2.1.1.4(6)   |
| 1 | 9 | $P \wedge Q$                                       | $\wedge I(7, 8)$ |

□

**Theorem 2.5.4.5 (Negation der Implikation durch  $\downarrow$ ).**

$$(P \downarrow P) \downarrow Q \dashv\vdash \neg(P \rightarrow Q)$$

*Beweis.*

$\vdash$ :

|   |   |                                      |                |
|---|---|--------------------------------------|----------------|
| 1 | 1 | $(P \downarrow P) \downarrow Q$      | $A$            |
| 1 | 2 | $\neg(P \downarrow P) \wedge \neg Q$ | Th. 2.5.1.2(1) |
| 1 | 3 | $\neg(P \downarrow P)$               | $\wedge E1(2)$ |
| 1 | 4 | $\neg Q$                             | $\wedge E2(2)$ |
| 1 | 5 | $P \vee P$                           | Th. 2.5.1.3(3) |
| 1 | 6 | $P$                                  | Th. 2.1.1.4(5) |

|   |   |                         |                  |
|---|---|-------------------------|------------------|
| 1 | 7 | $P \wedge \neg Q$       | $\wedge I(6, 4)$ |
| 1 | 8 | $\neg(P \rightarrow Q)$ | Th. 2.2.5.5(7)   |

$\dashv$ :

|   |   |                                      |                  |
|---|---|--------------------------------------|------------------|
| 1 | 1 | $\neg(P \rightarrow Q)$              | $A$              |
| 1 | 2 | $P \wedge \neg Q$                    | Th. 2.2.5.5(1)   |
| 1 | 3 | $P$                                  | $\wedge E1(2)$   |
| 1 | 4 | $\neg Q$                             | $\wedge E2(2)$   |
| 1 | 5 | $P \vee P$                           | $\vee I1(3)$     |
| 1 | 6 | $\neg(P \downarrow P)$               | Th. 2.5.1.3(5)   |
| 1 | 7 | $\neg(P \downarrow P) \wedge \neg Q$ | $\wedge I(6, 4)$ |
| 1 | 8 | $(P \downarrow P) \downarrow Q$      | Th. 2.5.1.2(7)   |

□

**Theorem 2.5.4.6 (Implikation durch  $\downarrow$ ).**

$$P \rightarrow Q \dashv\vdash ((P \downarrow P) \downarrow Q) \downarrow ((P \downarrow P) \downarrow Q)$$

*Beweis.*

$\vdash$ :

|     |   |                                                                              |                 |
|-----|---|------------------------------------------------------------------------------|-----------------|
| 1   | 1 | $P \rightarrow Q$                                                            | $A$             |
| 2   | 2 | $(P \downarrow P) \downarrow Q$                                              | $A$             |
| 2   | 3 | $\neg(P \rightarrow Q)$                                                      | Th. 2.5.4.5(2)  |
| 1,2 | 4 | $\perp$                                                                      | $\perp I(3, 1)$ |
| 1   | 5 | $\neg((P \downarrow P) \downarrow Q)$                                        | $\neg I(2, 4)$  |
| 1   | 6 | $((P \downarrow P) \downarrow Q) \downarrow ((P \downarrow P) \downarrow Q)$ | Th. 2.5.4.2(5)  |

$\dashv$ :

|     |   |                                                                              |                 |
|-----|---|------------------------------------------------------------------------------|-----------------|
| 1   | 1 | $((P \downarrow P) \downarrow Q) \downarrow ((P \downarrow P) \downarrow Q)$ | $A$             |
| 1   | 2 | $\neg((P \downarrow P) \downarrow Q)$                                        | Th. 2.5.4.2(1)  |
| 3   | 3 | $\neg(P \rightarrow Q)$                                                      | $A$             |
| 3   | 4 | $(P \downarrow P) \downarrow Q$                                              | Th. 2.5.4.5(3)  |
| 1,3 | 5 | $\perp$                                                                      | $\perp I(2, 4)$ |
| 1   | 6 | $\neg\neg(P \rightarrow Q)$                                                  | $\neg I(3, 5)$  |
| 1   | 7 | $P \rightarrow Q$                                                            | Th. 2.1.6.1(6)  |

□

*Bemerkung.* Durch die Sätze Th. 2.5.4.2, Th. 2.5.4.3, Th. 2.5.4.4 und Th. 2.5.4.6 kann man Negation, Disjunktion, Konjunktion und Implikation allein mithilfe von  $\downarrow$  ausdrücken.

# Kapitel 6

## Halbaddierer

### 6.1 Definitionen und Grundregeln

Im Kapitel verwendete Symbole

Aussagevariablen:  $P, Q, R$

**Definition 2.6.1.1** (Summenbit des Halbaddierers).

$$S_{1/2}(P, Q) := P \vee Q$$

**Definition 2.6.1.2** (Übertragsbit des Halbaddierers).

$$C_{1/2}(P, Q) := P \wedge Q$$

**Theorem 2.6.1.1** (Definitionsäquivalenz für das Summenbit).

$$S_{1/2}(P, Q) \dashv\vdash P \vee Q$$

*Beweis.*

$\vdash$ :

|   |   |                 |                 |
|---|---|-----------------|-----------------|
| 1 | 1 | $S_{1/2}(P, Q)$ | $A$             |
| 1 | 2 | $P \vee Q$      | Def. 2.6.1.1(1) |

$\dashv$ :

|   |   |                 |                 |
|---|---|-----------------|-----------------|
| 1 | 1 | $P \vee Q$      | $A$             |
| 1 | 2 | $S_{1/2}(P, Q)$ | Def. 2.6.1.1(1) |

□

**Theorem 2.6.1.2** (Definitionsäquivalenz für das Übertragsbit).

$$C_{1/2}(P, Q) \dashv\vdash P \wedge Q$$

*Beweis.*

$\vdash$ :

|   |   |                 |                 |
|---|---|-----------------|-----------------|
| 1 | 1 | $C_{1/2}(P, Q)$ | $A$             |
| 1 | 2 | $P \wedge Q$    | Def. 2.6.1.2(1) |

$\dashv$ :

|   |   |                 |                 |
|---|---|-----------------|-----------------|
| 1 | 1 | $P \wedge Q$    | $A$             |
| 1 | 2 | $C_{1/2}(P, Q)$ | Def. 2.6.1.2(1) |

□

**Theorem 2.6.1.3 (Fallunterscheidung für das Summenbit).**

$$S_{1/2}(P, Q) \dashv\vdash (\neg P \wedge Q) \vee (P \wedge \neg Q)$$

*Beweis.*

$\vdash$ :

|   |   |                                            |                |
|---|---|--------------------------------------------|----------------|
| 1 | 1 | $S_{1/2}(P, Q)$                            | $A$            |
| 1 | 2 | $P \underline{\vee} Q$                     | Th. 2.6.1.1(1) |
| 1 | 3 | $(\neg P \wedge Q) \vee (P \wedge \neg Q)$ | Th. 2.3.1.2(2) |

$\dashv$ :

|   |   |                                            |                |
|---|---|--------------------------------------------|----------------|
| 1 | 1 | $(\neg P \wedge Q) \vee (P \wedge \neg Q)$ | $A$            |
| 1 | 2 | $P \underline{\vee} Q$                     | Th. 2.3.1.2(1) |
| 1 | 3 | $S_{1/2}(P, Q)$                            | Th. 2.6.1.1(2) |

□

**Theorem 2.6.1.4 (Negation des Summenbits).**

$$\neg S_{1/2}(P, Q) \dashv\vdash P \leftrightarrow Q$$

*Beweis.*

$\vdash$ :

|     |   |                              |                 |
|-----|---|------------------------------|-----------------|
| 1   | 1 | $\neg S_{1/2}(P, Q)$         | $A$             |
| 2   | 2 | $P \underline{\vee} Q$       | $A$             |
| 2   | 3 | $S_{1/2}(P, Q)$              | Th. 2.6.1.1(2)  |
| 1,2 | 4 | $\perp$                      | $\perp I(1, 3)$ |
| 1   | 5 | $\neg(P \underline{\vee} Q)$ | $\neg I(2, 4)$  |

|   |   |                       |                |
|---|---|-----------------------|----------------|
| 1 | 6 | $P \leftrightarrow Q$ | Th. 2.3.4.2(5) |
|---|---|-----------------------|----------------|

⊢:

|     |   |                       |                 |
|-----|---|-----------------------|-----------------|
| 1   | 1 | $P \leftrightarrow Q$ | $A$             |
| 1   | 2 | $\neg(P \vee Q)$      | Th. 2.3.4.2(1)  |
| 3   | 3 | $S_{1/2}(P, Q)$       | $A$             |
| 3   | 4 | $P \vee Q$            | Th. 2.6.1.1(3)  |
| 1,3 | 5 | $\perp$               | $\perp I(2, 4)$ |
| 1   | 6 | $\neg S_{1/2}(P, Q)$  | $\neg I(3, 5)$  |

□

**Theorem 2.6.1.5 (Negation des Übertragsbits).**

$$\neg C_{1/2}(P, Q) \dashv\vdash \neg P \vee \neg Q$$

*Beweis.*

⊢:

|     |   |                      |                 |
|-----|---|----------------------|-----------------|
| 1   | 1 | $\neg C_{1/2}(P, Q)$ | $A$             |
| 2   | 2 | $P \wedge Q$         | $A$             |
| 2   | 3 | $C_{1/2}(P, Q)$      | Th. 2.6.1.2(2)  |
| 1,2 | 4 | $\perp$              | $\perp I(1, 3)$ |
| 1   | 5 | $\neg(P \wedge Q)$   | $\neg I(2, 4)$  |
| 1   | 6 | $\neg P \vee \neg Q$ | Th. 2.1.8.2(5)  |

⊢:

|     |   |                      |                 |
|-----|---|----------------------|-----------------|
| 1   | 1 | $\neg P \vee \neg Q$ | $A$             |
| 1   | 2 | $\neg(P \wedge Q)$   | Th. 2.1.8.2(1)  |
| 3   | 3 | $C_{1/2}(P, Q)$      | $A$             |
| 3   | 4 | $P \wedge Q$         | Th. 2.6.1.2(3)  |
| 1,3 | 5 | $\perp$              | $\perp I(2, 4)$ |
| 1   | 6 | $\neg C_{1/2}(P, Q)$ | $\neg I(3, 5)$  |

□

## 6.2 Einführungsregeln

**Theorem 2.6.2.1 (Einführung des Summenbits, linker Fall).**

$$P, \neg Q \vdash S_{1/2}(P, Q)$$

|     |   |                 |                   |
|-----|---|-----------------|-------------------|
| 1   | 1 | $P$             | $A$               |
| 2   | 2 | $\neg Q$        | $A$               |
| 1,2 | 3 | $P \vee Q$      | Th. 2.3.2.1(1, 2) |
| 1,2 | 4 | $S_{1/2}(P, Q)$ | Th. 2.6.1.1(3)    |

**Theorem 2.6.2.2 (Einführung des Summenbits, rechter Fall).**

$$\neg P, Q \vdash S_{1/2}(P, Q)$$

|     |   |                 |                   |
|-----|---|-----------------|-------------------|
| 1   | 1 | $\neg P$        | $A$               |
| 2   | 2 | $Q$             | $A$               |
| 1,2 | 3 | $P \vee Q$      | Th. 2.3.2.2(1, 2) |
| 1,2 | 4 | $S_{1/2}(P, Q)$ | Th. 2.6.1.1(3)    |

**Theorem 2.6.2.3 (Einführung des Übertragsbits).**

$$P, Q \vdash C_{1/2}(P, Q)$$

|     |   |                 |                  |
|-----|---|-----------------|------------------|
| 1   | 1 | $P$             | $A$              |
| 2   | 2 | $Q$             | $A$              |
| 1,2 | 3 | $P \wedge Q$    | $\wedge I(1, 2)$ |
| 1,2 | 4 | $C_{1/2}(P, Q)$ | Th. 2.6.1.2(3)   |

## 6.3 Eliminationsregeln

**Theorem 2.6.3.1 (Elimination des Übertragsbits, linker Teil).**

$$C_{1/2}(P, Q) \vdash P$$

|   |   |                 |                |
|---|---|-----------------|----------------|
| 1 | 1 | $C_{1/2}(P, Q)$ | $A$            |
| 1 | 2 | $P \wedge Q$    | Th. 2.6.1.2(1) |
| 1 | 3 | $P$             | $\wedge E1(2)$ |

**Theorem 2.6.3.2 (Elimination des Übertragsbits, rechter Teil).**

$$C_{1/2}(P, Q) \vdash Q$$

|   |   |                 |                |
|---|---|-----------------|----------------|
| 1 | 1 | $C_{1/2}(P, Q)$ | $A$            |
| 1 | 2 | $P \wedge Q$    | Th. 2.6.1.2(1) |
| 1 | 3 | $Q$             | $\wedge E2(2)$ |

**Theorem 2.6.3.3 (Summenbit schließt Übertragsbit aus).**

$$S_{1/2}(P, Q) \vdash \neg C_{1/2}(P, Q)$$

|   |    |                                            |                         |
|---|----|--------------------------------------------|-------------------------|
| 1 | 1  | $S_{1/2}(P, Q)$                            | $A$                     |
| 1 | 2  | $(\neg P \wedge Q) \vee (P \wedge \neg Q)$ | Th. 2.6.1.3(1)          |
| 3 | 3  | $\neg P \wedge Q$                          | $A$                     |
| 3 | 4  | $\neg P$                                   | $\wedge E1(3)$          |
| 3 | 5  | $\neg P \vee \neg Q$                       | $\vee I1(4)$            |
| 6 | 6  | $P \wedge \neg Q$                          | $A$                     |
| 6 | 7  | $\neg Q$                                   | $\wedge E2(6)$          |
| 6 | 8  | $\neg P \vee \neg Q$                       | $\vee I2(7)$            |
| 1 | 9  | $\neg P \vee \neg Q$                       | $\vee E(2, 3, 5, 6, 8)$ |
| 1 | 10 | $\neg C_{1/2}(P, Q)$                       | Th. 2.6.1.5(9)          |

**Theorem 2.6.3.4 (Übertragsbit schließt Summenbit aus).**

$$C_{1/2}(P, Q) \vdash \neg S_{1/2}(P, Q)$$

|     |   |                      |                 |
|-----|---|----------------------|-----------------|
|     | 1 | $C_{1/2}(P, Q)$      | $A$             |
|     | 2 | $S_{1/2}(P, Q)$      | $A$             |
|     | 2 | $\neg C_{1/2}(P, Q)$ | Th. 2.6.3.3(2)  |
| 1,2 | 4 | $\perp$              | $\perp I(3, 1)$ |
| 1   | 5 | $\neg S_{1/2}(P, Q)$ | $\neg I(2, 4)$  |

**Theorem 2.6.3.5 (Widerspruch von Summenbit und Übertragsbit).**

$$S_{1/2}(P, Q) \wedge C_{1/2}(P, Q) \vdash R$$

|   |   |                                      |                   |
|---|---|--------------------------------------|-------------------|
| 1 | 1 | $S_{1/2}(P, Q) \wedge C_{1/2}(P, Q)$ | $A$               |
| 1 | 2 | $S_{1/2}(P, Q)$                      | $\wedge E1(1)$    |
| 1 | 3 | $C_{1/2}(P, Q)$                      | $\wedge E2(1)$    |
| 1 | 4 | $\neg C_{1/2}(P, Q)$                 | Th. 2.6.3.3(2)    |
| 1 | 5 | $R$                                  | Th. 2.1.7.3(3, 4) |

## 6.4 Strukturgesetze

**Theorem 2.6.4.1 (Kommutativität des Summenbits).**

$$S_{1/2}(P, Q) \dashv\vdash S_{1/2}(Q, P)$$

*Beweis.*

$\vdash$ :

|   |   |                        |                |
|---|---|------------------------|----------------|
| 1 | 1 | $S_{1/2}(P, Q)$        | $A$            |
| 1 | 2 | $P \underline{\vee} Q$ | Th. 2.6.1.1(1) |
| 1 | 3 | $Q \underline{\vee} P$ | Th. 2.3.4.1(2) |
| 1 | 4 | $S_{1/2}(Q, P)$        | Th. 2.6.1.1(3) |

$\dashv$ :

|   |   |                        |                |
|---|---|------------------------|----------------|
| 1 | 1 | $S_{1/2}(Q, P)$        | $A$            |
| 1 | 2 | $Q \underline{\vee} P$ | Th. 2.6.1.1(1) |
| 1 | 3 | $P \underline{\vee} Q$ | Th. 2.3.4.1(2) |
| 1 | 4 | $S_{1/2}(P, Q)$        | Th. 2.6.1.1(3) |

□

**Theorem 2.6.4.2 (Kommutativität des Übertragsbits).**

$$C_{1/2}(P, Q) \dashv\vdash C_{1/2}(Q, P)$$

*Beweis.*

$\vdash$ :

|   |   |                 |                |
|---|---|-----------------|----------------|
| 1 | 1 | $C_{1/2}(P, Q)$ | $A$            |
| 1 | 2 | $P \wedge Q$    | Th. 2.6.1.2(1) |
| 1 | 3 | $Q \wedge P$    | Th. 2.1.2.4(2) |
| 1 | 4 | $C_{1/2}(Q, P)$ | Th. 2.6.1.2(3) |

$\dashv$ :

|   |   |                 |                |
|---|---|-----------------|----------------|
| 1 | 1 | $C_{1/2}(Q, P)$ | $A$            |
| 1 | 2 | $Q \wedge P$    | Th. 2.6.1.2(1) |
| 1 | 3 | $P \wedge Q$    | Th. 2.1.2.4(2) |
| 1 | 4 | $C_{1/2}(P, Q)$ | Th. 2.6.1.2(3) |

□

## 6.5 Wahrheitstafel des Halbaddierers

**Theorem 2.6.5.1 (Halbaddierer bei zwei falschen Eingängen).**

$$\neg P, \neg Q \vdash \neg S_{1/2}(P, Q) \wedge \neg C_{1/2}(P, Q)$$

*Beweis.*

Summenbit:

|     |   |                       |                           |
|-----|---|-----------------------|---------------------------|
| 1   | 1 | $\neg P$              | $A$                       |
| 2   | 2 | $\neg Q$              | $A$                       |
| 1   | 3 | $P \rightarrow Q$     | Th. 2.2.7.12(1)           |
| 2   | 4 | $Q \rightarrow P$     | Th. 2.2.7.12(2)           |
| 1,2 | 5 | $P \leftrightarrow Q$ | $\leftrightarrow I(3, 4)$ |
| 1,2 | 6 | $\neg S_{1/2}(P, Q)$  | Th. 2.6.1.4(5)            |

Übertragsbit:

|   |   |                      |                |
|---|---|----------------------|----------------|
| 1 | 7 | $\neg P \vee \neg Q$ | $\vee I1(1)$   |
| 1 | 8 | $\neg C_{1/2}(P, Q)$ | Th. 2.6.1.5(7) |

Zusammenführung:

|     |   |                                                |                  |
|-----|---|------------------------------------------------|------------------|
| 1,2 | 9 | $\neg S_{1/2}(P, Q) \wedge \neg C_{1/2}(P, Q)$ | $\wedge I(6, 8)$ |
|-----|---|------------------------------------------------|------------------|

□

**Theorem 2.6.5.2 (Halbaddierer bei linkem wahren Eingang).**

$$P, \neg Q \vdash S_{1/2}(P, Q) \wedge \neg C_{1/2}(P, Q)$$

*Beweis.*

Summenbit:

|     |   |                 |                   |
|-----|---|-----------------|-------------------|
| 1   | 1 | $P$             | $A$               |
| 2   | 2 | $\neg Q$        | $A$               |
| 1,2 | 3 | $S_{1/2}(P, Q)$ | Th. 2.6.2.1(1, 2) |

Übertragsbit:

|   |   |                      |                |
|---|---|----------------------|----------------|
| 2 | 4 | $\neg P \vee \neg Q$ | $\vee I2(2)$   |
| 2 | 5 | $\neg C_{1/2}(P, Q)$ | Th. 2.6.1.5(4) |

Zusammenführung:

|     |   |                                           |                  |
|-----|---|-------------------------------------------|------------------|
| 1,2 | 6 | $S_{1/2}(P, Q) \wedge \neg C_{1/2}(P, Q)$ | $\wedge I(3, 5)$ |
|-----|---|-------------------------------------------|------------------|

□

**Theorem 2.6.5.3 (Halbaddierer bei rechtem wahren Eingang).**

$$\neg P, Q \vdash S_{1/2}(P, Q) \wedge \neg C_{1/2}(P, Q)$$

*Beweis.*

Summenbit:

|     |   |                 |                   |
|-----|---|-----------------|-------------------|
| 1   | 1 | $\neg P$        | $A$               |
| 2   | 2 | $Q$             | $A$               |
| 1,2 | 3 | $S_{1/2}(P, Q)$ | Th. 2.6.2.2(1, 2) |

Übertragsbit:

|   |   |                      |                |
|---|---|----------------------|----------------|
| 1 | 4 | $\neg P \vee \neg Q$ | $\vee I1(1)$   |
| 1 | 5 | $\neg C_{1/2}(P, Q)$ | Th. 2.6.1.5(4) |

Zusammenführung:

|     |   |                                           |                  |
|-----|---|-------------------------------------------|------------------|
| 1,2 | 6 | $S_{1/2}(P, Q) \wedge \neg C_{1/2}(P, Q)$ | $\wedge I(3, 5)$ |
|-----|---|-------------------------------------------|------------------|

□

**Theorem 2.6.5.4 (Halbaddierer bei zwei wahren Eingängen).**

$$P, Q \vdash \neg S_{1/2}(P, Q) \wedge C_{1/2}(P, Q)$$

*Beweis.*

Summenbit:

|     |   |                       |                           |
|-----|---|-----------------------|---------------------------|
| 1   | 1 | $P$                   | $A$                       |
| 2   | 2 | $Q$                   | $A$                       |
| 2   | 3 | $P \rightarrow Q$     | Th. 2.2.7.11(2)           |
| 1   | 4 | $Q \rightarrow P$     | Th. 2.2.7.11(1)           |
| 1,2 | 5 | $P \leftrightarrow Q$ | $\leftrightarrow I(3, 4)$ |
| 1,2 | 6 | $\neg S_{1/2}(P, Q)$  | Th. 2.6.1.4(5)            |

Übertragsbit:

|     |   |                 |                   |
|-----|---|-----------------|-------------------|
| 1,2 | 7 | $C_{1/2}(P, Q)$ | Th. 2.6.2.3(1, 2) |
|-----|---|-----------------|-------------------|

Zusammenführung:

|     |   |                                           |                  |
|-----|---|-------------------------------------------|------------------|
| 1,2 | 8 | $\neg S_{1/2}(P, Q) \wedge C_{1/2}(P, Q)$ | $\wedge I(6, 7)$ |
|-----|---|-------------------------------------------|------------------|

□

# Kapitel 7

## Volladdierer

### 7.1 Definitionen und Grundregeln

Im Kapitel verwendete Symbole

Aussagevariablen:  $P, Q, R$

**Definition 2.7.1.1** (Summenbit des Volladdierers).

$$S_{FA}(P, Q, R) := S_{1/2}(S_{1/2}(P, Q), R)$$

**Definition 2.7.1.2** (Übertragsbit des Volladdierers).

$$C_{FA}(P, Q, R) := C_{1/2}(P, Q) \vee C_{1/2}(S_{1/2}(P, Q), R)$$

**Theorem 2.7.1.1** (Definitionsäquivalenz für das Summenbit des Volladdierers).

$$S_{FA}(P, Q, R) \dashv\vdash S_{1/2}(S_{1/2}(P, Q), R)$$

*Beweis.*

$\vdash$ :

|   |   |                             |                 |
|---|---|-----------------------------|-----------------|
| 1 | 1 | $S_{FA}(P, Q, R)$           | $A$             |
| 1 | 2 | $S_{1/2}(S_{1/2}(P, Q), R)$ | Def. 2.7.1.1(1) |

$\dashv$ :

|   |   |                             |                 |
|---|---|-----------------------------|-----------------|
| 1 | 1 | $S_{1/2}(S_{1/2}(P, Q), R)$ | $A$             |
| 1 | 2 | $S_{FA}(P, Q, R)$           | Def. 2.7.1.1(1) |

□

**Theorem 2.7.1.2 (Definitionsäquivalenz für das Übertragsbit des Volladdierers).**

$$C_{FA}(P, Q, R) \dashv\vdash C_{1/2}(P, Q) \vee C_{1/2}(S_{1/2}(P, Q), R)$$

*Beweis.*

$\vdash$ :

|   |   |                                                                         |                 |
|---|---|-------------------------------------------------------------------------|-----------------|
| 1 | 1 | C <sub>FA</sub> (P, Q, R)                                               | A               |
| 1 | 2 | C <sub>1/2</sub> (P, Q) ∨ C <sub>1/2</sub> (S <sub>1/2</sub> (P, Q), R) | Def. 2.7.1.2(1) |

$\dashv$ :

|   |   |                                                                         |                 |
|---|---|-------------------------------------------------------------------------|-----------------|
| 1 | 1 | C <sub>1/2</sub> (P, Q) ∨ C <sub>1/2</sub> (S <sub>1/2</sub> (P, Q), R) | A               |
| 1 | 2 | C <sub>FA</sub> (P, Q, R)                                               | Def. 2.7.1.2(1) |

□

**Theorem 2.7.1.3 (Summenbit des Volladdierers).**

$$S_{FA}(P, Q, R) \dashv\vdash (P \vee Q) \vee R$$

*Beweis.*

$\vdash$ :

|   |   |                                               |                     |
|---|---|-----------------------------------------------|---------------------|
| 1 | 1 | S <sub>FA</sub> (P, Q, R)                     | A                   |
| 1 | 2 | S <sub>1/2</sub> (S <sub>1/2</sub> (P, Q), R) | Th. 2.7.1.1(1)      |
| 1 | 3 | S <sub>1/2</sub> (P, Q) ∨ R                   | Th. 2.6.1.1(2)      |
| 1 | 4 | (P ∨ Q) ∨ R                                   | ↔ S(Th. 2.6.1.1, 3) |

$\dashv$ :

|   |   |                                               |                     |
|---|---|-----------------------------------------------|---------------------|
| 1 | 1 | (P ∨ Q) ∨ R                                   | A                   |
| 1 | 2 | S <sub>1/2</sub> (P, Q) ∨ R                   | ↔ S(Th. 2.6.1.1, 1) |
| 1 | 3 | S <sub>1/2</sub> (S <sub>1/2</sub> (P, Q), R) | Th. 2.6.1.1(2)      |
| 1 | 4 | S <sub>FA</sub> (P, Q, R)                     | Th. 2.7.1.1(3)      |

□

**Theorem 2.7.1.4 (Übertragsbit des Volladdierers).**

$$C_{FA}(P, Q, R) \dashv\vdash (P \wedge Q) \vee ((P \vee Q) \wedge R)$$

*Beweis.*

$\vdash$ :

|   |   |                                                |                                            |
|---|---|------------------------------------------------|--------------------------------------------|
| 1 | 1 | $C_{FA}(P, Q, R)$                              | $A$                                        |
| 1 | 2 | $C_{1/2}(P, Q) \vee C_{1/2}(S_{1/2}(P, Q), R)$ | Th. 2.7.1.2(1)                             |
| 1 | 3 | $(P \wedge Q) \vee C_{1/2}(S_{1/2}(P, Q), R)$  | $\leftrightarrow S(\text{Th. 2.6.1.2, 2})$ |
| 1 | 4 | $(P \wedge Q) \vee (S_{1/2}(P, Q) \wedge R)$   | $\leftrightarrow S(\text{Th. 2.6.1.2, 3})$ |
| 1 | 5 | $(P \wedge Q) \vee ((P \vee Q) \wedge R)$      | $\leftrightarrow S(\text{Th. 2.6.1.1, 4})$ |

$\dashv$ :

|   |   |                                                |                                            |
|---|---|------------------------------------------------|--------------------------------------------|
| 1 | 1 | $(P \wedge Q) \vee ((P \vee Q) \wedge R)$      | $A$                                        |
| 1 | 2 | $(P \wedge Q) \vee (S_{1/2}(P, Q) \wedge R)$   | $\leftrightarrow S(\text{Th. 2.6.1.1, 1})$ |
| 1 | 3 | $(P \wedge Q) \vee C_{1/2}(S_{1/2}(P, Q), R)$  | $\leftrightarrow S(\text{Th. 2.6.1.2, 2})$ |
| 1 | 4 | $C_{1/2}(P, Q) \vee C_{1/2}(S_{1/2}(P, Q), R)$ | $\leftrightarrow S(\text{Th. 2.6.1.2, 3})$ |
| 1 | 5 | $C_{FA}(P, Q, R)$                              | Th. 2.7.1.2(4)                             |

□

**Theorem 2.7.1.5 (Metamath-Form des Übertragsbits des Vollad-  
dierers).**

$$C_{FA}(P, Q, R) \dashv\vdash (P \wedge Q) \vee (R \wedge (P \vee Q))$$

*Beweis.*

$\vdash$ :

|   |   |                                           |                                            |
|---|---|-------------------------------------------|--------------------------------------------|
| 1 | 1 | $C_{FA}(P, Q, R)$                         | $A$                                        |
| 1 | 2 | $(P \wedge Q) \vee ((P \vee Q) \wedge R)$ | Th. 2.7.1.4(1)                             |
| 1 | 3 | $(P \wedge Q) \vee (R \wedge (P \vee Q))$ | $\leftrightarrow S(\text{Th. 2.1.2.4, 2})$ |

$\dashv$ :

|   |   |                                           |                                            |
|---|---|-------------------------------------------|--------------------------------------------|
| 1 | 1 | $(P \wedge Q) \vee (R \wedge (P \vee Q))$ | $A$                                        |
| 1 | 2 | $(P \wedge Q) \vee ((P \vee Q) \wedge R)$ | $\leftrightarrow S(\text{Th. 2.1.2.4, 1})$ |
| 1 | 3 | $C_{FA}(P, Q, R)$                         | Th. 2.7.1.4(2)                             |

□

# Kapitel 8

## Wahrheitswerte

### 8.1 Definition und Grundregeln

Im Kapitel verwendete Symbole

Aussagevariablen:  $P, Q, R$

**Definition 2.8.1.1 (Wahrheitswert wahr).**

$$\top := \neg \perp$$

**Theorem 2.8.1.1 (Definitionsäquivalenz für  $\top$ ).**

$$\top \dashv\vdash \neg \perp$$

*Beweis.*

$\vdash$ :

|   |   |              |                 |
|---|---|--------------|-----------------|
| 1 | 1 | $\top$       | $A$             |
| 1 | 2 | $\neg \perp$ | Def. 2.8.1.1(1) |

$\dashv$ :

|   |   |              |                 |
|---|---|--------------|-----------------|
| 1 | 1 | $\neg \perp$ | $A$             |
| 1 | 2 | $\top$       | Def. 2.8.1.1(1) |

□

**Theorem 2.8.1.2 (Wahrheit).**

$$\top$$

|   |                |                |
|---|----------------|----------------|
| 1 | 1 $\perp$      | $A$            |
|   | 2 $\neg \perp$ | $\neg I(1,1)$  |
|   | 3 $\top$       | Th. 2.8.1.1(2) |

**Theorem 2.8.1.3 (Negation von wahr).**

$$\neg \top \dashv\vdash \perp$$

*Beweis.*

$\vdash$ :

|   |               |                |
|---|---------------|----------------|
| 1 | 1 $\neg \top$ | $A$            |
|   | 2 $\top$      | Th. 2.8.1.2    |
| 1 | 3 $\perp$     | $\perp I(1,2)$ |

$\dashv$ :

|     |                |                |
|-----|----------------|----------------|
| 1   | 1 $\perp$      | $A$            |
| 2   | 2 $\top$       | $A$            |
| 2   | 3 $\neg \perp$ | Th. 2.8.1.1(2) |
| 1,2 | 4 $\perp$      | $\perp I(3,1)$ |
| 1   | 5 $\neg \top$  | $\neg I(2,4)$  |

□

**Theorem 2.8.1.4 (Ausgeschlossenes Drittes als Wahrheit).**

$$P \vee \neg P \vdash \top$$

|   |                   |             |
|---|-------------------|-------------|
| 1 | 1 $P \vee \neg P$ | $A$         |
|   | 2 $\top$          | Th. 2.8.1.2 |

## 8.2 Wahrheitswerte der Negation

**Theorem 2.8.2.1.**

$$\neg \perp \dashv\vdash \top$$

*Beweis.*

$\vdash$ :

|   |                |                |
|---|----------------|----------------|
| 1 | 1 $\neg \perp$ | $A$            |
| 1 | 2 $\top$       | Th. 2.8.1.1(1) |

$\dashv$ :

|   |   |              |
|---|---|--------------|
| 1 | 1 | $\top$       |
| 1 | 2 | $\neg \perp$ |

$A$

Th. 2.8.1.1(1)

□

### 8.3 Wahrheitswerte der Konjunktion

**Theorem 2.8.3.1.**

$\top \wedge \top \dashv\vdash \top$

*Beweis.*

$\vdash$ :

|   |   |                    |
|---|---|--------------------|
| 1 | 1 | $\top \wedge \top$ |
| 1 | 2 | $\top$             |

$A$

$\wedge E1(1)$

$\dashv$ :

|   |   |                    |
|---|---|--------------------|
| 1 | 1 | $\top$             |
| 1 | 2 | $\top \wedge \top$ |

$A$

$\wedge I^*(Wdh_2(1))$

□

**Theorem 2.8.3.2.**

$\top \wedge \perp \dashv\vdash \perp$

*Beweis.*

$\vdash$ :

|   |   |                     |
|---|---|---------------------|
| 1 | 1 | $\top \wedge \perp$ |
| 1 | 2 | $\perp$             |

$A$

$\wedge E2(1)$

$\dashv$ :

|   |   |                     |
|---|---|---------------------|
| 1 | 1 | $\perp$             |
|   | 2 | $\top$              |
| 1 | 3 | $\top \wedge \perp$ |

$A$

Th. 2.8.1.2

$\wedge I(2,1)$

□

**Theorem 2.8.3.3.**

$$\perp \wedge \top \dashv\vdash \perp$$

*Beweis.*

$\vdash$ :

$$\begin{array}{l} 1 \quad 1 \quad \perp \wedge \top \\ 1 \quad 2 \quad \perp \end{array}$$

$$\begin{array}{l} A \\ \wedge E1(1) \end{array}$$

$\dashv$ :

$$\begin{array}{l} 1 \quad 1 \quad \perp \\ \quad 2 \quad \top \\ 1 \quad 3 \quad \perp \wedge \top \end{array}$$

$$\begin{array}{l} A \\ \text{Th. 2.8.1.2} \\ \wedge I(1, 2) \end{array}$$

□

**Theorem 2.8.3.4.**

$$\perp \wedge \perp \dashv\vdash \perp$$

*Beweis.*

$\vdash$ :

$$\begin{array}{l} 1 \quad 1 \quad \perp \wedge \perp \\ 1 \quad 2 \quad \perp \end{array}$$

$$\begin{array}{l} A \\ \wedge E1(1) \end{array}$$

$\dashv$ :

$$\begin{array}{l} 1 \quad 1 \quad \perp \\ 1 \quad 2 \quad \perp \wedge \perp \end{array}$$

$$\begin{array}{l} A \\ \wedge I^*(\text{Wdh}_2(1)) \end{array}$$

□

## 8.4 Wahrheitswerte der Disjunktion

**Theorem 2.8.4.1.**

$$\top \vee \top \dashv\vdash \top$$

*Beweis.*

$\vdash$ :

|   |   |                  |
|---|---|------------------|
| 1 | 1 | $\top \vee \top$ |
| 2 | 2 | $\top$           |
| 1 | 3 | $\top$           |

|                         |
|-------------------------|
| $A$                     |
| $A$                     |
| $\vee E(1, 2, 2, 2, 2)$ |

$\vdash$ :

|   |   |                  |
|---|---|------------------|
| 1 | 1 | $\top$           |
| 1 | 2 | $\top \vee \top$ |

|              |
|--------------|
| $A$          |
| $\vee I1(1)$ |

□

### Theorem 2.8.4.2.

$$\top \vee \perp \dashv\vdash \top$$

*Beweis.*

$\vdash$ :

|   |   |                   |
|---|---|-------------------|
| 1 | 1 | $\top \vee \perp$ |
| 2 | 2 | $\top$            |
| 3 | 3 | $\perp$           |
|   | 4 | $\top$            |
| 1 | 5 | $\top$            |

|                         |
|-------------------------|
| $A$                     |
| $A$                     |
| $A$                     |
| Th. 2.8.1.2             |
| $\vee E(1, 2, 2, 3, 4)$ |

$\vdash$ :

|   |   |                   |
|---|---|-------------------|
| 1 | 1 | $\top$            |
| 1 | 2 | $\top \vee \perp$ |

|              |
|--------------|
| $A$          |
| $\vee I1(1)$ |

□

### Theorem 2.8.4.3.

$$\perp \vee \top \dashv\vdash \top$$

*Beweis.*

$\vdash$ :

|   |   |                   |
|---|---|-------------------|
| 1 | 1 | $\perp \vee \top$ |
| 2 | 2 | $\perp$           |
|   | 3 | $\top$            |
| 4 | 4 | $\top$            |
| 1 | 5 | $\top$            |

|                         |
|-------------------------|
| $A$                     |
| $A$                     |
| Th. 2.8.1.2             |
| $A$                     |
| $\vee E(1, 2, 3, 4, 4)$ |

$\vdash$ :

|   |   |                   |              |
|---|---|-------------------|--------------|
| 1 | 1 | $\top$            | $A$          |
| 1 | 2 | $\perp \vee \top$ | $\vee I2(1)$ |

□

**Theorem 2.8.4.4.**

$\perp \vee \perp \dashv\vdash \perp$

*Beweis.*

$\vdash$ :

|   |   |                    |                         |
|---|---|--------------------|-------------------------|
| 1 | 1 | $\perp \vee \perp$ | $A$                     |
| 2 | 2 | $\perp$            | $A$                     |
| 1 | 3 | $\perp$            | $\vee E(1, 2, 2, 2, 2)$ |

$\dashv$ :

|   |   |                    |              |
|---|---|--------------------|--------------|
| 1 | 1 | $\perp$            | $A$          |
| 1 | 2 | $\perp \vee \perp$ | $\vee I1(1)$ |

□

## 8.5 Wahrheitswerte der Implikation

**Theorem 2.8.5.1.**

$\top \rightarrow \top \dashv\vdash \top$

*Beweis.*

$\vdash$ :

|   |   |                         |             |
|---|---|-------------------------|-------------|
| 1 | 1 | $\top \rightarrow \top$ | $A$         |
|   | 2 | $\top$                  | Th. 2.8.1.2 |

$\dashv$ :

|   |   |                         |             |
|---|---|-------------------------|-------------|
| 1 | 1 | $\top$                  | $A$         |
|   | 2 | $\top \rightarrow \top$ | Th. 2.1.1.2 |

□

**Theorem 2.8.5.2.**

$\top \rightarrow \perp \dashv\vdash \perp$

*Beweis.*

$\vdash$ :

1   1  $\top \rightarrow \perp$   
      2  $\top$   
 1   3  $\perp$

$A$   
 Th. 2.8.1.2  
 $\rightarrow E(1, 2)$

$\dashv$ :

1   1  $\perp$   
 1   2  $\top \rightarrow \perp$

$A$   
 Th. 2.2.7.11(1)

□

**Theorem 2.8.5.3.**

$$\perp \rightarrow \top \dashv\vdash \top$$

*Beweis.*

$\vdash$ :

1   1  $\perp \rightarrow \top$   
      2  $\top$

$A$   
 Th. 2.8.1.2

$\dashv$ :

1   1  $\top$   
 1   2  $\perp \rightarrow \top$

$A$   
 Th. 2.2.7.11(1)

□

**Theorem 2.8.5.4.**

$$\perp \rightarrow \perp \dashv\vdash \top$$

*Beweis.*

$\vdash$ :

1   1  $\perp \rightarrow \perp$   
      2  $\top$

$A$   
 Th. 2.8.1.2

$\dashv$ :

1   1  $\top$

$A$

$$2 \perp \rightarrow \perp$$

Th. 2.1.1.2

□

## 8.6 Wahrheitswerte des Bikonditionals

**Theorem 2.8.6.1.**

$$\top \leftrightarrow \top \dashv\vdash \top$$

*Beweis.*

$\vdash$ :

$$\begin{array}{ll} 1 & 1 \top \leftrightarrow \top \\ & 2 \top \end{array} \quad \begin{array}{l} A \\ \text{Th. 2.8.1.2} \end{array}$$

$\dashv$ :

$$\begin{array}{ll} 1 & 1 \top \\ & 2 \top \leftrightarrow \top \end{array} \quad \begin{array}{l} A \\ \text{Th. 2.1.1.3} \end{array}$$

□

**Theorem 2.8.6.2.**

$$\top \leftrightarrow \perp \dashv\vdash \perp$$

*Beweis.*

$\vdash$ :

$$\begin{array}{ll} 1 & 1 \top \leftrightarrow \perp \\ & 2 \top \\ 1 & 3 \perp \end{array} \quad \begin{array}{l} A \\ \text{Th. 2.8.1.2} \\ \text{Th. 2.2.4.1(1, 2)} \end{array}$$

$\dashv$ :

$$\begin{array}{ll} 1 & 1 \perp \\ 1 & 2 \top \rightarrow \perp \\ & 3 \top \\ & 4 \perp \rightarrow \top \\ 1 & 5 \top \leftrightarrow \perp \end{array} \quad \begin{array}{l} A \\ \text{Th. 2.2.7.11(1)} \\ \text{Th. 2.8.1.2} \\ \text{Th. 2.2.7.11(3)} \\ \leftrightarrow I(2, 4) \end{array}$$

□

**Theorem 2.8.6.3.**

$$\perp \leftrightarrow \top \dashv\vdash \perp$$

*Beweis.*

$\vdash$ :

$$\begin{array}{ll} 1 & 1 \perp \leftrightarrow \top \\ & 2 \top \\ 1 & 3 \perp \end{array}$$

$$\begin{array}{l} A \\ \text{Th. 2.8.1.2} \\ \text{Th. 2.2.4.2(1, 2)} \end{array}$$

$\dashv$ :

$$\begin{array}{ll} 1 & 1 \perp \\ & 2 \top \\ & 3 \perp \rightarrow \top \\ 1 & 4 \top \rightarrow \perp \\ 1 & 5 \perp \leftrightarrow \top \end{array}$$

$$\begin{array}{l} A \\ \text{Th. 2.8.1.2} \\ \text{Th. 2.2.7.11(2)} \\ \text{Th. 2.2.7.11(1)} \\ \leftrightarrow I(3, 4) \end{array}$$

□

**Theorem 2.8.6.4.**

$$\perp \leftrightarrow \perp \dashv\vdash \top$$

*Beweis.*

$\vdash$ :

$$\begin{array}{ll} 1 & 1 \perp \leftrightarrow \perp \\ & 2 \top \end{array}$$

$$\begin{array}{l} A \\ \text{Th. 2.8.1.2} \end{array}$$

$\dashv$ :

$$\begin{array}{ll} 1 & 1 \top \\ & 2 \perp \leftrightarrow \perp \end{array}$$

$$\begin{array}{l} A \\ \text{Th. 2.1.1.3} \end{array}$$

□

## 8.7 Wahrheitswerte des exklusiven Oder

**Theorem 2.8.7.1.**

$$\top \vee \top \dashv\vdash \perp$$

*Beweis.*

$\vdash$ :

|   |   |                  |                   |
|---|---|------------------|-------------------|
| 1 | 1 | $\top \vee \top$ | $A$               |
|   | 2 | $\top$           | Th. 2.8.1.2       |
| 1 | 3 | $\neg \top$      | Th. 2.3.3.1(1, 2) |
| 1 | 4 | $\perp$          | $\perp I(3, 2)$   |

$\dashv$ :

|   |   |                  |                   |
|---|---|------------------|-------------------|
| 1 | 1 | $\perp$          | $A$               |
|   | 2 | $\top$           | Th. 2.8.1.2       |
| 1 | 3 | $\neg \top$      | Th. 2.8.1.3(1)    |
| 1 | 4 | $\top \vee \top$ | Th. 2.1.7.3(2, 3) |

□

**Theorem 2.8.7.2.**

$$\top \vee \perp \dashv\vdash \top$$

*Beweis.*

$\vdash$ :

|   |   |                   |             |
|---|---|-------------------|-------------|
| 1 | 1 | $\top \vee \perp$ | $A$         |
|   | 2 | $\top$            | Th. 2.8.1.2 |

$\dashv$ :

|   |   |                   |                   |
|---|---|-------------------|-------------------|
| 1 | 1 | $\top$            | $A$               |
| 1 | 2 | $\neg \perp$      | Th. 2.8.1.1(1)    |
| 1 | 3 | $\top \vee \perp$ | Th. 2.3.2.1(1, 2) |

□

**Theorem 2.8.7.3.**

$$\perp \vee \top \dashv\vdash \top$$

*Beweis.*

$\vdash$ :

|   |   |                   |             |
|---|---|-------------------|-------------|
| 1 | 1 | $\perp \vee \top$ | $A$         |
|   | 2 | $\top$            | Th. 2.8.1.2 |

$\dashv$ :

|   |   |                   |                   |
|---|---|-------------------|-------------------|
| 1 | 1 | $\top$            | $A$               |
| 1 | 2 | $\neg \perp$      | Th. 2.8.1.1(1)    |
| 1 | 3 | $\perp \vee \top$ | Th. 2.3.2.2(2, 1) |

□

**Theorem 2.8.7.4.**

$$\perp \vee \perp \dashv \vdash \perp$$

*Beweis.*

$\vdash$ :

|   |   |                          |                 |
|---|---|--------------------------|-----------------|
| 1 | 1 | $\perp \vee \perp$       | $A$             |
|   | 2 | $\neg(\perp \vee \perp)$ | Th. 2.3.4.4     |
| 1 | 3 | $\perp$                  | $\perp I(2, 1)$ |

$\dashv$ :

|   |   |                    |                   |
|---|---|--------------------|-------------------|
| 1 | 1 | $\perp$            | $A$               |
|   | 2 | $\top$             | Th. 2.8.1.2       |
| 1 | 3 | $\neg \top$        | Th. 2.8.1.3(1)    |
| 1 | 4 | $\perp \vee \perp$ | Th. 2.1.7.3(2, 3) |

□

## 8.8 Wahrheitswerte von NAND

**Theorem 2.8.8.1.**

$$\top \uparrow \top \dashv \vdash \perp$$

*Beweis.*

$\vdash$ :

$$\begin{array}{l}
1 \quad 1 \quad \top \uparrow \top \\
1 \quad 2 \quad \neg(\top \wedge \top) \\
\quad 3 \quad \top \\
\quad 4 \quad \top \wedge \top \\
1 \quad 5 \quad \perp
\end{array}$$

$A$   
 Th. 2.4.1.1(1)  
 Th. 2.8.1.2  
 $\wedge I^*(\text{Wdh}_2(3))$   
 $\perp I(2, 4)$

$\dashv$ :

$$\begin{array}{l}
1 \quad 1 \quad \perp \\
\quad 2 \quad \top \\
1 \quad 3 \quad \neg \top \\
1 \quad 4 \quad \top \uparrow \top
\end{array}$$

$A$   
 Th. 2.8.1.2  
 Th. 2.8.1.3(1)  
 Th. 2.1.7.3(2, 3)

□

### Theorem 2.8.8.2.

$$\top \uparrow \perp \dashv \vdash \top$$

*Beweis.*

$\vdash$ :

$$\begin{array}{l}
1 \quad 1 \quad \top \uparrow \perp \\
\quad 2 \quad \top
\end{array}$$

$A$   
 Th. 2.8.1.2

$\dashv$ :

$$\begin{array}{l}
1 \quad 1 \quad \top \\
1 \quad 2 \quad \neg \perp \\
1 \quad 3 \quad \top \uparrow \perp
\end{array}$$

$A$   
 Th. 2.8.1.1(1)  
 Th. 2.4.2.2(2)

□

### Theorem 2.8.8.3.

$$\perp \uparrow \top \dashv \vdash \top$$

*Beweis.*

$\vdash$ :

$$\begin{array}{l}
1 \quad 1 \quad \perp \uparrow \top \\
\quad 2 \quad \top
\end{array}$$

$A$   
 Th. 2.8.1.2

$\dashv$ :

$$\begin{array}{l} 1 \quad 1 \quad \top \\ 1 \quad 2 \quad \neg \perp \\ 1 \quad 3 \quad \perp \uparrow \top \end{array}$$

$$\begin{array}{l} A \\ \text{Th. 2.8.1.1(1)} \\ \text{Th. 2.4.2.1(2)} \end{array}$$

□

#### Theorem 2.8.8.4.

$$\perp \uparrow \perp \dashv\vdash \top$$

*Beweis.*

$\vdash$ :

$$\begin{array}{l} 1 \quad 1 \quad \perp \uparrow \perp \\ \quad 2 \quad \top \end{array}$$

$$\begin{array}{l} A \\ \text{Th. 2.8.1.2} \end{array}$$

$\dashv$ :

$$\begin{array}{l} 1 \quad 1 \quad \top \\ 1 \quad 2 \quad \neg \perp \\ 1 \quad 3 \quad \perp \uparrow \perp \end{array}$$

$$\begin{array}{l} A \\ \text{Th. 2.8.1.1(1)} \\ \text{Th. 2.4.2.1(2)} \end{array}$$

□

## 8.9 Wahrheitswerte von NOR

#### Theorem 2.8.9.1.

$$\top \downarrow \top \dashv\vdash \perp$$

*Beweis.*

$\vdash$ :

$$\begin{array}{l} 1 \quad 1 \quad \top \downarrow \top \\ 1 \quad 2 \quad \neg \top \\ \quad 3 \quad \top \\ 1 \quad 4 \quad \perp \end{array}$$

$$\begin{array}{l} A \\ \text{Th. 2.5.3.1(1)} \\ \text{Th. 2.8.1.2} \\ \perp I(2,3) \end{array}$$

$\dashv$ :

$$\begin{array}{l} 1 \quad 1 \quad \perp \\ \quad 2 \quad \top \\ 1 \quad 3 \quad \neg \top \\ 1 \quad 4 \quad \top \downarrow \top \end{array}$$

$$\begin{array}{l} A \\ \text{Th. 2.8.1.2} \\ \text{Th. 2.8.1.3(1)} \\ \text{Th. 2.1.7.3(2,3)} \end{array}$$

□

**Theorem 2.8.9.2.**

$$\top \downarrow \perp \dashv\vdash \perp$$

*Beweis.*

$\vdash$ :

$$\begin{array}{ll} 1 & 1 \top \downarrow \perp \\ 1 & 2 \neg \top \\ & 3 \top \\ 1 & 4 \perp \end{array}$$

$$\begin{array}{l} A \\ \text{Th. 2.5.3.1(1)} \\ \text{Th. 2.8.1.2} \\ \perp I(2,3) \end{array}$$

$\dashv$ :

$$\begin{array}{ll} 1 & 1 \perp \\ & 2 \top \\ 1 & 3 \neg \top \\ 1 & 4 \top \downarrow \perp \end{array}$$

$$\begin{array}{l} A \\ \text{Th. 2.8.1.2} \\ \text{Th. 2.8.1.3(1)} \\ \text{Th. 2.1.7.3(2,3)} \end{array}$$

□

**Theorem 2.8.9.3.**

$$\perp \downarrow \top \dashv\vdash \perp$$

*Beweis.*

$\vdash$ :

$$\begin{array}{ll} 1 & 1 \perp \downarrow \top \\ 1 & 2 \neg \top \\ & 3 \top \\ 1 & 4 \perp \end{array}$$

$$\begin{array}{l} A \\ \text{Th. 2.5.3.2(1)} \\ \text{Th. 2.8.1.2} \\ \perp I(2,3) \end{array}$$

$\dashv$ :

$$\begin{array}{ll} 1 & 1 \perp \\ & 2 \top \\ 1 & 3 \neg \top \\ 1 & 4 \perp \downarrow \top \end{array}$$

$$\begin{array}{l} A \\ \text{Th. 2.8.1.2} \\ \text{Th. 2.8.1.3(1)} \\ \text{Th. 2.1.7.3(2,3)} \end{array}$$

□

**Theorem 2.8.9.4.**

$$\perp \downarrow \perp \dashv\vdash \top$$

*Beweis.*

$\vdash$ :

|   |   |                          |             |
|---|---|--------------------------|-------------|
| 1 | 1 | $\perp \downarrow \perp$ | $A$         |
|   | 2 | $\top$                   | Th. 2.8.1.2 |

$\dashv$ :

|   |   |                          |                   |
|---|---|--------------------------|-------------------|
| 1 | 1 | $\top$                   | $A$               |
| 1 | 2 | $\neg \perp$             | Th. 2.8.1.1(1)    |
| 1 | 3 | $\perp \downarrow \perp$ | Th. 2.5.2.1(2, 2) |

□

## 8.10 Wahrheitstafel des Volladdierers

**Theorem 2.8.10.1** (Volladdierer bei 000).

$$\neg P, \neg Q, \neg R \vdash \neg S_{\text{FA}}(P, Q, R) \wedge \neg C_{\text{FA}}(P, Q, R)$$

*Beweis.*

Summenbit:

|       |   |                                                                        |                                            |
|-------|---|------------------------------------------------------------------------|--------------------------------------------|
| 1     | 1 | $\neg P$                                                               | $A$                                        |
| 2     | 2 | $\neg Q$                                                               | $A$                                        |
| 3     | 3 | $\neg R$                                                               | $A$                                        |
| 1,2   | 4 | $\neg S_{1/2}(P, Q) \wedge \neg C_{1/2}(P, Q)$                         | Th. 2.6.5.1(1, 2)                          |
| 1,2   | 5 | $\neg S_{1/2}(P, Q)$                                                   | $\wedge E1(4)$                             |
| 1,2   | 6 | $\neg C_{1/2}(P, Q)$                                                   | $\wedge E2(4)$                             |
| 1,2,3 | 7 | $\neg S_{1/2}(S_{1/2}(P, Q), R) \wedge \neg C_{1/2}(S_{1/2}(P, Q), R)$ | Th. 2.6.5.1(5, 3)                          |
| 1,2,3 | 8 | $\neg S_{1/2}(S_{1/2}(P, Q), R)$                                       | $\wedge E1(7)$                             |
| 1,2,3 | 9 | $\neg S_{\text{FA}}(P, Q, R)$                                          | $\leftrightarrow S(\text{Th. 2.7.1.1, 8})$ |

Übertragsbit:

|       |    |                                                            |                                             |
|-------|----|------------------------------------------------------------|---------------------------------------------|
| 1,2,3 | 10 | $\neg C_{1/2}(S_{1/2}(P, Q), R)$                           | $\wedge E2(7)$                              |
| 1,2,3 | 11 | $\neg C_{1/2}(P, Q) \wedge \neg C_{1/2}(S_{1/2}(P, Q), R)$ | $\wedge I(6, 10)$                           |
| 1,2,3 | 12 | $\neg(C_{1/2}(P, Q) \vee C_{1/2}(S_{1/2}(P, Q), R))$       | Th. 2.1.8.1(11)                             |
| 1,2,3 | 13 | $\neg C_{\text{FA}}(P, Q, R)$                              | $\leftrightarrow S(\text{Th. 2.7.1.2, 12})$ |

Zusammenführung:

$$1,2,3 \quad 14 \quad \neg S_{\text{FA}}(P, Q, R) \wedge \neg C_{\text{FA}}(P, Q, R) \quad \wedge I(9, 13)$$

□

**Theorem 2.8.10.2 (Volladdierer bei 100).**

$$P, \neg Q, \neg R \vdash S_{\text{FA}}(P, Q, R) \wedge \neg C_{\text{FA}}(P, Q, R)$$

*Beweis.*

Summenbit:

|       |   |                                                                   |                   |
|-------|---|-------------------------------------------------------------------|-------------------|
| 1     | 1 | $P$                                                               | $A$               |
| 2     | 2 | $\neg Q$                                                          | $A$               |
| 3     | 3 | $\neg R$                                                          | $A$               |
| 1,2   | 4 | $S_{1/2}(P, Q) \wedge \neg C_{1/2}(P, Q)$                         | Th. 2.6.5.2(1, 2) |
| 1,2   | 5 | $S_{1/2}(P, Q)$                                                   | $\wedge E1(4)$    |
| 1,2   | 6 | $\neg C_{1/2}(P, Q)$                                              | $\wedge E2(4)$    |
| 1,2,3 | 7 | $S_{1/2}(S_{1/2}(P, Q), R) \wedge \neg C_{1/2}(S_{1/2}(P, Q), R)$ | Th. 2.6.5.2(5, 3) |
| 1,2,3 | 8 | $S_{1/2}(S_{1/2}(P, Q), R)$                                       | $\wedge E1(7)$    |
| 1,2,3 | 9 | $S_{\text{FA}}(P, Q, R)$                                          | Th. 2.7.1.1(8)    |

Übertragsbit:

|       |    |                                                            |                                             |
|-------|----|------------------------------------------------------------|---------------------------------------------|
| 1,2,3 | 10 | $\neg C_{1/2}(S_{1/2}(P, Q), R)$                           | $\wedge E2(7)$                              |
| 1,2,3 | 11 | $\neg C_{1/2}(P, Q) \wedge \neg C_{1/2}(S_{1/2}(P, Q), R)$ | $\wedge I(6, 10)$                           |
| 1,2,3 | 12 | $\neg(C_{1/2}(P, Q) \vee C_{1/2}(S_{1/2}(P, Q), R))$       | Th. 2.1.8.1(11)                             |
| 1,2,3 | 13 | $\neg C_{\text{FA}}(P, Q, R)$                              | $\leftrightarrow S(\text{Th. 2.7.1.2, 12})$ |

Zusammenführung:

$$1,2,3 \quad 14 \quad S_{\text{FA}}(P, Q, R) \wedge \neg C_{\text{FA}}(P, Q, R) \quad \wedge I(9, 13)$$

□

**Theorem 2.8.10.3 (Volladdierer bei 010).**

$$\neg P, Q, \neg R \vdash S_{\text{FA}}(P, Q, R) \wedge \neg C_{\text{FA}}(P, Q, R)$$

*Beweis.*

Summenbit:

|   |   |          |     |
|---|---|----------|-----|
| 1 | 1 | $\neg P$ | $A$ |
| 2 | 2 | $Q$      | $A$ |
| 3 | 3 | $\neg R$ | $A$ |

|       |   |                                                                   |                   |
|-------|---|-------------------------------------------------------------------|-------------------|
| 1,2   | 4 | $S_{1/2}(P, Q) \wedge \neg C_{1/2}(P, Q)$                         | Th. 2.6.5.3(1, 2) |
| 1,2   | 5 | $S_{1/2}(P, Q)$                                                   | $\wedge E1(4)$    |
| 1,2   | 6 | $\neg C_{1/2}(P, Q)$                                              | $\wedge E2(4)$    |
| 1,2,3 | 7 | $S_{1/2}(S_{1/2}(P, Q), R) \wedge \neg C_{1/2}(S_{1/2}(P, Q), R)$ | Th. 2.6.5.2(5, 3) |
| 1,2,3 | 8 | $S_{1/2}(S_{1/2}(P, Q), R)$                                       | $\wedge E1(7)$    |
| 1,2,3 | 9 | $S_{FA}(P, Q, R)$                                                 | Th. 2.7.1.1(8)    |

Übertragsbit:

|       |    |                                                            |                                             |
|-------|----|------------------------------------------------------------|---------------------------------------------|
| 1,2,3 | 10 | $\neg C_{1/2}(S_{1/2}(P, Q), R)$                           | $\wedge E2(7)$                              |
| 1,2,3 | 11 | $\neg C_{1/2}(P, Q) \wedge \neg C_{1/2}(S_{1/2}(P, Q), R)$ | $\wedge I(6, 10)$                           |
| 1,2,3 | 12 | $\neg(C_{1/2}(P, Q) \vee C_{1/2}(S_{1/2}(P, Q), R))$       | Th. 2.1.8.1(11)                             |
| 1,2,3 | 13 | $\neg C_{FA}(P, Q, R)$                                     | $\leftrightarrow S(\text{Th. 2.7.1.2, 12})$ |

Zusammenführung:

|       |    |                                               |                   |
|-------|----|-----------------------------------------------|-------------------|
| 1,2,3 | 14 | $S_{FA}(P, Q, R) \wedge \neg C_{FA}(P, Q, R)$ | $\wedge I(9, 13)$ |
|-------|----|-----------------------------------------------|-------------------|

□

**Theorem 2.8.10.4 (Volladdierer bei 001).**

$$\neg P, \neg Q, R \vdash S_{FA}(P, Q, R) \wedge \neg C_{FA}(P, Q, R)$$

*Beweis.*

Summenbit:

|       |   |                                                                   |                   |
|-------|---|-------------------------------------------------------------------|-------------------|
| 1     | 1 | $\neg P$                                                          | $A$               |
| 2     | 2 | $\neg Q$                                                          | $A$               |
| 3     | 3 | $R$                                                               | $A$               |
| 1,2   | 4 | $\neg S_{1/2}(P, Q) \wedge \neg C_{1/2}(P, Q)$                    | Th. 2.6.5.1(1, 2) |
| 1,2   | 5 | $\neg S_{1/2}(P, Q)$                                              | $\wedge E1(4)$    |
| 1,2   | 6 | $\neg C_{1/2}(P, Q)$                                              | $\wedge E2(4)$    |
| 1,2,3 | 7 | $S_{1/2}(S_{1/2}(P, Q), R) \wedge \neg C_{1/2}(S_{1/2}(P, Q), R)$ | Th. 2.6.5.3(5, 3) |
| 1,2,3 | 8 | $S_{1/2}(S_{1/2}(P, Q), R)$                                       | $\wedge E1(7)$    |
| 1,2,3 | 9 | $S_{FA}(P, Q, R)$                                                 | Th. 2.7.1.1(8)    |

Übertragsbit:

|       |    |                                                            |                                             |
|-------|----|------------------------------------------------------------|---------------------------------------------|
| 1,2,3 | 10 | $\neg C_{1/2}(S_{1/2}(P, Q), R)$                           | $\wedge E2(7)$                              |
| 1,2,3 | 11 | $\neg C_{1/2}(P, Q) \wedge \neg C_{1/2}(S_{1/2}(P, Q), R)$ | $\wedge I(6, 10)$                           |
| 1,2,3 | 12 | $\neg(C_{1/2}(P, Q) \vee C_{1/2}(S_{1/2}(P, Q), R))$       | Th. 2.1.8.1(11)                             |
| 1,2,3 | 13 | $\neg C_{FA}(P, Q, R)$                                     | $\leftrightarrow S(\text{Th. 2.7.1.2, 12})$ |

Zusammenführung:

$$1,2,3 \quad 14 \quad S_{FA}(P, Q, R) \wedge \neg C_{FA}(P, Q, R) \quad \wedge I(9, 13)$$

□

**Theorem 2.8.10.5 (Volladdierer bei 110).**

$$P, Q, \neg R \vdash \neg S_{FA}(P, Q, R) \wedge C_{FA}(P, Q, R)$$

*Beweis.*

Summenbit:

|       |   |                                                                        |                                            |
|-------|---|------------------------------------------------------------------------|--------------------------------------------|
| 1     | 1 | P                                                                      | A                                          |
| 2     | 2 | Q                                                                      | A                                          |
| 3     | 3 | $\neg R$                                                               | A                                          |
| 1,2   | 4 | $\neg S_{1/2}(P, Q) \wedge C_{1/2}(P, Q)$                              | Th. 2.6.5.4(1, 2)                          |
| 1,2   | 5 | $\neg S_{1/2}(P, Q)$                                                   | $\wedge E1(4)$                             |
| 1,2   | 6 | $C_{1/2}(P, Q)$                                                        | $\wedge E2(4)$                             |
| 1,2,3 | 7 | $\neg S_{1/2}(S_{1/2}(P, Q), R) \wedge \neg C_{1/2}(S_{1/2}(P, Q), R)$ | Th. 2.6.5.1(5, 3)                          |
| 1,2,3 | 8 | $\neg S_{1/2}(S_{1/2}(P, Q), R)$                                       | $\wedge E1(7)$                             |
| 1,2,3 | 9 | $\neg S_{FA}(P, Q, R)$                                                 | $\leftrightarrow S(\text{Th. 2.7.1.1, 8})$ |

Übertragsbit:

|     |    |                                                                              |                 |
|-----|----|------------------------------------------------------------------------------|-----------------|
| 1,2 | 10 | C <sub>1/2</sub> (P, Q) $\vee$ C <sub>1/2</sub> (S <sub>1/2</sub> (P, Q), R) | $\vee I1(6)$    |
| 1,2 | 11 | C <sub>FA</sub> (P, Q, R)                                                    | Th. 2.7.1.2(10) |

Zusammenführung:

$$1,2,3 \quad 12 \quad \neg S_{FA}(P, Q, R) \wedge C_{FA}(P, Q, R) \quad \wedge I(9, 11)$$

□

**Theorem 2.8.10.6 (Volladdierer bei 101).**

$$P, \neg Q, R \vdash \neg S_{FA}(P, Q, R) \wedge C_{FA}(P, Q, R)$$

*Beweis.*

Summenbit:

|     |   |                                           |                   |
|-----|---|-------------------------------------------|-------------------|
| 1   | 1 | P                                         | A                 |
| 2   | 2 | $\neg Q$                                  | A                 |
| 3   | 3 | R                                         | A                 |
| 1,2 | 4 | $S_{1/2}(P, Q) \wedge \neg C_{1/2}(P, Q)$ | Th. 2.6.5.2(1, 2) |

|       |   |                                                                   |                                            |
|-------|---|-------------------------------------------------------------------|--------------------------------------------|
| 1,2   | 5 | $S_{1/2}(P, Q)$                                                   | $\wedge E1(4)$                             |
| 1,2   | 6 | $\neg C_{1/2}(P, Q)$                                              | $\wedge E2(4)$                             |
| 1,2,3 | 7 | $\neg S_{1/2}(S_{1/2}(P, Q), R) \wedge C_{1/2}(S_{1/2}(P, Q), R)$ | Th. 2.6.5.4(5, 3)                          |
| 1,2,3 | 8 | $\neg S_{1/2}(S_{1/2}(P, Q), R)$                                  | $\wedge E1(7)$                             |
| 1,2,3 | 9 | $\neg S_{FA}(P, Q, R)$                                            | $\leftrightarrow S(\text{Th. 2.7.1.1, 8})$ |

Übertragsbit:

|       |    |                                                |                 |
|-------|----|------------------------------------------------|-----------------|
| 1,2,3 | 10 | $C_{1/2}(S_{1/2}(P, Q), R)$                    | $\wedge E2(7)$  |
| 1,2,3 | 11 | $C_{1/2}(P, Q) \vee C_{1/2}(S_{1/2}(P, Q), R)$ | $\vee I2(10)$   |
| 1,2,3 | 12 | $C_{FA}(P, Q, R)$                              | Th. 2.7.1.2(11) |

Zusammenführung:

|       |    |                                               |                   |
|-------|----|-----------------------------------------------|-------------------|
| 1,2,3 | 13 | $\neg S_{FA}(P, Q, R) \wedge C_{FA}(P, Q, R)$ | $\wedge I(9, 12)$ |
|-------|----|-----------------------------------------------|-------------------|

□

**Theorem 2.8.10.7 (Volladdierer bei 011).**

$$\neg P, Q, R \vdash \neg S_{FA}(P, Q, R) \wedge C_{FA}(P, Q, R)$$

*Beweis.*

Summenbit:

|       |   |                                                                   |                                            |
|-------|---|-------------------------------------------------------------------|--------------------------------------------|
| 1     | 1 | $\neg P$                                                          | $A$                                        |
| 2     | 2 | $Q$                                                               | $A$                                        |
| 3     | 3 | $R$                                                               | $A$                                        |
| 1,2   | 4 | $S_{1/2}(P, Q) \wedge \neg C_{1/2}(P, Q)$                         | Th. 2.6.5.3(1, 2)                          |
| 1,2   | 5 | $S_{1/2}(P, Q)$                                                   | $\wedge E1(4)$                             |
| 1,2   | 6 | $\neg C_{1/2}(P, Q)$                                              | $\wedge E2(4)$                             |
| 1,2,3 | 7 | $\neg S_{1/2}(S_{1/2}(P, Q), R) \wedge C_{1/2}(S_{1/2}(P, Q), R)$ | Th. 2.6.5.4(5, 3)                          |
| 1,2,3 | 8 | $\neg S_{1/2}(S_{1/2}(P, Q), R)$                                  | $\wedge E1(7)$                             |
| 1,2,3 | 9 | $\neg S_{FA}(P, Q, R)$                                            | $\leftrightarrow S(\text{Th. 2.7.1.1, 8})$ |

Übertragsbit:

|       |    |                                                |                 |
|-------|----|------------------------------------------------|-----------------|
| 1,2,3 | 10 | $C_{1/2}(S_{1/2}(P, Q), R)$                    | $\wedge E2(7)$  |
| 1,2,3 | 11 | $C_{1/2}(P, Q) \vee C_{1/2}(S_{1/2}(P, Q), R)$ | $\vee I2(10)$   |
| 1,2,3 | 12 | $C_{FA}(P, Q, R)$                              | Th. 2.7.1.2(11) |

Zusammenführung:

$$1,2,3 \quad 13 \quad \neg S_{\text{FA}}(P, Q, R) \wedge C_{\text{FA}}(P, Q, R) \quad \wedge I(9, 12)$$

□

**Theorem 2.8.10.8 (Volladdierer bei 111).**

$$P, Q, R \vdash S_{\text{FA}}(P, Q, R) \wedge C_{\text{FA}}(P, Q, R)$$

*Beweis.*

Summenbit:

|       |   |                                                                   |                   |
|-------|---|-------------------------------------------------------------------|-------------------|
| 1     | 1 | $P$                                                               | $A$               |
| 2     | 2 | $Q$                                                               | $A$               |
| 3     | 3 | $R$                                                               | $A$               |
| 1,2   | 4 | $\neg S_{1/2}(P, Q) \wedge C_{1/2}(P, Q)$                         | Th. 2.6.5.4(1, 2) |
| 1,2   | 5 | $\neg S_{1/2}(P, Q)$                                              | $\wedge E1(4)$    |
| 1,2   | 6 | $C_{1/2}(P, Q)$                                                   | $\wedge E2(4)$    |
| 1,2,3 | 7 | $S_{1/2}(S_{1/2}(P, Q), R) \wedge \neg C_{1/2}(S_{1/2}(P, Q), R)$ | Th. 2.6.5.3(5, 3) |
| 1,2,3 | 8 | $S_{1/2}(S_{1/2}(P, Q), R)$                                       | $\wedge E1(7)$    |
| 1,2,3 | 9 | $S_{\text{FA}}(P, Q, R)$                                          | Th. 2.7.1.1(8)    |

Übertragsbit:

|     |    |                                                |                 |
|-----|----|------------------------------------------------|-----------------|
| 1,2 | 10 | $C_{1/2}(P, Q) \vee C_{1/2}(S_{1/2}(P, Q), R)$ | $\vee I1(6)$    |
| 1,2 | 11 | $C_{\text{FA}}(P, Q, R)$                       | Th. 2.7.1.2(10) |

Zusammenführung:

$$1,2,3 \quad 12 \quad S_{\text{FA}}(P, Q, R) \wedge C_{\text{FA}}(P, Q, R) \quad \wedge I(9, 11)$$

□

## Kapitel 9

# Identität und Ungleichheit

### 9.1 Grundgesetze der Identität

Im Kapitel verwendete Symbole

Konstantensymbole:  $a, b, c$

**Theorem 2.9.1.1 (Symmetrie der Identität).**

$$a = b \vdash b = a$$

|   |   |         |             |
|---|---|---------|-------------|
| 1 | 1 | $a = b$ | $A$         |
|   | 2 | $a = a$ | $= I$       |
| 1 | 3 | $b = a$ | $= E(1, 2)$ |

*Bemerkung.* Aus Th. 2.9.1.1 wissen wir, dass jede hergeleitete Gleichung auch in umgekehrter Richtung gilt. Um die Beweise übersichtlicher zu halten, vereinbaren wir daher: *Wenn wir einen Satz der Form  $\Gamma \vdash s = t$  zitieren, dürfen wir bei Bedarf auch die Form  $\Gamma \vdash t = s$  verwenden, ohne den Zwischenschritt über die Symmetrie explizit aufzuschreiben*

**Theorem 2.9.1.2 (Transitivität der Identität).**

$$a = b, b = c \vdash a = c$$

|     |   |         |             |
|-----|---|---------|-------------|
| 1   | 1 | $a = b$ | $A$         |
|     | 2 | $b = c$ | $A$         |
| 1,2 | 3 | $a = c$ | $= E(1, 2)$ |

**Theorem 2.9.1.3.**

$$a = b, c = b \vdash a = c$$

|     |   |         |             |
|-----|---|---------|-------------|
| 1   | 1 | $a = b$ | $A$         |
| 2   | 2 | $c = b$ | $A$         |
| 1,2 | 3 | $a = c$ | $= E(1, 2)$ |

**Theorem 2.9.1.4.**

$$a = b, a = c \vdash b = c$$

|     |   |         |                   |
|-----|---|---------|-------------------|
| 1   | 1 | $a = b$ | $A$               |
| 2   | 2 | $a = c$ | $A$               |
| 1   | 3 | $b = a$ | Th. 2.9.1.1(1)    |
| 1,2 | 4 | $b = c$ | Th. 2.9.1.2(3, 2) |

**Theorem 2.9.1.5.**

$$a = b, c = a, d = b \vdash c = d$$

|       |   |         |                   |
|-------|---|---------|-------------------|
| 1     | 1 | $a = b$ | $A$               |
| 2     | 2 | $c = a$ | $A$               |
| 3     | 3 | $d = b$ | $A$               |
| 2     | 4 | $a = c$ | Th. 2.9.1.1(2)    |
| 1,2   | 5 | $b = c$ | Th. 2.9.1.4(1, 4) |
| 1,2,3 | 6 | $d = c$ | Th. 2.9.1.2(3, 5) |
| 1,2,3 | 7 | $c = d$ | Th. 2.9.1.1(6)    |

**Theorem 2.9.1.6.**

$$a = b, a = c, b = d \vdash c = d$$

|       |   |         |                      |
|-------|---|---------|----------------------|
| 1     | 1 | $a = b$ | $A$                  |
| 2     | 2 | $a = c$ | $A$                  |
| 3     | 3 | $b = d$ | $A$                  |
| 2     | 4 | $c = a$ | Th. 2.9.1.1(2)       |
| 3     | 5 | $d = b$ | Th. 2.9.1.1(3)       |
| 1,2,3 | 6 | $c = d$ | Th. 2.9.1.5(1, 4, 5) |

**Theorem 2.9.1.7.**

**Mengen:**  $a, b, c, d$   
**Zweistellige Prädikate:**  $P$

$$a = b, c = d, P(b, d) \vdash P(a, c)$$

|   |       |           |                |
|---|-------|-----------|----------------|
| 1 | 1     | $a = b$   | $A$            |
|   | 2     | $c = d$   | $A$            |
|   | 3     | $P(b, d)$ | $A$            |
|   | 1     | $b = a$   | Th. 2.9.1.1(1) |
|   | 2     | $d = c$   | Th. 2.9.1.1(2) |
|   | 1,3   | $P(a, d)$ | $= E(4, 3)$    |
|   | 1,2,3 | $P(a, c)$ | $= E(5, 6)$    |

**Theorem 2.9.1.8 (Kongruenz im rechten Argument eines zweistelligen Funktionsterms).**

*Konstantensymbole:*  $a, b, c$   
*Zweistellige Funktionensymbole:*  $f$

$$a = b \vdash f(c, a) = f(c, b)$$

|   |   |                     |             |
|---|---|---------------------|-------------|
| 1 | 1 | $a = b$             | $A$         |
|   | 2 | $f(c, a) = f(c, a)$ | $= I$       |
|   | 1 | $f(c, a) = f(c, b)$ | $= E(1, 2)$ |

**Theorem 2.9.1.9 (Kongruenz im linken Argument eines zweistelligen Funktionsterms).**

*Konstantensymbole:*  $a, b, c$   
*Zweistellige Funktionensymbole:*  $f$

$$a = b \vdash f(a, c) = f(b, c)$$

|   |   |                     |             |
|---|---|---------------------|-------------|
| 1 | 1 | $a = b$             | $A$         |
|   | 2 | $f(a, c) = f(a, c)$ | $= I$       |
|   | 1 | $f(a, c) = f(b, c)$ | $= E(1, 2)$ |

**Theorem 2.9.1.10.**

$$a = c, b = c, d = a \vee d = b \vdash d = c$$

|  |     |                    |                   |
|--|-----|--------------------|-------------------|
|  | 1   | $a = c$            | $A$               |
|  | 2   | $b = c$            | $A$               |
|  | 3   | $d = a \vee d = b$ | $A$               |
|  | 4   | $d = a$            | $A$               |
|  | 1,4 | $d = c$            | Th. 2.9.1.2(4, 1) |
|  | 6   | $d = b$            | $A$               |

2,6 7  $d = c$   
 1,2,3 8  $d = c$

Th. 2.9.1.2(6, 2)  
 $\vee E(3, 4, 5, 6, 7)$

## 9.2 Identität und Prädikate

**Im Kapitel verwendete Symbole**

**Prädikatensymbole:**  $F, P$

**Variablen:**  $x$

**Konstantensymbole:**  $a, b, c$

**Theorem 2.9.2.1.**

$$Fa \dashv\vdash \exists x(x = a \wedge Fx)$$

|   |   |                              |                      |
|---|---|------------------------------|----------------------|
| 1 | 1 | $Fa$                         | $A$                  |
|   | 2 | $a = a$                      | $= I$                |
| 1 | 3 | $a = a \wedge Fa$            | $\wedge I(2, 1)$     |
| 1 | 4 | $\exists x(x = a \wedge Fx)$ | $\exists I(3)$       |
|   |   |                              |                      |
| 1 | 1 | $\exists x(x = a \wedge Fx)$ | $A$                  |
| 2 | 2 | $b = a \wedge Fb$            | $A$                  |
| 2 | 3 | $b = a$                      | $\wedge E1(2)$       |
| 2 | 4 | $Fb$                         | $\wedge E2(2)$       |
| 2 | 5 | $Fa$                         | $= E(3, 4)$          |
| 1 | 6 | $Fa$                         | $\exists E(1, 2, 5)$ |

**Theorem 2.9.2.2.**

$$\exists c((c = a \vee c = b) \wedge P(c)) \dashv\vdash P(a) \vee P(b)$$

*Beweis.*

$\vdash$ :

|     |   |                                             |                |
|-----|---|---------------------------------------------|----------------|
| 1   | 1 | $\exists c((c = a \vee c = b) \wedge P(c))$ | $A$            |
| 2   | 2 | $(c = a \vee c = b) \wedge P(c)$            | $A$            |
| 2   | 3 | $c = a \vee c = b$                          | $\wedge E1(2)$ |
| 2   | 4 | $P(c)$                                      | $\wedge E2(2)$ |
| 5   | 5 | $c = a$                                     | $A$            |
| 2,5 | 6 | $P(a)$                                      | $= E(5, 4)$    |
| 2,5 | 7 | $P(a) \vee P(b)$                            | $\vee I1(6)$   |
| 8   | 8 | $c = b$                                     | $A$            |
| 2,8 | 9 | $P(b)$                                      | $= E(8, 4)$    |

|     |    |                  |                          |
|-----|----|------------------|--------------------------|
| 2,8 | 10 | $P(a) \vee P(b)$ | $\vee I 2(9)$            |
| 2   | 11 | $P(a) \vee P(b)$ | $\vee E(3, 5, 7, 8, 10)$ |
| 1   | 12 | $P(a) \vee P(b)$ | $\exists E(1, 2, 11)$    |

$\neg$ :

|   |    |                                             |                          |
|---|----|---------------------------------------------|--------------------------|
| 1 | 1  | $P(a) \vee P(b)$                            | $A$                      |
| 2 | 2  | $P(a)$                                      | $A$                      |
|   | 3  | $a = a$                                     | $= I$                    |
|   | 4  | $a = a \vee a = b$                          | $\vee I 1(3)$            |
| 2 | 5  | $(a = a \vee a = b) \wedge P(a)$            | $\wedge I(4, 2)$         |
| 2 | 6  | $\exists c((c = a \vee c = b) \wedge P(c))$ | $\exists I(5)$           |
| 7 | 7  | $P(b)$                                      | $A$                      |
|   | 8  | $b = b$                                     | $= I$                    |
|   | 9  | $b = a \vee b = b$                          | $\vee I 2(8)$            |
| 7 | 10 | $(b = a \vee b = b) \wedge P(b)$            | $\wedge I(9, 7)$         |
| 7 | 11 | $\exists c((c = a \vee c = b) \wedge P(c))$ | $\exists I(10)$          |
| 1 | 12 | $\exists c((c = a \vee c = b) \wedge P(c))$ | $\vee E(1, 2, 6, 7, 11)$ |

□

### 9.3 Theoreme mit dem Nicht-Gleichheitszeichen

Im Kapitel verwendete Symbole

Konstantensymbole:  $a, b$

Theorem 2.9.3.1 (Symmetrie der Ungleichheit).

$$a \neq b \vdash b \neq a$$

|     |   |            |                 |
|-----|---|------------|-----------------|
| 1   | 1 | $a \neq b$ | $A$             |
| 2   | 2 | $b = a$    | $A$             |
| 2   | 3 | $a = b$    | Th. 2.9.1.1(2)  |
| 1,2 | 4 | $\perp$    | $\perp I(1, 3)$ |
| 1   | 5 | $b \neq a$ | $\neg I(2, 4)$  |

# Kapitel 10

## Quantorenlogik

### 10.1 Strukturgesetze für $\rightarrow$ und $\leftrightarrow$ unter $\forall$

Im Kapitel verwendete Symbole

Prädikatensymbole:  $P, Q, R$

Variablen:  $x$

**Theorem 2.10.1.1.**

$$\forall x(P(x) \leftrightarrow Q(x)) \vdash \forall x(Q(x) \leftrightarrow P(x))$$

|   |   |                                        |                |
|---|---|----------------------------------------|----------------|
| 1 | 1 | $\forall x(P(x) \leftrightarrow Q(x))$ | $A$            |
| 1 | 2 | $P(a) \leftrightarrow Q(a)$            | $\forall E(1)$ |
| 1 | 3 | $Q(a) \leftrightarrow P(a)$            | Th. 2.1.2.5(2) |
| 1 | 4 | $\forall x(Q(x) \leftrightarrow P(x))$ | $\forall I(3)$ |

**Theorem 2.10.1.2 (Implikationstransitivität).**

$$\forall x(P(x) \rightarrow Q(x)), \forall x(Q(x) \rightarrow R(x)) \vdash \forall x(P(x) \rightarrow R(x))$$

|     |   |                                    |                   |
|-----|---|------------------------------------|-------------------|
| 1   | 1 | $\forall x(P(x) \rightarrow Q(x))$ | $A$               |
| 2   | 2 | $\forall x(Q(x) \rightarrow R(x))$ | $A$               |
| 1   | 3 | $P(a) \rightarrow Q(a)$            | $\forall E(1)$    |
| 2   | 4 | $Q(a) \rightarrow R(a)$            | $\forall E(2)$    |
| 1,2 | 5 | $P(a) \rightarrow R(a)$            | Th. 2.2.3.2(3, 4) |
| 1,2 | 6 | $\forall x(P(x) \rightarrow R(x))$ | $\forall I(5)$    |

**Theorem 2.10.1.3 (Äquivalenztransitivität).**

$$\forall x(P(x) \leftrightarrow Q(x)), \forall x(Q(x) \leftrightarrow R(x)) \vdash \forall x(P(x) \leftrightarrow R(x))$$

|     |   |                                        |                   |
|-----|---|----------------------------------------|-------------------|
| 1   | 1 | $\forall x(P(x) \leftrightarrow Q(x))$ | $A$               |
| 2   | 2 | $\forall x(Q(x) \leftrightarrow R(x))$ | $A$               |
| 1   | 3 | $P(a) \leftrightarrow Q(a)$            | $\forall E(1)$    |
| 2   | 4 | $Q(a) \leftrightarrow R(a)$            | $\forall E(2)$    |
| 1,2 | 5 | $P(a) \leftrightarrow R(a)$            | Th. 2.2.3.5(3, 4) |
| 1,2 | 6 | $\forall x(P(x) \leftrightarrow R(x))$ | $\forall I(5)$    |

**Theorem 2.10.1.4 (Äquivalenztransitivität).**

$$\forall x(P(x) \leftrightarrow Q(x)), \forall x(R(x) \leftrightarrow Q(x)) \vdash \forall x(P(x) \leftrightarrow R(x))$$

|     |   |                                        |                    |
|-----|---|----------------------------------------|--------------------|
| 1   | 1 | $\forall x(P(x) \leftrightarrow Q(x))$ | $A$                |
| 2   | 2 | $\forall x(R(x) \leftrightarrow Q(x))$ | $A$                |
| 2   | 3 | $\forall x(Q(x) \leftrightarrow R(x))$ | Th. 2.10.1.1(2)    |
| 1,2 | 4 | $\forall x(P(x) \leftrightarrow R(x))$ | Th. 2.10.1.3(1, 2) |

## 10.2 Quantorenlogische Implikationen (Spezialfälle)

**Im Kapitel verwendete Symbole**

**Prädikatensymbole:**  $P, Q, R$

**Variablen:**  $x$

**Theorem 2.10.2.1.**

$$\forall x(P(x) \rightarrow Q(x)) \vdash \forall x((P(x) \wedge Q(x)) \leftrightarrow P(x))$$

|   |   |                                                      |                |
|---|---|------------------------------------------------------|----------------|
| 1 | 1 | $\forall x(P(x) \rightarrow Q(x))$                   | $A$            |
| 1 | 2 | $P(a) \rightarrow Q(a)$                              | $\forall E(1)$ |
| 1 | 3 | $(P(a) \wedge Q(a)) \leftrightarrow P(a)$            | Th. 2.2.1.5(2) |
| 1 | 4 | $\forall x((P(x) \wedge Q(x)) \leftrightarrow P(x))$ | $\forall I(3)$ |

**Theorem 2.10.2.2.**

$$\forall x(P(x) \rightarrow Q(x)) \vdash \forall x((Q(x) \wedge P(x)) \leftrightarrow P(x))$$

|   |   |                                                      |                |
|---|---|------------------------------------------------------|----------------|
| 1 | 1 | $\forall x(P(x) \rightarrow Q(x))$                   | $A$            |
| 1 | 2 | $P(a) \rightarrow Q(a)$                              | $\forall E(1)$ |
| 1 | 3 | $(Q(a) \wedge P(a)) \leftrightarrow P(a)$            | Th. 2.2.1.6(2) |
| 1 | 4 | $\forall x((Q(x) \wedge P(x)) \leftrightarrow P(x))$ | $\forall I(3)$ |

**Theorem 2.10.2.3.**

$$\forall x(P(x) \rightarrow Q(x)) \dashv\vdash \forall x(\neg P(x) \vee Q(x))$$

*Beweis.* $\vdash$ :

|   |   |                                    |                |
|---|---|------------------------------------|----------------|
| 1 | 1 | $\forall x(P(x) \rightarrow Q(x))$ | $A$            |
| 1 | 2 | $P(a) \rightarrow Q(a)$            | $\forall E(1)$ |
| 1 | 3 | $\neg P(a) \vee Q(a)$              | Th. 2.2.5.1(2) |
| 1 | 4 | $\forall x(\neg P(x) \vee Q(x))$   | $\forall I(3)$ |

 $\dashv$ :

|   |   |                                    |                |
|---|---|------------------------------------|----------------|
| 1 | 1 | $\forall x(\neg P(x) \vee Q(x))$   | $A$            |
| 1 | 2 | $\neg P(a) \vee Q(a)$              | $\forall E(1)$ |
| 1 | 3 | $P(a) \rightarrow Q(a)$            | Th. 2.2.5.1(2) |
| 1 | 4 | $\forall x(P(x) \rightarrow Q(x))$ | $\forall I(3)$ |

□

**10.3 Bikonditional mit Quantoren****Im Kapitel verwendete Symbole****Prädikatensymbole:**  $P, Q$ **Variablen:**  $x$ **Theorem 2.10.3.1.**

$$\forall x(P(x) \leftrightarrow Q(x)) \dashv\vdash \forall x(\neg P(x) \leftrightarrow \neg Q(x))$$

*Beweis.* $\vdash$ :

|   |   |                                                  |                |
|---|---|--------------------------------------------------|----------------|
| 1 | 1 | $\forall x(P(x) \leftrightarrow Q(x))$           | $A$            |
| 1 | 2 | $P(a) \leftrightarrow Q(a)$                      | $\forall E(1)$ |
| 1 | 3 | $\neg P(a) \leftrightarrow \neg Q(a)$            | Th. 2.2.6.1(2) |
| 1 | 4 | $\forall x(\neg P(x) \leftrightarrow \neg Q(x))$ | $\forall I(3)$ |

 $\dashv$ :

|   |   |                                                  |                |
|---|---|--------------------------------------------------|----------------|
| 1 | 1 | $\forall x(\neg P(x) \leftrightarrow \neg Q(x))$ | $A$            |
| 1 | 2 | $\neg P(a) \leftrightarrow \neg Q(a)$            | $\forall E(1)$ |

|   |   |                                        |                |
|---|---|----------------------------------------|----------------|
| 1 | 3 | $P(a) \leftrightarrow Q(a)$            | Th. 2.2.6.1(2) |
| 1 | 4 | $\forall x(P(x) \leftrightarrow Q(x))$ | $\forall I(3)$ |

□

**Theorem 2.10.3.2.**

$$\forall x(P(x) \leftrightarrow Q(x)) \dashv\vdash \forall x(P(x) \rightarrow Q(x)) \wedge \forall x(Q(x) \rightarrow P(x))$$

*Beweis.*

$\vdash$ :

|   |   |                                                                            |                         |
|---|---|----------------------------------------------------------------------------|-------------------------|
| 1 | 1 | $\forall x(P(x) \leftrightarrow Q(x))$                                     | $A$                     |
| 1 | 2 | $P(a) \leftrightarrow Q(a)$                                                | $\forall E(1)$          |
| 1 | 3 | $P(a) \rightarrow Q(a)$                                                    | $\leftrightarrow E1(2)$ |
| 1 | 4 | $Q(a) \rightarrow P(a)$                                                    | $\leftrightarrow E2(2)$ |
| 1 | 5 | $\forall x(P(x) \rightarrow Q(x))$                                         | $\forall I(3)$          |
| 1 | 6 | $\forall x(Q(x) \rightarrow P(x))$                                         | $\forall I(4)$          |
| 1 | 7 | $\forall x(P(x) \rightarrow Q(x)) \wedge \forall x(Q(x) \rightarrow P(x))$ | $\wedge I(5, 6)$        |

$\dashv\vdash$ :

|   |   |                                                                            |                           |
|---|---|----------------------------------------------------------------------------|---------------------------|
| 1 | 1 | $\forall x(P(x) \rightarrow Q(x)) \wedge \forall x(Q(x) \rightarrow P(x))$ | $A$                       |
| 1 | 2 | $\forall x(P(x) \rightarrow Q(x))$                                         | $\wedge E1(1)$            |
| 1 | 3 | $\forall x(Q(x) \rightarrow P(x))$                                         | $\wedge E2(1)$            |
| 1 | 4 | $P(a) \rightarrow Q(a)$                                                    | $\forall E(2)$            |
| 1 | 5 | $Q(a) \rightarrow P(a)$                                                    | $\forall E(3)$            |
| 1 | 6 | $P(a) \leftrightarrow Q(a)$                                                | $\leftrightarrow I(4, 5)$ |
| 1 | 7 | $\forall x(P(x) \leftrightarrow Q(x))$                                     | $\forall I(6)$            |

□

**Theorem 2.10.3.3.**

$$\forall x(P(x) \leftrightarrow Q(x)) \dashv\vdash \forall x(\neg P(x) \vee Q(x)) \wedge \forall x(\neg Q(x) \vee P(x))$$

*Beweis.*

$\vdash$ :

|   |   |                                                                            |                 |
|---|---|----------------------------------------------------------------------------|-----------------|
| 1 | 1 | $\forall x(P(x) \leftrightarrow Q(x))$                                     | $A$             |
| 1 | 2 | $\forall x(P(x) \rightarrow Q(x)) \wedge \forall x(Q(x) \rightarrow P(x))$ | Th. 2.10.3.2(1) |
| 1 | 3 | $\forall x(P(x) \rightarrow Q(x))$                                         | $\wedge E1(2)$  |
| 1 | 4 | $\forall x(Q(x) \rightarrow P(x))$                                         | $\wedge E2(2)$  |
| 1 | 5 | $\forall x(\neg P(x) \vee Q(x))$                                           | Th. 2.10.2.3(3) |

|   |   |                                                                        |                  |
|---|---|------------------------------------------------------------------------|------------------|
| 1 | 6 | $\forall x(\neg Q(x) \vee P(x))$                                       | Th. 2.10.2.3(4)  |
| 1 | 7 | $\forall x(\neg P(x) \vee Q(x)) \wedge \forall x(\neg Q(x) \vee P(x))$ | $\wedge I(5, 6)$ |

$\dashv$ :

|   |   |                                                                            |                  |
|---|---|----------------------------------------------------------------------------|------------------|
| 1 | 1 | $\forall x(\neg P(x) \vee Q(x)) \wedge \forall x(\neg Q(x) \vee P(x))$     | $A$              |
| 1 | 2 | $\forall x(\neg P(x) \vee Q(x))$                                           | $\wedge E1(1)$   |
| 1 | 3 | $\forall x(\neg Q(x) \vee P(x))$                                           | $\wedge E2(1)$   |
| 1 | 4 | $\forall x(P(x) \rightarrow Q(x))$                                         | Th. 2.10.2.3(2)  |
| 1 | 5 | $\forall x(Q(x) \rightarrow P(x))$                                         | Th. 2.10.2.3(3)  |
| 1 | 6 | $\forall x(P(x) \rightarrow Q(x)) \wedge \forall x(Q(x) \rightarrow P(x))$ | $\wedge I(4, 5)$ |
| 1 | 7 | $\forall x(P(x) \leftrightarrow Q(x))$                                     | Th. 2.10.3.2(6)  |

□

## 10.4 Negation von $\forall$ und $\exists$

### Im Kapitel verwendete Symbole

**Prädikatsymbole:**  $P, Q$

**Variablen:**  $x$

#### Theorem 2.10.4.1.

$$\exists x(P(x)) \dashv\vdash \neg \forall x(\neg P(x))$$

*Beweis.*

$\vdash$ :

|     |   |                             |                      |
|-----|---|-----------------------------|----------------------|
| 1   | 1 | $\exists x(P(x))$           | $A$                  |
| 2   | 2 | $P(a)$                      | $A$                  |
| 3   | 3 | $\forall x(\neg P(x))$      | $A$                  |
| 3   | 4 | $\neg P(a)$                 | $\forall E(3)$       |
| 2,3 | 5 | $\perp$                     | $\perp I(2, 4)$      |
| 2   | 6 | $\neg \forall x(\neg P(x))$ | $\neg I(3, 5)$       |
| 1   | 7 | $\neg \forall x(\neg P(x))$ | $\exists E(1, 2, 6)$ |

$\dashv$ :

|     |   |                             |                 |
|-----|---|-----------------------------|-----------------|
| 1   | 1 | $\neg \forall x(\neg P(x))$ | $A$             |
| 2   | 2 | $\neg \exists x(P(x))$      | $A$             |
| 3   | 3 | $P(a)$                      | $A$             |
| 3   | 4 | $\exists x(P(x))$           | $\exists I(3)$  |
| 2,3 | 5 | $\perp$                     | $\perp I(2, 4)$ |

|     |   |                        |                 |
|-----|---|------------------------|-----------------|
| 2   | 6 | $\neg P(a)$            | $\neg I(3, 5)$  |
| 2   | 7 | $\forall x(\neg P(x))$ | $\forall I(6)$  |
| 1,2 | 8 | $\perp$                | $\perp I(1, 7)$ |
| 1   | 9 | $\exists x(P(x))$      | $\neg E(2, 8)$  |

□

**Theorem 2.10.4.2.**

$$\exists x(\neg P(x)) \dashv\vdash \neg \forall x(P(x))$$

*Beweis.*

⊢:

|     |   |                        |                      |
|-----|---|------------------------|----------------------|
| 1   | 1 | $\exists x(\neg P(x))$ | $A$                  |
| 2   | 2 | $\neg P(a)$            | $A$                  |
| 3   | 3 | $\forall x(P(x))$      | $A$                  |
| 3   | 4 | $P(a)$                 | $\forall E(3)$       |
| 2,3 | 5 | $\perp$                | $\perp I(2, 4)$      |
| 2   | 6 | $\neg \forall x(P(x))$ | $\neg I(3, 5)$       |
| 1   | 7 | $\neg \forall x(P(x))$ | $\exists E(1, 2, 6)$ |

⊢:

|     |   |                             |                 |
|-----|---|-----------------------------|-----------------|
| 1   | 1 | $\neg \forall x(P(x))$      | $A$             |
| 2   | 2 | $\neg \exists x(\neg P(x))$ | $A$             |
| 3   | 3 | $\neg P(a)$                 | $A$             |
| 3   | 4 | $\exists x(\neg P(x))$      | $\exists I(3)$  |
| 2,3 | 5 | $\perp$                     | $\perp I(2, 4)$ |
| 2   | 6 | $P(a)$                      | $\neg E(3, 5)$  |
| 2   | 7 | $\forall x(P(x))$           | $\forall I(6)$  |
| 1,2 | 8 | $\perp$                     | $\perp I(1, 7)$ |
| 1   | 9 | $\exists x(\neg P(x))$      | $\neg E(2, 8)$  |

□

**Theorem 2.10.4.3.**

$$\neg \forall x(P(x) \rightarrow \neg Q(x)) \dashv\vdash \exists x(P(x) \wedge Q(x))$$

*Beweis.*

⊢:

|   |   |                                              |     |
|---|---|----------------------------------------------|-----|
| 1 | 1 | $\neg \forall x(P(x) \rightarrow \neg Q(x))$ | $A$ |
|---|---|----------------------------------------------|-----|

|   |    |                                              |                      |
|---|----|----------------------------------------------|----------------------|
| 1 | 2  | $\exists x \neg(P(x) \rightarrow \neg Q(x))$ | Th. 2.10.4.2(1)      |
| 3 | 3  | $\neg(P(a) \rightarrow \neg Q(a))$           | $A$                  |
| 3 | 4  | $P(a) \wedge \neg \neg Q(a)$                 | Th. 2.2.5.5(3)       |
| 3 | 5  | $P(a)$                                       | $\wedge E1(4)$       |
| 3 | 6  | $\neg \neg Q(a)$                             | $\wedge E2(4)$       |
| 3 | 7  | $Q(a)$                                       | Th. 2.1.6.1(6)       |
| 3 | 8  | $P(a) \wedge Q(a)$                           | $\wedge I(5, 7)$     |
| 3 | 9  | $\exists x(P(x) \wedge Q(x))$                | $\exists I(8)$       |
| 1 | 10 | $\exists x(P(x) \wedge Q(x))$                | $\exists E(2, 3, 9)$ |

$\vdash$ :

|   |    |                                               |                      |
|---|----|-----------------------------------------------|----------------------|
| 1 | 1  | $\exists x(P(x) \wedge Q(x))$                 | $A$                  |
| 2 | 2  | $P(a) \wedge Q(a)$                            | $A$                  |
| 2 | 3  | $P(a)$                                        | $\wedge E1(2)$       |
| 2 | 4  | $Q(a)$                                        | $\wedge E2(2)$       |
| 2 | 5  | $\neg \neg Q(a)$                              | Th. 2.1.6.1(4)       |
| 2 | 6  | $P(a) \wedge \neg \neg Q(a)$                  | $\wedge I(3, 5)$     |
| 2 | 7  | $\neg(P(a) \rightarrow \neg Q(a))$            | Th. 2.2.5.5(6)       |
| 2 | 8  | $\exists x(\neg(P(x) \rightarrow \neg Q(x)))$ | $\exists I(7)$       |
| 2 | 9  | $\neg \forall x(P(x) \rightarrow \neg Q(x))$  | Th. 2.10.4.2(8)      |
| 1 | 10 | $\neg \forall x(P(x) \rightarrow \neg Q(x))$  | $\exists E(1, 2, 9)$ |

□

**Theorem 2.10.4.4.**

$$\neg \forall x(P(x) \rightarrow Q(x)) \not\vdash \exists x(P(x) \wedge \neg Q(x))$$

*Beweis.*

$\vdash$ :

|   |   |                                         |                      |
|---|---|-----------------------------------------|----------------------|
| 1 | 1 | $\neg \forall x(P(x) \rightarrow Q(x))$ | $A$                  |
| 1 | 2 | $\exists x \neg(P(x) \rightarrow Q(x))$ | Th. 2.10.4.2(1)      |
| 3 | 3 | $\neg(P(a) \rightarrow Q(a))$           | $A$                  |
| 3 | 4 | $P(a) \wedge \neg Q(a)$                 | Th. 2.2.5.5(3)       |
| 3 | 5 | $\exists x(P(x) \wedge \neg Q(x))$      | $\exists I(4)$       |
| 1 | 6 | $\exists x(P(x) \wedge \neg Q(x))$      | $\exists E(2, 3, 5)$ |

$\vdash$ :

|   |   |                                    |                |
|---|---|------------------------------------|----------------|
| 1 | 1 | $\exists x(P(x) \wedge \neg Q(x))$ | $A$            |
| 2 | 2 | $P(a) \wedge \neg Q(a)$            | $A$            |
| 2 | 3 | $\neg(P(a) \rightarrow Q(a))$      | Th. 2.2.5.5(2) |

|   |   |                                          |                      |
|---|---|------------------------------------------|----------------------|
| 2 | 4 | $\exists x(\neg(P(x) \rightarrow Q(x)))$ | $\exists I(3)$       |
| 2 | 5 | $\neg\forall x(P(x) \rightarrow Q(x))$   | Th. 2.10.4.2(4)      |
| 1 | 6 | $\neg\forall x(P(x) \rightarrow Q(x))$   | $\exists E(1, 2, 5)$ |

□

**Theorem 2.10.4.5.**

$$\forall x(P(x)) \dashv\vdash \neg\exists x\neg(P(x))$$

*Beweis.*

⊢:

|     |   |                            |                      |
|-----|---|----------------------------|----------------------|
| 1   | 1 | $\forall x(P(x))$          | $A$                  |
| 2   | 2 | $\exists x(\neg P(x))$     | $A$                  |
| 3   | 3 | $\neg P(a)$                | $A$                  |
| 1   | 4 | $P(a)$                     | $\forall E(1)$       |
| 1,3 | 5 | $\perp$                    | $\perp I(3, 4)$      |
| 3   | 6 | $\neg\forall x(P(x))$      | $\neg I(1, 5)$       |
| 2   | 7 | $\neg\forall x(P(x))$      | $\exists E(2, 3, 6)$ |
| 1,2 | 8 | $\perp$                    | $\perp I(1, 7)$      |
| 1   | 9 | $\neg\exists x(\neg P(x))$ | $\neg I(2, 8)$       |

⊢:

|     |   |                            |                 |
|-----|---|----------------------------|-----------------|
| 1   | 1 | $\neg\exists x(\neg P(x))$ | $A$             |
| 2   | 2 | $\neg P(a)$                | $A$             |
| 2   | 3 | $\exists x(\neg P(x))$     | $\exists I(2)$  |
| 1,2 | 4 | $\perp$                    | $\perp I(1, 3)$ |
| 1   | 5 | $P(a)$                     | $\neg E(2, 4)$  |
| 1   | 6 | $\forall x(P(x))$          | $\forall I(5)$  |

□

**Theorem 2.10.4.6.**

$$\neg\forall x(P(x)) \dashv\vdash \exists x(\neg P(x))$$

|   |                                                           |                |
|---|-----------------------------------------------------------|----------------|
| 1 | $\forall x(P(x)) \leftrightarrow \neg\exists x\neg(P(x))$ | Th. 2.10.4.5   |
| 2 | $\neg\forall x(P(x)) \leftrightarrow \exists x\neg(P(x))$ | Th. 2.2.6.2(1) |

**Theorem 2.10.4.7.**

$$\forall x(\neg P(x)) \dashv\vdash \neg\exists x(P(x))$$

*Beweis.*

$\vdash$ :

|     |   |                          |                    |
|-----|---|--------------------------|--------------------|
| 1   | 1 | $\forall x(\neg P(x))$   | $A$                |
|     | 2 | $\exists x(P(x))$        | $A$                |
|     | 3 | $P(a)$                   | $A$                |
|     | 1 | 4 $\neg P(a)$            | $\forall E(1)$     |
| 1,3 | 5 | $\perp$                  | $\perp I(3,4)$     |
| 1,2 | 6 | $\perp$                  | $\exists E(2,3,5)$ |
|     | 1 | 7 $\neg \exists x(P(x))$ | $\neg I(2,6)$      |

$\dashv$ :

|     |   |                          |                |
|-----|---|--------------------------|----------------|
|     | 1 | 1 $\neg \exists x(P(x))$ | $A$            |
|     | 2 | 2 $P(a)$                 | $A$            |
|     | 2 | 3 $\exists x(P(x))$      | $\exists I(2)$ |
| 1,2 | 4 | $\perp$                  | $\perp I(1,3)$ |
|     | 1 | 5 $\neg P(a)$            | $\neg I(2,4)$  |
|     | 1 | 6 $\forall x(\neg P(x))$ | $\forall I(5)$ |

□

**Theorem 2.10.4.8.**

$$\forall x(P(x) \rightarrow Q(x)) \dashv\vdash \neg \exists x(P(x) \wedge \neg Q(x))$$

*Beweis.*

$\vdash$ :

|     |   |                                            |                      |
|-----|---|--------------------------------------------|----------------------|
|     | 1 | 1 $\forall x(P(x) \rightarrow Q(x))$       | $A$                  |
|     | 2 | 2 $\exists x(P(x) \wedge \neg Q(x))$       | $A$                  |
|     | 3 | 3 $P(a) \wedge \neg Q(a)$                  | $A$                  |
|     | 3 | 4 $P(a)$                                   | $\wedge E1(3)$       |
|     | 3 | 5 $\neg Q(a)$                              | $\wedge E2(3)$       |
|     | 1 | 6 $P(a) \rightarrow Q(a)$                  | $\forall E(1)$       |
| 1,3 | 7 | $Q(a)$                                     | $\rightarrow E(4,6)$ |
| 1,3 | 8 | $\perp$                                    | $\perp I(5,7)$       |
| 1,2 | 9 | $\perp$                                    | $\exists E(2,3,8)$   |
|     | 1 | 10 $\neg \exists x(P(x) \wedge \neg Q(x))$ | $\neg I(2,9)$        |

$\dashv$ :

|  |   |                                            |                 |
|--|---|--------------------------------------------|-----------------|
|  | 1 | 1 $\neg \exists x(P(x) \wedge \neg Q(x))$  | $A$             |
|  | 1 | 2 $\forall x(\neg(P(x) \wedge \neg Q(x)))$ | Th. 2.10.4.7(1) |

|   |    |                                    |                         |
|---|----|------------------------------------|-------------------------|
| 1 | 3  | $\neg(P(a) \wedge \neg Q(a))$      | $\forall E(2)$          |
| 1 | 4  | $\neg P(a) \vee \neg \neg Q(a)$    | Th. 2.1.8.2(3)          |
| 5 | 5  | $\neg P(a)$                        | $A$                     |
| 5 | 6  | $\neg P(a) \vee Q(a)$              | $\vee I1(5)$            |
| 7 | 7  | $\neg \neg Q(a)$                   | $A$                     |
| 7 | 8  | $Q(a)$                             | Th. 2.1.6.1(7)          |
| 7 | 9  | $\neg P(a) \vee Q(a)$              | $\vee I2(8)$            |
| 1 | 10 | $\neg P(a) \vee Q(a)$              | $\vee E(4, 5, 6, 7, 9)$ |
| 1 | 11 | $P(a) \rightarrow Q(a)$            | Th. 2.2.5.1(10)         |
| 1 | 12 | $\forall x(P(x) \rightarrow Q(x))$ | $\forall I(11)$         |

□

**Theorem 2.10.4.9.**

$$\forall x(P(x) \rightarrow \neg Q(x)) \dashv\vdash \neg \exists x(P(x) \wedge Q(x))$$

*Beweis.*

$\vdash$ :

|     |    |                                         |                       |
|-----|----|-----------------------------------------|-----------------------|
| 1   | 1  | $\forall x(P(x) \rightarrow \neg Q(x))$ | $A$                   |
| 2   | 2  | $\exists x(P(x) \wedge Q(x))$           | $A$                   |
| 3   | 3  | $P(a) \wedge Q(a)$                      | $A$                   |
| 3   | 4  | $P(a)$                                  | $\wedge E1(3)$        |
| 3   | 5  | $Q(a)$                                  | $\wedge E2(3)$        |
| 1   | 6  | $P(a) \rightarrow \neg Q(a)$            | $\forall E(1)$        |
| 1,3 | 7  | $\neg Q(a)$                             | $\rightarrow E(4, 6)$ |
| 1,3 | 8  | $\perp$                                 | $\perp I(5, 7)$       |
| 1,2 | 9  | $\perp$                                 | $\exists E(2, 3, 8)$  |
| 1   | 10 | $\neg \exists x(P(x) \wedge Q(x))$      | $\neg I(2, 9)$        |

$\dashv$ :

|   |   |                                         |                 |
|---|---|-----------------------------------------|-----------------|
| 1 | 1 | $\neg \exists x(P(x) \wedge Q(x))$      | $A$             |
| 1 | 2 | $\forall x(\neg(P(x) \wedge Q(x)))$     | Th. 2.10.4.7(1) |
| 1 | 3 | $\neg(P(a) \wedge Q(a))$                | $\forall E(2)$  |
| 1 | 4 | $\neg P(a) \vee \neg Q(a)$              | Th. 2.1.8.2(3)  |
| 1 | 5 | $P(a) \rightarrow \neg Q(a)$            | Th. 2.2.5.1(4)  |
| 1 | 6 | $\forall x(P(x) \rightarrow \neg Q(x))$ | $\forall I(5)$  |

□

**Theorem 2.10.4.10.**

$$\forall x(P(x) \leftrightarrow Q(x)) \dashv\vdash \neg \exists x(P(x) \wedge \neg Q(x)) \wedge \neg \exists x(Q(x) \wedge \neg P(x))$$

*Beweis.*

$\vdash$ :

|   |   |                                                                                      |                  |
|---|---|--------------------------------------------------------------------------------------|------------------|
| 1 | 1 | $\forall x(P(x) \leftrightarrow Q(x))$                                               | $A$              |
| 1 | 2 | $\forall x(P(x) \rightarrow Q(x)) \wedge \forall x(Q(x) \rightarrow P(x))$           | Th. 2.10.3.2(1)  |
| 1 | 3 | $\forall x(P(x) \rightarrow Q(x))$                                                   | $\wedge E1(2)$   |
| 1 | 4 | $\forall x(Q(x) \rightarrow P(x))$                                                   | $\wedge E2(2)$   |
| 1 | 5 | $\neg \exists x(P(x) \wedge \neg Q(x))$                                              | Th. 2.10.4.8(3)  |
| 1 | 6 | $\neg \exists x(Q(x) \wedge \neg P(x))$                                              | Th. 2.10.4.8(4)  |
| 1 | 7 | $\neg \exists x(P(x) \wedge \neg Q(x)) \wedge \neg \exists x(Q(x) \wedge \neg P(x))$ | $\wedge I(5, 6)$ |

$\dashv$ :

|   |   |                                                                                      |                  |
|---|---|--------------------------------------------------------------------------------------|------------------|
| 1 | 1 | $\neg \exists x(P(x) \wedge \neg Q(x)) \wedge \neg \exists x(Q(x) \wedge \neg P(x))$ | $A$              |
| 1 | 2 | $\neg \exists x(P(x) \wedge \neg Q(x))$                                              | $\wedge E1(1)$   |
| 1 | 3 | $\neg \exists x(Q(x) \wedge \neg P(x))$                                              | $\wedge E2(1)$   |
| 1 | 4 | $\forall x(P(x) \rightarrow Q(x))$                                                   | Th. 2.10.4.8(2)  |
| 1 | 5 | $\forall x(Q(x) \rightarrow P(x))$                                                   | Th. 2.10.4.8(3)  |
| 1 | 6 | $\forall x(P(x) \rightarrow Q(x)) \wedge \forall x(Q(x) \rightarrow P(x))$           | $\wedge I(4, 5)$ |
| 1 | 7 | $\forall x(P(x) \leftrightarrow Q(x))$                                               | Th. 2.10.3.2(6)  |

□

**Theorem 2.10.4.11.**

$$\neg \forall x(P(x) \leftrightarrow Q(x)) \dashv \vdash \exists x(P(x) \wedge \neg Q(x)) \vee \exists x(Q(x) \wedge \neg P(x))$$

*Beweis.*

$\vdash$ :

|   |   |                                                                                              |                   |
|---|---|----------------------------------------------------------------------------------------------|-------------------|
| 1 | 1 | $\neg \forall x(P(x) \leftrightarrow Q(x))$                                                  | $A$               |
| 1 | 2 | $\forall x(P(x) \leftrightarrow Q(x)) \leftrightarrow \neg \exists x(P(x) \wedge \neg Q(x))$ | Th. 2.10.4.10     |
|   |   | $\wedge \neg \exists x(Q(x) \wedge \neg P(x))$                                               |                   |
| 1 | 3 | $\neg(\neg \exists x(P(x) \wedge \neg Q(x))$                                                 | Th. 2.2.2.4(2, 1) |
|   |   | $\wedge \neg \exists x(Q(x) \wedge \neg P(x)))$                                              |                   |
| 1 | 4 | $\exists x(P(x) \wedge \neg Q(x))$                                                           | Th. 2.1.8.4(3)    |
|   |   | $\vee \exists x(Q(x) \wedge \neg P(x))$                                                      |                   |

$\dashv$ :

|   |   |                                                 |                |
|---|---|-------------------------------------------------|----------------|
| 1 | 1 | $\exists x(P(x) \wedge \neg Q(x))$              | $A$            |
|   |   | $\vee \exists x(Q(x) \wedge \neg P(x))$         |                |
| 1 | 2 | $\neg(\neg \exists x(P(x) \wedge \neg Q(x))$    | Th. 2.1.8.4(1) |
|   |   | $\wedge \neg \exists x(Q(x) \wedge \neg P(x)))$ |                |

|   |   |                                                                                              |                   |
|---|---|----------------------------------------------------------------------------------------------|-------------------|
| 1 | 3 | $\forall x(P(x) \leftrightarrow Q(x)) \leftrightarrow \neg \exists x(P(x) \wedge \neg Q(x))$ | Th. 2.10.4.10     |
|   |   | $\wedge \neg \exists x(Q(x) \wedge \neg P(x))$                                               |                   |
| 1 | 4 | $\neg \forall x(P(x) \leftrightarrow Q(x))$                                                  | Th. 2.2.2.3(3, 2) |

□

## 10.5 Distribution von Quantoren über Junktoren

Im Kapitel verwendete Symbole

Aussagenvariablen:  $P$   
 Prädikatensymbole:  $F, G, P, Q$   
 Variablen:  $x, y$   
 Konstantensymbole:  $a$

**Theorem 2.10.5.1.**

$$P(a), \forall x(P(x) \rightarrow Q(x)) \vdash Q(a)$$

|     |   |                                    |                       |
|-----|---|------------------------------------|-----------------------|
| 1   | 1 | $P(a)$                             | $A$                   |
| 2   | 2 | $\forall x(P(x) \rightarrow Q(x))$ | $A$                   |
| 2   | 3 | $P(a) \rightarrow Q(a)$            | $\forall E(2)$        |
| 1,2 | 4 | $Q(a)$                             | $\rightarrow E(1, 3)$ |

**Theorem 2.10.5.2.**

$$P(a), \forall x(P(x) \leftrightarrow Q(x)) \vdash Q(a)$$

|     |   |                                        |                   |
|-----|---|----------------------------------------|-------------------|
| 1   | 1 | $P(a)$                                 | $A$               |
| 2   | 2 | $\forall x(P(x) \leftrightarrow Q(x))$ | $A$               |
| 2   | 3 | $P(a) \leftrightarrow Q(a)$            | $\forall E(2)$    |
| 1,2 | 4 | $Q(a)$                                 | Th. 2.2.4.1(3, 1) |

**Theorem 2.10.5.3.**

$$Q(a), \forall x(P(x) \leftrightarrow Q(x)) \vdash P(a)$$

|     |   |                                        |                   |
|-----|---|----------------------------------------|-------------------|
| 1   | 1 | $Q(a)$                                 | $A$               |
| 2   | 2 | $\forall x(P(x) \leftrightarrow Q(x))$ | $A$               |
| 2   | 3 | $P(a) \leftrightarrow Q(a)$            | $\forall E(2)$    |
| 1,2 | 4 | $P(a)$                                 | Th. 2.2.4.2(3, 1) |

**Theorem 2.10.5.4.**

$$\forall x(P \rightarrow Q(x)) \dashv\vdash P \rightarrow \forall x(Q(x))$$

*Beweis.*

$\vdash$ :

|     |   |                                 |                      |
|-----|---|---------------------------------|----------------------|
| 1   | 1 | $\forall x(P \rightarrow Q(x))$ | $A$                  |
| 2   | 2 | $P$                             | $A$                  |
| 1   | 3 | $P \rightarrow Q(x)$            | $\forall E(1)$       |
| 1,2 | 4 | $Q(x)$                          | $\rightarrow E(2,3)$ |
| 1,2 | 5 | $\forall x(Q(x))$               | $\forall I(4)$       |
| 1   | 6 | $P \rightarrow \forall x(Q(x))$ | $\rightarrow I(2,5)$ |

$\dashv$ :

|     |   |                                 |                      |
|-----|---|---------------------------------|----------------------|
| 1   | 1 | $P \rightarrow \forall x(Q(x))$ | $A$                  |
| 2   | 2 | $P$                             | $A$                  |
| 1,2 | 3 | $\forall x(Q(x))$               | $\rightarrow E(1,2)$ |
| 1,2 | 4 | $Q(x)$                          | $\forall E(3)$       |
| 1   | 5 | $P \rightarrow Q(x)$            | $\rightarrow I(2,4)$ |
| 1   | 6 | $\forall x(P \rightarrow Q(x))$ | $\forall I(5)$       |

□

**Theorem 2.10.5.5.**

$$\forall x(P \wedge Q(x)) \dashv\vdash P \wedge \forall x(Q(x))$$

*Beweis.*

$\vdash$ :

|   |   |                            |                 |
|---|---|----------------------------|-----------------|
| 1 | 1 | $\forall x(P \wedge Q(x))$ | $A$             |
| 1 | 2 | $P \wedge Q(x)$            | $\forall E(1)$  |
| 1 | 3 | $P$                        | $\wedge E1(2)$  |
| 1 | 4 | $Q(x)$                     | $\wedge E2(2)$  |
| 1 | 5 | $\forall x(Q(x))$          | $\forall I(4)$  |
| 1 | 6 | $P \wedge \forall x(Q(x))$ | $\wedge I(3,5)$ |

$\dashv$ :

|   |   |                            |                 |
|---|---|----------------------------|-----------------|
| 1 | 1 | $P \wedge \forall x(Q(x))$ | $A$             |
| 1 | 2 | $P$                        | $\wedge E1(1)$  |
| 1 | 3 | $\forall x(Q(x))$          | $\wedge E2(1)$  |
| 1 | 4 | $Q(x)$                     | $\forall E(3)$  |
| 1 | 5 | $P \wedge Q(x)$            | $\wedge I(2,4)$ |
| 1 | 6 | $\forall x(P \wedge Q(x))$ | $\forall I(5)$  |

□

**Theorem 2.10.5.6.**

$$\exists x(P \wedge Q(x)) \dashv\vdash P \wedge \exists x(Q(x))$$

*Beweis.*

$\vdash$ :

|   |   |                            |                      |
|---|---|----------------------------|----------------------|
| 1 | 1 | $\exists x(P \wedge Q(x))$ | $A$                  |
| 2 | 2 | $P \wedge Q(a)$            | $A$                  |
| 2 | 3 | $P$                        | $\wedge E1(2)$       |
| 2 | 4 | $Q(a)$                     | $\wedge E2(2)$       |
| 2 | 5 | $\exists x(Q(x))$          | $\exists I(4)$       |
| 2 | 6 | $P \wedge \exists x(Q(x))$ | $\wedge I(3, 5)$     |
| 1 | 7 | $P \wedge \exists x(Q(x))$ | $\exists E(1, 2, 6)$ |

$\dashv$ :

|     |   |                            |                      |
|-----|---|----------------------------|----------------------|
| 1   | 1 | $P \wedge \exists x(Q(x))$ | $A$                  |
| 1   | 2 | $P$                        | $\wedge E1(1)$       |
| 1   | 3 | $\exists x(Q(x))$          | $\wedge E2(1)$       |
| 4   | 4 | $Q(a)$                     | $A$                  |
| 1,4 | 5 | $P \wedge Q(a)$            | $\wedge I(2, 4)$     |
| 1,4 | 6 | $\exists x(P \wedge Q(x))$ | $\exists I(5)$       |
| 1   | 7 | $\exists x(P \wedge Q(x))$ | $\exists E(3, 4, 6)$ |

□

**Theorem 2.10.5.7.**

$$\forall x(P \vee Q(x)) \dashv\vdash P \vee \forall x(Q(x))$$

*Beweis.*

$\vdash$ :

|     |    |                                      |                 |
|-----|----|--------------------------------------|-----------------|
| 1   | 1  | $\forall x(P \vee Q(x))$             | $A$             |
| 2   | 2  | $\neg(P \vee \forall x(Q(x)))$       | $A$             |
| 2   | 3  | $\neg P \wedge \neg \forall x(Q(x))$ | Th. 2.1.8.1(2)  |
| 2   | 4  | $\neg P$                             | $\wedge E1(3)$  |
| 2   | 5  | $\neg \forall x(Q(x))$               | $\wedge E2(3)$  |
| 2   | 6  | $\exists x(\neg Q(x))$               | Th. 2.10.4.6(5) |
| 7   | 7  | $\neg Q(a)$                          | $A$             |
| 1   | 8  | $P \vee Q(a)$                        | $\vee E(1)$     |
| 9   | 9  | $P$                                  | $A$             |
| 2,9 | 10 | $\perp$                              | $\perp I(4, 9)$ |

|       |    |                          |                            |
|-------|----|--------------------------|----------------------------|
| 11    | 11 | $Q(a)$                   | $A$                        |
| 7,11  | 12 | $\perp$                  | $\perp I(11, 7)$           |
| 1,2,7 | 13 | $\perp$                  | $\vee E(8, 9, 10, 11, 12)$ |
| 1,2   | 14 | $\perp$                  | $\exists E(6, 7, 13)$      |
| 1     | 15 | $P \vee \forall x(Q(x))$ | $\neg E(2, 14)$            |

$\dashv$ :

|   |   |                          |                         |
|---|---|--------------------------|-------------------------|
| 1 | 1 | $P \vee \forall x(Q(x))$ | $A$                     |
| 2 | 2 | $P$                      | $A$                     |
| 2 | 3 | $P \vee Q(x)$            | $\vee I(2)$             |
| 2 | 4 | $\forall x(P \vee Q(x))$ | $\forall I(3)$          |
| 5 | 5 | $\forall x(Q(x))$        | $A$                     |
| 5 | 6 | $Q(x)$                   | $\forall E(5)$          |
| 5 | 7 | $P \vee Q(x)$            | $\vee I(6)$             |
| 5 | 8 | $\forall x(P \vee Q(x))$ | $\forall I(7)$          |
| 1 | 9 | $\forall x(P \vee Q(x))$ | $\vee E(1, 2, 4, 5, 8)$ |

□

**Theorem 2.10.5.8.**

$$\exists x(P \vee F(x)) \dashv\vdash P \vee \exists x(F(x))$$

*Beweis.*

$\vdash$ :

|   |   |                          |                         |
|---|---|--------------------------|-------------------------|
| 1 | 1 | $\exists x(P \vee F(x))$ | $A$                     |
| 2 | 2 | $P \vee F(a)$            | $A$                     |
| 3 | 3 | $P$                      | $A$                     |
| 3 | 4 | $P \vee \exists x(F(x))$ | $\vee I(3)$             |
| 5 | 5 | $F(a)$                   | $A$                     |
| 5 | 6 | $\exists x(F(x))$        | $\exists I(5)$          |
| 5 | 7 | $P \vee \exists x(F(x))$ | $\vee I(6)$             |
| 2 | 8 | $P \vee \exists x(F(x))$ | $\vee E(2, 3, 4, 5, 7)$ |
| 1 | 9 | $P \vee \exists x(F(x))$ | $\exists E(1, 2, 8)$    |

$\dashv$ :

|   |   |                          |                |
|---|---|--------------------------|----------------|
| 1 | 1 | $P \vee \exists x(F(x))$ | $A$            |
| 2 | 2 | $P$                      | $A$            |
| 2 | 3 | $P \vee F(x)$            | $\vee I(2)$    |
| 2 | 4 | $\exists x(P \vee F(x))$ | $\exists I(3)$ |
| 5 | 5 | $\exists x(F(x))$        | $A$            |

|   |    |                          |                         |
|---|----|--------------------------|-------------------------|
| 6 | 6  | $F(a)$                   | $A$                     |
| 6 | 7  | $P \vee F(a)$            | $\vee I2(6)$            |
| 6 | 8  | $\exists x(P \vee F(x))$ | $\exists I(7)$          |
| 5 | 9  | $\exists x(P \vee F(x))$ | $\exists E(5, 6, 8)$    |
| 1 | 10 | $\exists x(P \vee F(x))$ | $\vee E(1, 2, 4, 5, 9)$ |

□

**Theorem 2.10.5.9.**

$$\exists x(F(x) \rightarrow P) \dashv\vdash \forall x(F(x)) \rightarrow P$$

*Beweis.*

$\vdash$ :

|     |   |                                 |                       |
|-----|---|---------------------------------|-----------------------|
| 1   | 1 | $\exists x(F(x) \rightarrow P)$ | $A$                   |
| 2   | 2 | $F(a) \rightarrow P$            | $A$                   |
| 3   | 3 | $\forall x(F(x))$               | $A$                   |
| 3   | 4 | $F(a)$                          | $\forall E(3)$        |
| 2,3 | 5 | $P$                             | $\rightarrow E(2, 4)$ |
| 2   | 6 | $\forall x(F(x)) \rightarrow P$ | $\rightarrow I(3, 5)$ |
| 1   | 7 | $\forall x(F(x)) \rightarrow P$ | $\exists E(1, 2, 6)$  |

$\dashv\vdash$ :

|     |    |                                 |                          |
|-----|----|---------------------------------|--------------------------|
| 1   | 1  | $\forall x(F(x)) \rightarrow P$ | $A$                      |
|     | 2  | $P \vee \neg P$                 | Th. 2.1.7.1              |
| 3   | 3  | $P$                             | $A$                      |
| 3   | 4  | $F(a) \rightarrow P$            | Th. 2.2.7.11(3)          |
| 3   | 5  | $\exists x(F(x) \rightarrow P)$ | $\exists I(4)$           |
| 6   | 6  | $\neg P$                        | $A$                      |
| 1,6 | 7  | $\neg \forall x(F(x))$          | Th. 2.2.2.1(1, 6)        |
| 1,6 | 8  | $\exists x(\neg F(x))$          | Th. 2.10.4.6(7)          |
| 9   | 9  | $\neg F(a)$                     | $A$                      |
| 9   | 10 | $F(a) \rightarrow P$            | Th. 2.2.7.12(9)          |
| 9   | 11 | $\exists x(F(x) \rightarrow P)$ | $\exists I(10)$          |
| 1,6 | 12 | $\exists x(F(x) \rightarrow P)$ | $\exists E(8, 9, 11)$    |
| 1   | 13 | $\exists x(F(x) \rightarrow P)$ | $\vee E(2, 3, 5, 6, 12)$ |

□

**Theorem 2.10.5.10.**

$$\forall x(F(x) \wedge G(x)) \dashv\vdash \forall x(F(x)) \wedge \forall x(G(x))$$

*Beweis.*

$\vdash$ :

|   |   |                                          |                  |
|---|---|------------------------------------------|------------------|
| 1 | 1 | $\forall x(F(x) \wedge G(x))$            | $A$              |
| 1 | 2 | $F(x) \wedge G(x)$                       | $\forall E(1)$   |
| 1 | 3 | $F(x)$                                   | $\wedge E1(2)$   |
| 1 | 4 | $\forall x(F(x))$                        | $\forall I(3)$   |
| 1 | 5 | $G(x)$                                   | $\wedge E2(2)$   |
| 1 | 6 | $\forall x(G(x))$                        | $\forall I(5)$   |
| 1 | 7 | $\forall x(F(x)) \wedge \forall x(G(x))$ | $\wedge I(4, 6)$ |

$\dashv$ :

|   |   |                                          |                  |
|---|---|------------------------------------------|------------------|
| 1 | 1 | $\forall x(F(x)) \wedge \forall x(G(x))$ | $A$              |
| 1 | 2 | $\forall x(F(x))$                        | $\wedge E1(1)$   |
| 1 | 3 | $F(x)$                                   | $\forall E(2)$   |
| 1 | 4 | $\forall x(G(x))$                        | $\wedge E2(1)$   |
| 1 | 5 | $G(x)$                                   | $\forall E(4)$   |
| 1 | 6 | $F(x) \wedge G(x)$                       | $\wedge I(3, 5)$ |
| 1 | 7 | $\forall x(F(x) \wedge G(x))$            | $\forall I(6)$   |

□

**Theorem 2.10.5.11.**

$$\forall x(F(x)) \vee \forall x(G(x)) \vdash \forall x(F(x) \vee G(x))$$

|   |    |                                        |                         |
|---|----|----------------------------------------|-------------------------|
| 1 | 1  | $\forall x(F(x)) \vee \forall x(G(x))$ | $A$                     |
| 2 | 2  | $\forall x(F(x))$                      | $A$                     |
| 2 | 3  | $F(a)$                                 | $\forall E(2)$          |
| 2 | 4  | $F(a) \vee G(a)$                       | $\vee I1(3)$            |
| 2 | 5  | $\forall x(F(x) \vee G(x))$            | $\forall I(4)$          |
| 6 | 6  | $\forall x(G(x))$                      | $A$                     |
| 6 | 7  | $G(a)$                                 | $\forall E(6)$          |
| 6 | 8  | $F(a) \vee G(a)$                       | $\vee I2(7)$            |
| 6 | 9  | $\forall x(F(x) \vee G(x))$            | $\forall I(8)$          |
| 1 | 10 | $\forall x(F(x) \vee G(x))$            | $\vee E(1, 2, 5, 6, 9)$ |

**Theorem 2.10.5.12.**

$$\exists x(F(x) \vee G(x)) \dashv\vdash \exists x(F(x)) \vee \exists x(G(x))$$

*Beweis.*

$\vdash$ :

|   |    |                                        |                         |
|---|----|----------------------------------------|-------------------------|
| 1 | 1  | $\exists x(F(x) \vee G(x))$            | $A$                     |
| 2 | 2  | $F(a) \vee G(a)$                       | $A$                     |
| 3 | 3  | $F(a)$                                 | $A$                     |
| 3 | 4  | $\exists x(F(x))$                      | $\exists I(3)$          |
| 3 | 5  | $\exists x(F(x)) \vee \exists x(G(x))$ | $\vee I1(4)$            |
| 6 | 6  | $G(a)$                                 | $A$                     |
| 6 | 7  | $\exists x(G(x))$                      | $\exists I(6)$          |
| 6 | 8  | $\exists x(F(x)) \vee \exists x(G(x))$ | $\vee I2(7)$            |
| 2 | 9  | $\exists x(F(x)) \vee \exists x(G(x))$ | $\vee E(2, 3, 5, 6, 8)$ |
| 1 | 10 | $\exists x(F(x)) \vee \exists x(G(x))$ | $\exists E(1, 2, 9)$    |

$\dashv$ :

|   |    |                                        |                          |
|---|----|----------------------------------------|--------------------------|
| 1 | 1  | $\exists x(F(x)) \vee \exists x(G(x))$ | $A$                      |
| 2 | 2  | $\exists x(F(x))$                      | $A$                      |
| 3 | 3  | $F(a)$                                 | $A$                      |
| 3 | 4  | $F(a) \vee G(a)$                       | $\vee I1(3)$             |
| 3 | 5  | $\exists x(F(x) \vee G(x))$            | $\exists I(4)$           |
| 2 | 6  | $\exists x(F(x) \vee G(x))$            | $\exists E(2, 3, 5)$     |
| 7 | 7  | $\exists x(G(x))$                      | $A$                      |
| 8 | 8  | $G(a)$                                 | $A$                      |
| 8 | 9  | $F(a) \vee G(a)$                       | $\vee I2(8)$             |
| 8 | 10 | $\exists x(F(x) \vee G(x))$            | $\exists I(9)$           |
| 7 | 11 | $\exists x(F(x) \vee G(x))$            | $\exists E(7, 8, 10)$    |
| 1 | 12 | $\exists x(F(x) \vee G(x))$            | $\vee E(1, 2, 6, 7, 11)$ |

□

**Theorem 2.10.5.13.**

$$\exists x \exists y F(x, y) \vdash \exists y \exists x F(x, y)$$

|   |   |                               |                      |
|---|---|-------------------------------|----------------------|
| 1 | 1 | $\exists x \exists y F(x, y)$ | $A$                  |
| 2 | 2 | $\exists y F(a, y)$           | $A$                  |
| 3 | 3 | $F(a, b)$                     | $A$                  |
| 3 | 4 | $\exists x F(x, b)$           | $\exists I(3)$       |
| 3 | 5 | $\exists y \exists x F(x, y)$ | $\exists I(4)$       |
| 2 | 6 | $\exists y \exists x F(x, y)$ | $\exists E(2, 3, 5)$ |
| 1 | 7 | $\exists y \exists x F(x, y)$ | $\exists E(1, 2, 6)$ |

**Theorem 2.10.5.14.**

$$\forall x \forall y F(x, y) \vdash \forall y \forall x F(x, y)$$

|   |   |                               |                |
|---|---|-------------------------------|----------------|
| 1 | 1 | $\forall x \forall y F(x, y)$ | $A$            |
| 1 | 2 | $\forall y F(a, y)$           | $\forall E(1)$ |
| 1 | 3 | $F(a, b)$                     | $\forall E(2)$ |
| 1 | 4 | $\forall x F(x, b)$           | $\forall I(3)$ |
| 1 | 5 | $\forall y \forall x F(x, y)$ | $\forall I(4)$ |

## 10.6 Invarianzen gegenüber universellen Aussagen

### Im Kapitel verwendete Symbole

**Prädikatensymbole:**  $P, Q$

**Variablen:**  $x$

**Theorem 2.10.6.1.**

$$\forall x (P(x)) \vdash \forall x (Q(x) \leftrightarrow P(x) \wedge Q(x))$$

|     |    |                                                       |                           |
|-----|----|-------------------------------------------------------|---------------------------|
| 1   | 1  | $\forall x (P(x))$                                    | $A$                       |
| 2   | 2  | $Q(a)$                                                | $A$                       |
| 1   | 3  | $P(a)$                                                | $\forall E(1)$            |
| 1,2 | 4  | $P(a) \wedge Q(a)$                                    | $\wedge I(3, 2)$          |
| 1   | 5  | $Q(a) \rightarrow (P(a) \wedge Q(a))$                 | $\rightarrow I(2, 4)$     |
| 6   | 6  | $P(a) \wedge Q(a)$                                    | $A$                       |
| 6   | 7  | $Q(a)$                                                | $\wedge E(6)$             |
|     | 8  | $(P(a) \wedge Q(a)) \rightarrow Q(a)$                 | $\rightarrow I(6, 7)$     |
| 1   | 9  | $Q(a) \leftrightarrow (P(a) \wedge Q(a))$             | $\leftrightarrow I(5, 8)$ |
| 1   | 10 | $\forall x (Q(x) \leftrightarrow (P(x) \wedge Q(x)))$ | $\forall I(9)$            |

**Theorem 2.10.6.2.**

$$\forall x (\neg Q(x)) \vdash P \leftrightarrow P \vee Q(a)$$

*Beweis.*

Hinrichtung:

|   |   |                         |             |
|---|---|-------------------------|-------------|
| 1 | 1 | $\forall x (\neg Q(x))$ | $A$         |
| 2 | 2 | $P$                     | $A$         |
| 2 | 3 | $P \vee Q(a)$           | $\vee I(2)$ |

Rückrichtung:

|     |   |               |                    |
|-----|---|---------------|--------------------|
| 4   | 4 | $P \vee Q(a)$ | $A$                |
| 1   | 5 | $\neg Q(a)$   | $\forall E(1)$     |
| 1,4 | 6 | $P$           | Th. 2.2.7.10(4, 5) |

Zusammenführung:

|   |   |                                 |                                 |
|---|---|---------------------------------|---------------------------------|
| 1 | 7 | $P \leftrightarrow P \vee Q(a)$ | $\leftrightarrow I(2, 3, 4, 6)$ |
|---|---|---------------------------------|---------------------------------|

□

## 10.7 Umgang mit = in Quantoren

**Im Kapitel verwendete Symbole**

**Prädikatensymbole:**  $F, P, Q, R$

**Variablen:**  $x, u$

### 10.7.1 Transitivität über Quantoren

**Theorem 2.10.7.1.**

$$\forall x P(x) = Q(x), \forall x P(x) = R(x) \vdash \forall x Q(x) = R(x)$$

|     |   |                         |                   |
|-----|---|-------------------------|-------------------|
| 1   | 1 | $\forall x P(x) = Q(x)$ | $A$               |
| 2   | 2 | $\forall x P(x) = R(x)$ | $A$               |
| 1   | 3 | $P(u) = Q(u)$           | $\forall E(1)$    |
| 2   | 4 | $P(u) = R(u)$           | $\forall E(2)$    |
| 1,2 | 5 | $Q(u) = R(u)$           | Th. 2.9.1.4(3, 4) |
| 1,2 | 6 | $\forall x Q(x) = R(x)$ | $\forall I(5)$    |

**Theorem 2.10.7.2.**

**Prädikatensymbole:**  $P, Q, R$

$$\begin{aligned} &\forall x F(x) \rightarrow P(x) = Q(x) \\ &\forall x F(x) \rightarrow P(x) = R(x) \\ &\vdash \forall x F(x) \rightarrow Q(x) = R(x) \end{aligned}$$

|   |   |                                          |                |
|---|---|------------------------------------------|----------------|
| 1 | 1 | $\forall x F(x) \rightarrow P(x) = Q(x)$ | $A$            |
| 2 | 2 | $\forall x F(x) \rightarrow P(x) = R(x)$ | $A$            |
| 1 | 3 | $F(u) \rightarrow P(u) = Q(u)$           | $\forall E(1)$ |

|       |    |                                          |                       |
|-------|----|------------------------------------------|-----------------------|
| 2     | 4  | $F(u) \rightarrow P(u) = R(u)$           | $\forall E(2)$        |
| 5     | 5  | $F(u)$                                   | $A$                   |
| 1,5   | 6  | $P(u) = Q(u)$                            | $\rightarrow E(5, 3)$ |
| 2,5   | 7  | $P(u) = R(u)$                            | $\rightarrow E(5, 4)$ |
| 1,2,5 | 8  | $Q(u) = R(u)$                            | Th. 2.9.1.4(6, 7)     |
| 1,2   | 9  | $F(u) \rightarrow Q(u) = R(u)$           | $\rightarrow I(5, 8)$ |
| 1,2   | 10 | $\forall x F(x) \rightarrow Q(x) = R(x)$ | $\forall I(9)$        |

## 10.8 Höchstens-ein-Quantor

Im Kapitel verwendete Symbole

Prädikatensymbole:  $P, Q$

Variablen:  $x, y, z, a, b$

Quantoren:  $\exists^{\leq 1}$

**Definition 2.10.8.1 (Höchstens-ein-Quantor).**

$$\exists^{\leq 1} x P(x) := \forall y \forall z ((P(y) \wedge P(z)) \rightarrow y = z)$$

**Theorem 2.10.8.1 (Definitionsäquivalenz für  $\exists^{\leq 1}$ ).**

$$\exists^{\leq 1} x P(x) \dashv\vdash \forall y \forall z ((P(y) \wedge P(z)) \rightarrow y = z)$$

*Beweis.*

$\vdash$ :

|   |   |                                                              |                  |
|---|---|--------------------------------------------------------------|------------------|
| 1 | 1 | $\exists^{\leq 1} x P(x)$                                    | $A$              |
| 1 | 2 | $\forall y \forall z ((P(y) \wedge P(z)) \rightarrow y = z)$ | Def. 2.10.8.1(1) |

$\dashv$ :

|   |   |                                                              |                  |
|---|---|--------------------------------------------------------------|------------------|
| 1 | 1 | $\forall y \forall z ((P(y) \wedge P(z)) \rightarrow y = z)$ | $A$              |
| 1 | 2 | $\exists^{\leq 1} x P(x)$                                    | Def. 2.10.8.1(1) |

□

**Theorem 2.10.8.2 (Elimination des Höchstens-ein-Quantors).**

$$\exists^{\leq 1} x P(x), P(a), P(b) \vdash a = b$$

|   |   |                                                              |                 |
|---|---|--------------------------------------------------------------|-----------------|
| 1 | 1 | $\exists^{\leq 1} x P(x)$                                    | $A$             |
| 2 | 2 | $P(a)$                                                       | $A$             |
| 3 | 3 | $P(b)$                                                       | $A$             |
| 1 | 4 | $\forall y \forall z ((P(y) \wedge P(z)) \rightarrow y = z)$ | Th. 2.10.8.1(1) |

|       |   |                                                   |                      |
|-------|---|---------------------------------------------------|----------------------|
| 1     | 5 | $\forall z((P(a) \wedge P(z)) \rightarrow a = z)$ | $\forall E(4)$       |
| 1     | 6 | $(P(a) \wedge P(b)) \rightarrow a = b$            | $\forall E(5)$       |
| 2,3   | 7 | $P(a) \wedge P(b)$                                | $\wedge I(2,3)$      |
| 1,2,3 | 8 | $a = b$                                           | $\rightarrow E(7,6)$ |

**Theorem 2.10.8.3 (Leerer Fall).**

$$\forall x \neg P(x) \vdash \exists^{\leq 1} x P(x)$$

|     |   |                                                             |                      |
|-----|---|-------------------------------------------------------------|----------------------|
| 1   | 1 | $\forall x \neg P(x)$                                       | $A$                  |
| 2   | 2 | $P(a) \wedge P(b)$                                          | $A$                  |
| 2   | 3 | $P(a)$                                                      | $\wedge E1(2)$       |
| 1   | 4 | $\neg P(a)$                                                 | $\forall E(1)$       |
| 1,2 | 5 | $a = b$                                                     | Th. 2.1.7.3(3,4)     |
| 1   | 6 | $(P(a) \wedge P(b)) \rightarrow a = b$                      | $\rightarrow I(2,5)$ |
| 1   | 7 | $\forall z((P(a) \wedge P(z)) \rightarrow a = z)$           | $\forall I(6)$       |
| 1   | 8 | $\forall y \forall z((P(y) \wedge P(z)) \rightarrow y = z)$ | $\forall I(7)$       |
| 1   | 9 | $\exists^{\leq 1} x P(x)$                                   | Th. 2.10.8.1(8)      |

**Theorem 2.10.8.4 (Kontrapositionsform).**

$$\exists^{\leq 1} x P(x), a \neq b \vdash \neg(P(a) \wedge P(b))$$

|       |   |                           |                     |
|-------|---|---------------------------|---------------------|
| 1     | 1 | $\exists^{\leq 1} x P(x)$ | $A$                 |
| 2     | 2 | $a \neq b$                | $A$                 |
| 3     | 3 | $P(a) \wedge P(b)$        | $A$                 |
| 3     | 4 | $P(a)$                    | $\wedge E1(3)$      |
| 3     | 5 | $P(b)$                    | $\wedge E2(3)$      |
| 1,3   | 6 | $a = b$                   | Th. 2.10.8.2(1,4,5) |
| 1,2,3 | 7 | $\perp$                   | $\perp I(2,6)$      |
| 1,2   | 8 | $\neg(P(a) \wedge P(b))$  | $\neg I(3,7)$       |

**Theorem 2.10.8.5 (Monotonie des Höchstens-ein-Quantors).**

$$\exists^{\leq 1} x P(x), \forall x(Q(x) \rightarrow P(x)) \vdash \exists^{\leq 1} x Q(x)$$

|   |   |                                    |                |
|---|---|------------------------------------|----------------|
| 1 | 1 | $\exists^{\leq 1} x P(x)$          | $A$            |
| 2 | 2 | $\forall x(Q(x) \rightarrow P(x))$ | $A$            |
| 3 | 3 | $Q(a) \wedge Q(b)$                 | $A$            |
| 3 | 4 | $Q(a)$                             | $\wedge E1(3)$ |
| 3 | 5 | $Q(b)$                             | $\wedge E2(3)$ |
| 2 | 6 | $Q(a) \rightarrow P(a)$            | $\forall E(2)$ |

|       |    |                                                             |                        |
|-------|----|-------------------------------------------------------------|------------------------|
| 2     | 7  | $Q(b) \rightarrow P(b)$                                     | $\forall E(2)$         |
| 2,3   | 8  | $P(a)$                                                      | $\rightarrow E(4, 6)$  |
| 2,3   | 9  | $P(b)$                                                      | $\rightarrow E(5, 7)$  |
| 1,2,3 | 10 | $a = b$                                                     | Th. 2.10.8.2(1, 8, 9)  |
| 1,2   | 11 | $(Q(a) \wedge Q(b)) \rightarrow a = b$                      | $\rightarrow I(3, 10)$ |
| 1,2   | 12 | $\forall z((Q(a) \wedge Q(z)) \rightarrow a = z)$           | $\forall I(11)$        |
| 1,2   | 13 | $\forall y \forall z((Q(y) \wedge Q(z)) \rightarrow y = z)$ | $\forall I(12)$        |
| 1,2   | 14 | $\exists^{\leq 1} x Q(x)$                                   | Th. 2.10.8.1(13)       |

**Theorem 2.10.8.6 (Invarianz unter prädikatenlogischer Äquivalenz).**

$$\exists^{\leq 1} x P(x), \forall x(P(x) \leftrightarrow Q(x)) \vdash \exists^{\leq 1} x Q(x)$$

|     |   |                                                                            |                    |
|-----|---|----------------------------------------------------------------------------|--------------------|
| 1   | 1 | $\exists^{\leq 1} x P(x)$                                                  | $A$                |
| 2   | 2 | $\forall x(P(x) \leftrightarrow Q(x))$                                     | $A$                |
| 2   | 3 | $\forall x(P(x) \rightarrow Q(x)) \wedge \forall x(Q(x) \rightarrow P(x))$ | Th. 2.10.3.2(2)    |
| 2   | 4 | $\forall x(Q(x) \rightarrow P(x))$                                         | $\wedge E2(3)$     |
| 1,2 | 5 | $\exists^{\leq 1} x Q(x)$                                                  | Th. 2.10.8.5(1, 4) |

## 10.9 Eindeutigkeitsquantor

**Im Kapitel verwendete Symbole**

**Prädikatensymbole:**  $P$   
**Variablen:**  $x, y, z, a, b, c$   
**Quantoren:**  $\exists!, \exists^{\leq 1}$   
**Kennzeichnungsoperator:**  $\iota$

**Theorem 2.10.9.1 (Eindeutigkeit des  $\exists!$ -Zeugen).**

$$\exists! x P(x), P(a), P(b) \vdash a = b$$

|       |    |                                     |                           |
|-------|----|-------------------------------------|---------------------------|
| 1     | 1  | $\exists! x P(x)$                   | $A$                       |
| 2     | 2  | $P(c)$                              | $A$                       |
| 3     | 3  | $\forall y(P(y) \rightarrow c = y)$ | $A$                       |
| 4     | 4  | $P(a)$                              | $A$                       |
| 5     | 5  | $P(b)$                              | $A$                       |
| 3     | 6  | $P(a) \rightarrow c = a$            | $\forall E(3)$            |
| 3     | 7  | $P(b) \rightarrow c = b$            | $\forall E(3)$            |
| 3,4   | 8  | $c = a$                             | $\rightarrow E(4, 6)$     |
| 3,5   | 9  | $c = b$                             | $\rightarrow E(5, 7)$     |
| 3,4,5 | 10 | $a = b$                             | Th. 2.9.1.4(8, 9)         |
| 1,4,5 | 11 | $a = b$                             | $\exists! E(1, 2, 3, 10)$ |

**Theorem 2.10.9.2 (Existenzanteil von  $\exists!$ ).**

$$\exists!x P(x) \vdash \exists x P(x)$$

|   |   |                                     |                         |
|---|---|-------------------------------------|-------------------------|
| 1 | 1 | $\exists!x P(x)$                    | $A$                     |
| 2 | 2 | $P(a)$                              | $A$                     |
| 3 | 3 | $\forall y(P(y) \rightarrow a = y)$ | $A$                     |
| 2 | 4 | $\exists x P(x)$                    | $\exists I(2)$          |
| 1 | 5 | $\exists x P(x)$                    | $\exists!E(1, 2, 3, 4)$ |

**Theorem 2.10.9.3 (Höchstens-ein-Anteil von  $\exists!$ ).**

$$\exists!x P(x) \vdash \exists^{\leq 1} x P(x)$$

|     |   |                                                            |                       |
|-----|---|------------------------------------------------------------|-----------------------|
| 1   | 1 | $\exists!x P(x)$                                           | $A$                   |
| 2   | 2 | $P(a) \wedge P(b)$                                         | $A$                   |
| 2   | 3 | $P(a)$                                                     | $\wedge E1(2)$        |
| 2   | 4 | $P(b)$                                                     | $\wedge E2(2)$        |
| 1,2 | 5 | $a = b$                                                    | Th. 2.10.9.1(1, 3, 4) |
| 1   | 6 | $(P(a) \wedge P(b)) \rightarrow a = b$                     | $\rightarrow I(2, 5)$ |
| 1   | 7 | $\forall z((P(a) \wedge P(z)) \rightarrow a = z)$          | $\forall I(6)$        |
| 1   | 8 | $\forall y\forall z((P(y) \wedge P(z)) \rightarrow y = z)$ | $\forall I(7)$        |
| 1   | 9 | $\exists^{\leq 1} x P(x)$                                  | Th. 2.10.8.1(8)       |

**Theorem 2.10.9.4 (Eindeutige Existenz aus Existenz und Höchstens-eins).**

$$\exists x P(x), \exists^{\leq 1} x P(x) \vdash \exists!x P(x)$$

*Beweis.*

Existenz:

|   |   |                           |     |
|---|---|---------------------------|-----|
| 1 | 1 | $\exists x P(x)$          | $A$ |
| 2 | 2 | $\exists^{\leq 1} x P(x)$ | $A$ |

Eindeutigkeit:

|       |   |         |                       |
|-------|---|---------|-----------------------|
| 3     | 3 | $P(a)$  | $A$                   |
| 4     | 4 | $P(b)$  | $A$                   |
| 2,3,4 | 5 | $a = b$ | Th. 2.10.8.2(2, 3, 4) |

Eindeutige Existenz:

$$1,2 \quad 6 \quad \exists!x P(x)$$

$$\exists!I(1, 3, 4, 5)$$

□

**Theorem 2.10.9.5 (Zerlegung der eindeutigen Existenz).**

$$\exists!x P(x) \dashv\vdash \exists x P(x) \wedge \exists^{\leq 1} x P(x)$$

*Beweis.*

$\vdash$ :

|   |   |                                                 |                  |
|---|---|-------------------------------------------------|------------------|
| 1 | 1 | $\exists!x P(x)$                                | $A$              |
| 1 | 2 | $\exists x P(x)$                                | Th. 2.10.9.2(1)  |
| 1 | 3 | $\exists^{\leq 1} x P(x)$                       | Th. 2.10.9.3(1)  |
| 1 | 4 | $\exists x P(x) \wedge \exists^{\leq 1} x P(x)$ | $\wedge I(2, 3)$ |

$\dashv$ :

|   |   |                                                 |                    |
|---|---|-------------------------------------------------|--------------------|
| 1 | 1 | $\exists x P(x) \wedge \exists^{\leq 1} x P(x)$ | $A$                |
| 1 | 2 | $\exists x P(x)$                                | $\wedge E1(1)$     |
| 1 | 3 | $\exists^{\leq 1} x P(x)$                       | $\wedge E2(1)$     |
| 1 | 4 | $\exists!x P(x)$                                | Th. 2.10.9.4(2, 3) |

□

**Theorem 2.10.9.6 (Geschachtelte eindeutige Existenz aus gemeinsamer Eindeutigkeit).**

**Prädikate:**  $F$

**Variablen:**  $X, Y, x, x', y, y', a, c, d, e, f$

$$\begin{aligned} & \exists x \in X \exists y \in Y F(x, y), \\ & \forall x, x' \in X \forall y, y' \in Y \left( (F(x, y) \wedge F(x', y')) \rightarrow (x = x' \wedge y = y') \right) \\ & \vdash \exists!x \in X \exists!y \in Y F(x, y) \end{aligned}$$

*Beweis.*

Existenz eines ersten Zeugen:

|   |   |                                                                                                                        |     |
|---|---|------------------------------------------------------------------------------------------------------------------------|-----|
| 1 | 1 | $\exists x \in X \exists y \in Y F(x, y)$                                                                              | $A$ |
| 2 | 2 | $\forall x, x' \in X \forall y, y' \in Y \left( (F(x, y) \wedge F(x', y')) \rightarrow (x = x' \wedge y = y') \right)$ | $A$ |
| 3 | 3 | $a \in X \wedge \exists y \in Y F(a, y)$                                                                               | $A$ |

|   |   |                           |                |
|---|---|---------------------------|----------------|
| 3 | 4 | $a \in X$                 | $\wedge E1(3)$ |
| 3 | 5 | $\exists y \in Y F(a, y)$ | $\wedge E2(3)$ |

Eindeutigkeit des zweiten Zeugen:

|         |    |                                                             |                                                                                                                          |
|---------|----|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 6       | 6  | $c \in Y \wedge F(a, c)$                                    | $A$                                                                                                                      |
| 6       | 7  | $c \in Y$                                                   | $\wedge E1(6)$                                                                                                           |
| 6       | 8  | $F(a, c)$                                                   | $\wedge E2(6)$                                                                                                           |
| 9       | 9  | $d \in Y \wedge F(a, d)$                                    | $A$                                                                                                                      |
| 9       | 10 | $d \in Y$                                                   | $\wedge E1(9)$                                                                                                           |
| 9       | 11 | $F(a, d)$                                                   | $\wedge E2(9)$                                                                                                           |
| 2,3,6,9 | 12 | $(F(a, c) \wedge F(a, d)) \rightarrow (a = a \wedge c = d)$ | $\rightarrow E(10, \forall E(\rightarrow E(7, \forall E(\rightarrow E(4, \forall E(\rightarrow E(4, \forall E(2))))))))$ |
| 6,9     | 13 | $F(a, c) \wedge F(a, d)$                                    | $\wedge I(8, 11)$                                                                                                        |
| 2,3,6,9 | 14 | $a = a \wedge c = d$                                        | $\rightarrow E(13, 12)$                                                                                                  |
| 2,3,6,9 | 15 | $c = d$                                                     | $\wedge E2(14)$                                                                                                          |
| 2,3     | 16 | $\exists! y \in Y F(a, y)$                                  | $\exists! I(5, 6, 9, 15)$                                                                                                |

Existenz mit einzigem zweiten Zeugen:

|     |    |                                            |                       |
|-----|----|--------------------------------------------|-----------------------|
| 2,3 | 17 | $a \in X \wedge \exists! y \in Y F(a, y)$  | $\wedge I(4, 16)$     |
| 2,3 | 18 | $\exists x \in X \exists! y \in Y F(x, y)$ | $\exists I(17)$       |
| 1,2 | 19 | $\exists x \in X \exists! y \in Y F(x, y)$ | $\exists E(1, 3, 18)$ |

Eindeutigkeit des ersten Zeugen:

|    |    |                                           |                  |
|----|----|-------------------------------------------|------------------|
| 20 | 20 | $c \in X \wedge \exists! y \in Y F(c, y)$ | $A$              |
| 21 | 21 | $d \in X \wedge \exists! y \in Y F(d, y)$ | $A$              |
| 20 | 22 | $c \in X$                                 | $\wedge E1(20)$  |
| 20 | 23 | $\exists! y \in Y F(c, y)$                | $\wedge E2(20)$  |
| 21 | 24 | $d \in X$                                 | $\wedge E1(21)$  |
| 21 | 25 | $\exists! y \in Y F(d, y)$                | $\wedge E2(21)$  |
| 20 | 26 | $\exists y \in Y F(c, y)$                 | Th. 2.10.9.2(23) |
| 21 | 27 | $\exists y \in Y F(d, y)$                 | Th. 2.10.9.2(25) |
| 28 | 28 | $e \in Y \wedge F(c, e)$                  | $A$              |
| 28 | 29 | $e \in Y$                                 | $\wedge E1(28)$  |
| 28 | 30 | $F(c, e)$                                 | $\wedge E2(28)$  |
| 31 | 31 | $f \in Y \wedge F(d, f)$                  | $A$              |
| 31 | 32 | $f \in Y$                                 | $\wedge E1(31)$  |
| 31 | 33 | $F(d, f)$                                 | $\wedge E2(31)$  |

|                   |    |                                                             |                                                                                                                                                |
|-------------------|----|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 2,20,21,<br>28,31 | 34 | $(F(c, e) \wedge F(d, f)) \rightarrow (c = d \wedge e = f)$ | $\rightarrow E(32, \forall E(\rightarrow E$<br>$(29, \forall E(\rightarrow E(24, \forall E$<br>$(\rightarrow E(22, \forall E$<br>$(2)))))))))$ |
| 28,31             | 35 | $F(c, e) \wedge F(d, f)$                                    | $\wedge I(30, 33)$                                                                                                                             |
| 2,20,21,<br>28,31 | 36 | $c = d \wedge e = f$                                        | $\rightarrow E(35, 34)$                                                                                                                        |
| 2,20,21,<br>28,31 | 37 | $c = d$                                                     | $\wedge E1(36)$                                                                                                                                |
| 2,20,21,<br>28    | 38 | $c = d$                                                     | $\exists E(27, 31, 37)$                                                                                                                        |
| 2,20,21           | 39 | $c = d$                                                     | $\exists E(26, 28, 38)$                                                                                                                        |

Geschachtelte eindeutige Existenz:

$$1,2 \quad 40 \quad \exists!x \in X \exists!y \in Y F(x, y) \quad \exists!I(19, 20, 21, 39)$$

□

**Theorem 2.10.9.7 (Eindeutige Existenz aus einem festen Zeugen).**

**Variablen:**  $a, b, c, x$   
**Prädikat:**  $P$

$$P(c), \forall x (P(x) \rightarrow x = c) \vdash \exists!x P(x)$$

|       |   |                                      |                                  |
|-------|---|--------------------------------------|----------------------------------|
| 1     | 1 | $P(c)$                               | $A$                              |
| 2     | 2 | $\forall x (P(x) \rightarrow x = c)$ | $A$                              |
| 1     | 3 | $\exists x P(x)$                     | $\exists I(1)$                   |
| 4     | 4 | $P(a)$                               | $A$                              |
| 5     | 5 | $P(b)$                               | $A$                              |
| 2,4   | 6 | $a = c$                              | $\rightarrow E(4, \forall E(2))$ |
| 2,5   | 7 | $b = c$                              | $\rightarrow E(5, \forall E(2))$ |
| 2,4,5 | 8 | $a = b$                              | Th. 2.9.1.3(6, 7)                |
| 1,2   | 9 | $\exists!x P(x)$                     | $\exists!I(3, 4, 5, 8)$          |

## 10.10 Konditionaler Aussagenoperator $\text{if}^-(P, Q, R)$

**Definition 2.10.10.1 (Konditionaler Aussagenoperator).**

**Aussagenvariablen:**  $P, Q, R$   
**Neue Symbole:**  $\text{if}^-(P, Q, R)$

$$\text{if}^-(P, Q, R) := (P \wedge Q) \vee (\neg P \wedge R)$$

**Theorem 2.10.10.1 (Implikationsform des konditionalen Aussagenoperators).**

**Aussagenvariablen:**  $P, Q, R$

$$\text{if}^-(P, Q, R) \dashv\vdash (P \rightarrow Q) \wedge (\neg P \rightarrow R)$$

*Beweis.*

Wahrer Bedingungsfall:

|     |    |                                       |                         |
|-----|----|---------------------------------------|-------------------------|
| 1   | 1  | $\text{if}^-(P, Q, R)$                | $A$                     |
| 1   | 2  | $(P \wedge Q) \vee (\neg P \wedge R)$ | Def. 2.10.10.1(1)       |
| 3   | 3  | $P$                                   | $A$                     |
| 4   | 4  | $P \wedge Q$                          | $A$                     |
| 4   | 5  | $Q$                                   | $\wedge E2(4)$          |
| 6   | 6  | $\neg P \wedge R$                     | $A$                     |
| 6   | 7  | $\neg P$                              | $\wedge E1(6)$          |
| 3,6 | 8  | $Q$                                   | Th. 2.1.7.3(3, 7)       |
| 1,3 | 9  | $Q$                                   | $\vee E(2, 4, 5, 6, 8)$ |
| 1   | 10 | $P \rightarrow Q$                     | $\rightarrow I(3, 9)$   |

Falscher Bedingungsfall:

|       |    |                        |                             |
|-------|----|------------------------|-----------------------------|
| 11    | 11 | $\neg P$               | $A$                         |
| 12    | 12 | $P \wedge Q$           | $A$                         |
| 12    | 13 | $P$                    | $\wedge E1(12)$             |
| 11,12 | 14 | $R$                    | Th. 2.1.7.3(13, 11)         |
| 15    | 15 | $\neg P \wedge R$      | $A$                         |
| 15    | 16 | $R$                    | $\wedge E2(15)$             |
| 1,11  | 17 | $R$                    | $\vee E(2, 12, 14, 15, 16)$ |
| 1     | 18 | $\neg P \rightarrow R$ | $\rightarrow I(11, 17)$     |

Zusammenführung der Hinrichtung:

|   |    |                                                   |                    |
|---|----|---------------------------------------------------|--------------------|
| 1 | 19 | $(P \rightarrow Q) \wedge (\neg P \rightarrow R)$ | $\wedge I(10, 18)$ |
|---|----|---------------------------------------------------|--------------------|

$\dashv\vdash$ :

|   |   |                                                   |                |
|---|---|---------------------------------------------------|----------------|
| 1 | 1 | $(P \rightarrow Q) \wedge (\neg P \rightarrow R)$ | $A$            |
| 1 | 2 | $P \rightarrow Q$                                 | $\wedge E1(1)$ |
| 1 | 3 | $\neg P \rightarrow R$                            | $\wedge E2(1)$ |
|   | 4 | $P \vee \neg P$                                   | Th. 2.1.7.1    |
| 5 | 5 | $P$                                               | $A$            |

|     |    |                                       |                             |
|-----|----|---------------------------------------|-----------------------------|
| 1,5 | 6  | $Q$                                   | $\rightarrow E(2, 5)$       |
| 1,5 | 7  | $P \wedge Q$                          | $\wedge I(5, 6)$            |
| 1,5 | 8  | $(P \wedge Q) \vee (\neg P \wedge R)$ | $\vee I1(7)$                |
| 9   | 9  | $\neg P$                              | $A$                         |
| 1,9 | 10 | $R$                                   | $\rightarrow E(3, 9)$       |
| 1,9 | 11 | $\neg P \wedge R$                     | $\wedge I(9, 10)$           |
| 1,9 | 12 | $(P \wedge Q) \vee (\neg P \wedge R)$ | $\vee I2(11)$               |
| 1   | 13 | $(P \wedge Q) \vee (\neg P \wedge R)$ | $\vee E(4, 5, 8, 9, 12)$    |
| 1   | 14 | $\text{if}^-(P, Q, R)$                | $\text{Def. 2.10.10.1(13)}$ |

□

**Theorem 2.10.10.2 (Falltermformel als Instanz von  $\text{if}^-(P, Q, R)$ ).**

**Aussagenvariablen:**  $P$   
**Variablen:**  $a, b, u$

$$\text{if}^-(P, u = a, u = b) \dashv\vdash (P \rightarrow u = a) \wedge (\neg P \rightarrow u = b)$$

*Beweis.*

$\vdash$ :

|   |   |                                                           |                           |
|---|---|-----------------------------------------------------------|---------------------------|
| 1 | 1 | $\text{if}^-(P, u = a, u = b)$                            | $A$                       |
| 1 | 2 | $(P \rightarrow u = a) \wedge (\neg P \rightarrow u = b)$ | $\text{Th. 2.10.10.1(1)}$ |

$\dashv\vdash$ :

|   |   |                                                           |                           |
|---|---|-----------------------------------------------------------|---------------------------|
| 1 | 1 | $(P \rightarrow u = a) \wedge (\neg P \rightarrow u = b)$ | $A$                       |
| 1 | 2 | $\text{if}^-(P, u = a, u = b)$                            | $\text{Th. 2.10.10.1(1)}$ |

□

## 10.11 Fallterme

**Theorem 2.10.11.1 (Wohldefiniertheit).**

**Aussagen:**  $P$   
**Mengen:**  $a, b, u, v, w$

$$P \vee \neg P \vdash \exists! u \left( (P \rightarrow u = a) \wedge (\neg P \rightarrow u = b) \right)$$

*Beweis.*

Existenz:

|   |    |                                                                                    |                          |
|---|----|------------------------------------------------------------------------------------|--------------------------|
| 1 | 1  | $P \vee \neg P$                                                                    | $A$                      |
| 2 | 2  | $P$                                                                                | $A$                      |
|   | 3  | $a = a$                                                                            | $= I$                    |
| 2 | 4  | $P \wedge a = a$                                                                   | $\wedge I(2, 3)$         |
| 2 | 5  | $(P \wedge a = a) \vee (\neg P \wedge a = b)$                                      | $\vee I1(4)$             |
| 2 | 6  | $\text{if}^-(P, a = a, a = b)$                                                     | Def. 2.10.10.1(5)        |
| 2 | 7  | $(P \rightarrow a = a) \wedge (\neg P \rightarrow a = b)$                          | Th. 2.10.10.2(6)         |
| 2 | 8  | $\exists u \left( (P \rightarrow u = a) \wedge (\neg P \rightarrow u = b) \right)$ | $\exists I(7)$           |
| 9 | 9  | $\neg P$                                                                           | $A$                      |
|   | 10 | $b = b$                                                                            | $= I$                    |
| 9 | 11 | $\neg P \wedge b = b$                                                              | $\wedge I(9, 10)$        |
| 9 | 12 | $(P \wedge b = a) \vee (\neg P \wedge b = b)$                                      | $\vee I2(11)$            |
| 9 | 13 | $\text{if}^-(P, b = a, b = b)$                                                     | Def. 2.10.10.1(12)       |
| 9 | 14 | $(P \rightarrow b = a) \wedge (\neg P \rightarrow b = b)$                          | Th. 2.10.10.2(13)        |
| 9 | 15 | $\exists u \left( (P \rightarrow u = a) \wedge (\neg P \rightarrow u = b) \right)$ | $\exists I(14)$          |
| 1 | 16 | $\exists u \left( (P \rightarrow u = a) \wedge (\neg P \rightarrow u = b) \right)$ | $\vee E(1, 2, 8, 9, 15)$ |

Eindeutigkeit:

|          |    |                                                           |                             |
|----------|----|-----------------------------------------------------------|-----------------------------|
| 17       | 17 | $(P \rightarrow v = a) \wedge (\neg P \rightarrow v = b)$ | $A$                         |
| 18       | 18 | $(P \rightarrow w = a) \wedge (\neg P \rightarrow w = b)$ | $A$                         |
| 19       | 19 | $P$                                                       | $A$                         |
| 17       | 20 | $P \rightarrow v = a$                                     | $\wedge E1(17)$             |
| 17,19    | 21 | $v = a$                                                   | $\rightarrow E(19, 20)$     |
| 18       | 22 | $P \rightarrow w = a$                                     | $\wedge E1(18)$             |
| 18,19    | 23 | $w = a$                                                   | $\rightarrow E(19, 22)$     |
| 17,18,19 | 24 | $v = w$                                                   | Th. 2.9.1.3(21, 23)         |
| 25       | 25 | $\neg P$                                                  | $A$                         |
| 17       | 26 | $\neg P \rightarrow v = b$                                | $\wedge E2(17)$             |
| 17,25    | 27 | $v = b$                                                   | $\rightarrow E(25, 26)$     |
| 18       | 28 | $\neg P \rightarrow w = b$                                | $\wedge E2(18)$             |
| 18,25    | 29 | $w = b$                                                   | $\rightarrow E(25, 28)$     |
| 17,18,25 | 30 | $v = w$                                                   | Th. 2.9.1.3(27, 29)         |
| 1,17,18  | 31 | $v = w$                                                   | $\vee E(1, 19, 24, 25, 30)$ |

Eindeutige Existenz:

|   |    |                                                                                     |                              |
|---|----|-------------------------------------------------------------------------------------|------------------------------|
| 1 | 32 | $\exists! u \left( (P \rightarrow u = a) \wedge (\neg P \rightarrow u = b) \right)$ | $\exists! I(16, 17, 18, 31)$ |
|---|----|-------------------------------------------------------------------------------------|------------------------------|

□

**Definition 2.10.11.1 (Notation des Fallterms).**

**Aussagenvariablen:**  $P$   
**Variablen:**  $a, b, u$   
**Definition:**

**Fallterm:** 
$$\begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} := \iota u \left( (P \rightarrow u = a) \wedge (\neg P \rightarrow u = b) \right)$$

**Neue Symbole:**

$$\begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases}$$

**Theorem 2.10.11.2 (Charakterisierung des Fallterms).**

**Aussagen:**  $P$   
**Variablen:**  $a, b$

$$\begin{aligned} & \vdash (P \rightarrow \begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = a) \wedge (\neg P \rightarrow \begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = b) \\ & \quad 1 \quad (P \rightarrow \begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = a) \wedge (\neg P \rightarrow \begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = b) \quad \text{Def. 2.10.11.1} \end{aligned}$$

**Theorem 2.10.11.3.**

**Aussagen:**  $P$   
**Variablen:**  $a, b$

$$\begin{aligned} & P \vdash \begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = a \\ & \quad 1 \quad 1 \quad P \quad A \\ & \quad 2 \quad (P \rightarrow \begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = a) \wedge (\neg P \rightarrow \begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = b) \quad \text{Def. 2.10.11.1} \\ & \quad 3 \quad P \rightarrow \begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = a \quad \wedge E1(2) \\ & \quad 1 \quad 4 \quad \begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = a \quad \rightarrow E(1, 3) \end{aligned}$$

**Theorem 2.10.11.4.**

**Aussagen:**  $P$   
**Variablen:**  $a, b$

$$\neg P \vdash \begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = b$$

|   |   |                                                                                                                                                                                                               |                       |
|---|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 1 | 1 | $\neg P$                                                                                                                                                                                                      | $A$                   |
|   | 2 | $(P \rightarrow \begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = a) \wedge (\neg P \rightarrow \begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = b)$ | Def. 2.10.11.1        |
|   | 3 | $\neg P \rightarrow \begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = b$                                                                                                          | $\wedge E2(2)$        |
| 1 | 4 | $\begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = b$                                                                                                                             | $\rightarrow E(1, 3)$ |

**Theorem 2.10.11.5.**

**Aussagen:**  $P$

**Variablen:**  $a, b$

$$a = b \vdash \begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = a$$

|     |   |                                                                                   |                         |
|-----|---|-----------------------------------------------------------------------------------|-------------------------|
| 1   | 1 | $a = b$                                                                           | $A$                     |
|     | 2 | $P \vee \neg P$                                                                   | Th. 2.1.7.1             |
| 3   | 3 | $P$                                                                               | $A$                     |
| 3   | 4 | $\begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = a$ | Th. 2.10.11.3(3)        |
| 5   | 5 | $\neg P$                                                                          | $A$                     |
| 5   | 6 | $\begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = b$ | Th. 2.10.11.4(5)        |
| 1   | 7 | $b = a$                                                                           | Th. 2.9.1.1(1)          |
| 1,5 | 8 | $\begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = a$ | Th. 2.9.1.2(6, 7)       |
| 1   | 9 | $\begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = a$ | $\vee E(2, 3, 4, 5, 8)$ |

**Theorem 2.10.11.6.**

**Aussagen:**  $P$

**Variablen:**  $a, b$

$$a = b \vdash \begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = b$$

|   |   |                                                                                   |                  |
|---|---|-----------------------------------------------------------------------------------|------------------|
| 1 | 1 | $a = b$                                                                           | $A$              |
| 1 | 2 | $\begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = a$ | Th. 2.10.11.5(1) |

$$1 \quad 3 \quad \begin{cases} a, & \text{falls } P \\ b, & \text{falls } \neg P \end{cases} = b$$

Th. 2.9.1.2(2, 1)